

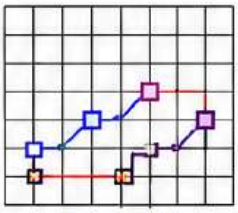
PHYSICAL DESIGN
ROUTING

Routing is the process of connecting all the placed standard cells using metal interconnect layers to form a complete, DRC clean and manufacturable design.

TWO STAGES OF ROUTING

1) GLOBAL ROUTING (GR)

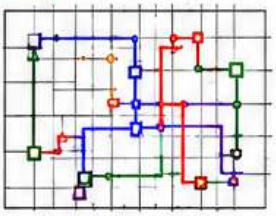
- Coarse routing to determine path for all nets.
- Provides initial guide for detailed routing.
- Estimates wirelength, congestion and timing.



Coarse Routes

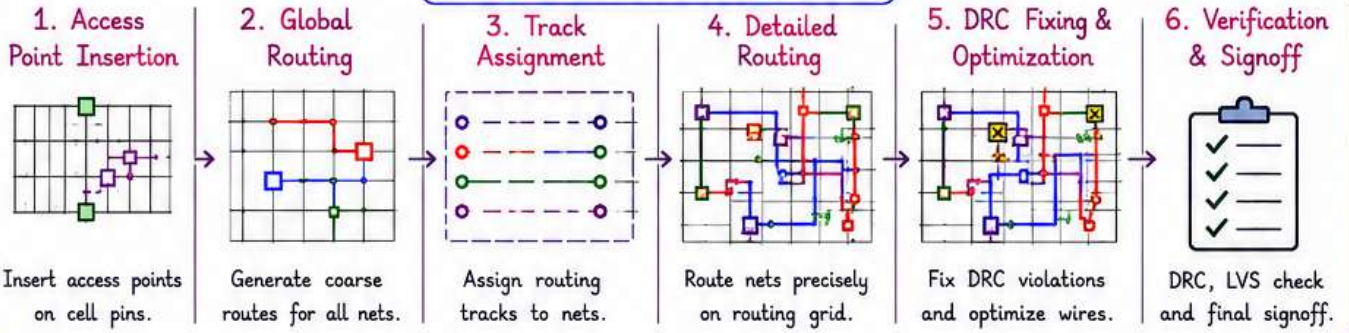
2) DETAILED ROUTING (DR)

- Creates exact routes on routing grid.
- Follows design rules accurately.
- Produces final, signoff ready layout.

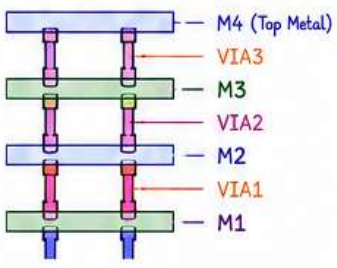


Detailed Routes

ROUTING STAGES (FLOW)



VIA STRUCTURE IN LAYERS



TYPES OF ROUTING

- By Abstraction
- By Layer Usage
- By Direction
- By Technology



IMPORTANT CHECKS

- DRC (Spacing, width, vias)
- Antenna checks
- Min area / Metal density
- LVS (Netlist match)
- Signal Integrity
- Timing impact

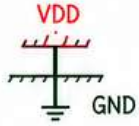
GOOD ROUTING DELIVERS

- ✓ Clean DRC & LVS
- ✓ Correct connectivity
- ✓ Optimized wirelength
- ✓ Better performance
- ✓ Reliable manufacturing
- ✓ On-time tapeout

KEY POINT

- Routing connects all cells using metal layers to form a complete, DRC clean and manufacturable design.

Good routing is not just about connection, it's about reliability, performance and silicon success!



Author :
Konda Tharun Vivekananda

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PD_Routing

1. INTRODUCTION

- Routing is the process of connecting all the placed standard cells and macros using available metal interconnect layers to form a complete, DRC clean and manufacturable design.

2. DEFINITION

- Routing is the physical implementation step that creates metal interconnections between pins/nets as per the design connectivity while satisfying DRC, timing, SI and reliability constraints.

3. GOALS

- Establish correct connectivity for all nets
- Meet timing (setup/hold) and minimize delay
- Avoid congestion and routing hotspots
- Satisfy DRC, antenna, SI and reliability rules
- Optimize wirelength, vias and overall QoR
- Ensure manufacturability and high yield

4. INPUTS REQUIRED

1. Placed DEF (cell locations & pin info)
2. LEF files (tech, layers, design rules)
3. Technology files (layer map, vias, rules)
4. Netlist (connectivity)
5. Timing libraries (for RC estimation)
6. Design rules (DRC, SI, antenna, EM limits)

5. CHECKS NEED BEFORE ROUTING

1. Floorplan & Power Plan finalized?
2. Placement legalized and optimized?
3. No major timing violations (critical nets analyzed)?
4. Congestion analysis passed?
5. DRC clean till placement stage?
6. Clock tree synthesized and buffers inserted?
7. Blockages / Regions defined correctly?
8. All macros have power and tap connections?
9. Routing layer stack and track allocation OK?
10. Antenna rules and SI constraints reviewed?

KEY POINT

Routing converts logical connectivity into physical interconnections using metal layers while meeting DRC, timing, SI and reliability constraints.

TWO STAGES OF ROUTING

1. Global Routing (GR)
   - Estimates routing resources on coarse grid
   - Finds routing paths and reserves tracks
   - Provides congestion and wirelength estimate
   - Output: Guide (preferred directions, tracks)
2. Detailed Routing (DR)
   - Routes each net precisely on actual grid
   - Adds vias, meets DRC and antenna rules
   - Optimizes wire shape, spacing and SI
   - Output: DRC clean routed DEF

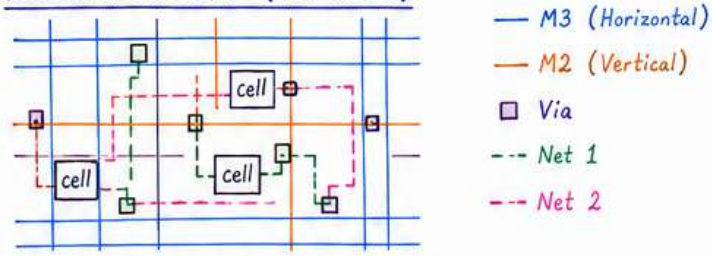
PRACTICAL OBSERVATION

- Better global routing → lower congestion
- Poor congestion → detours → more delay
- Good track allocation → higher routability

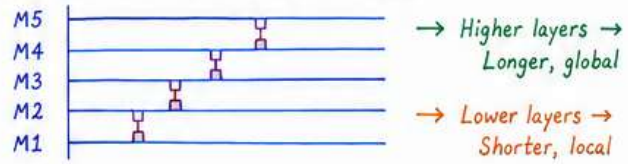
GLOBAL ROUTING vs DETAILED ROUTING

Parameter	Global Routing	Detailed Routing
Purpose	Estimate & plan routing	Create exact routes
Input	Placed DEF, Netlist, Tech files	GR guide, DEF, Tech files
Method	Coarse grid, fast congestion aware	Exact grid, DRC aware
Output	Routing guides, Track allocation	DRC clean routes, Final DEF
Main Concern	Congestion, Wirelength	DRC, SI, Antenna
Typical Issue	Over congestion	DRC violations, Antenna violation

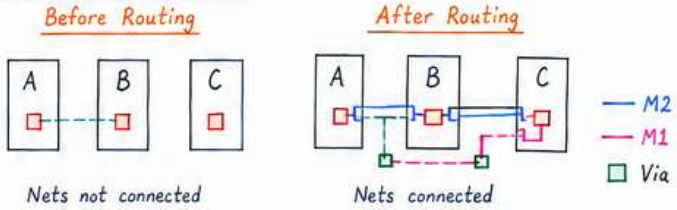
ROUTING CONCEPT (TOP VIEW)



METAL LAYER STACK (EXAMPLE)



ROUTING EXAMPLE



IMPORTANT CHECKS DURING ROUTING

- DRC (spacing, width, via rules)
- Antenna check
- Min area / Metal density
- Cut layer / Via rules
- Off-grid / design rule check
- Wirelength & delay
- Crosstalk / Coupling
- EM (current, IR drop)
- Signal integrity
- Timing impact

CONGESTION LEVELS



HIGH CONGESTION LEADS TO

Detours in routing
↓
More wirelength
↓
More delay
↓
Timing degradation
↓
Possible routing failure

INTERVIEW / VIVA POINT

Q. What is the difference between Global and Detailed routing?
→ Global routing estimates the routing resources and creates routing guides, while detailed routing creates exact, DRC clean routes using actual metal layers.

Q. Why is congestion important in routing?
→ High congestion causes detours, increases wirelength, delay and may even lead to routing failure.

SUMMARY / REVISION POINTS

1. Routing connects all nets using metal layers as per design rules.
2. Two main stages: Global Routing and Detailed Routing.
3. Routing must satisfy DRC, antenna and SI rules.
4. Congestion control is critical for successful routing.
5. Good routing improves timing, power and yield.
6. Final output is DRC clean routed DEF ready for sign-off.

SIGN-OFF CHECKLIST

- ✓ Timing Closure
- ✓ DRC Clean
- ✓ LVS Clean
- ✓ IR Drop
- ✓ Electromigration
- ✓ Antenna
- ✓ Signal Integrity
- ✓ Power Integrity
- ✓ Metal Density
- ✓ Final Verification

METRICS AFFECTED BY ROUTING

↑ Wirelength → Timing
↑ Vias → Resistance + Capacitance
↑ Congestion → Delay & Power
Violations → DRC / Ant / SI / EM

PD_Routing

1. ROUTING ORDER

☑ Routing order defines the sequence in which nets are routed. Proper order reduces congestion and improves routability.

Typical Routing Order (Most Critical First)

1. Power (VDD/GND) → 2. Clock Nets → 3. High Speed Nets → 4. Signal Nets → 5. Low Priority Nets

Why this order ?

- Power nets need wide wires & low IR-drop.
- Clock nets require low skew and good shielding.
- High speed nets are critical for timing closure.
- Remaining nets have lower priority.

2. TYPES OF ROUTING

A. By Abstraction
- Global Routing (GR) – Coarse, estimates paths on a grid.
- Detailed Routing (DR) – Exact routes on actual layers.

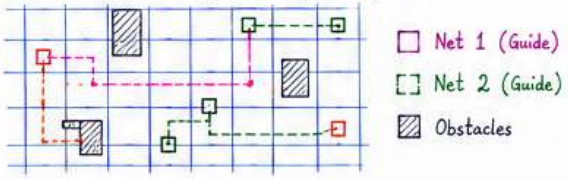
B. By Layer Usage
- Single Layer Routing – Uses only one metal layer.
- Multi Layer Routing – Uses multiple metal layers & vias.
- Mixed Mode Routing – Combination of both.

C. By Direction
- Horizontal Routing → Wires in horizontal direction.
- Vertical Routing → Wires in vertical direction.
- 45° Routing → Used in some advanced technologies.

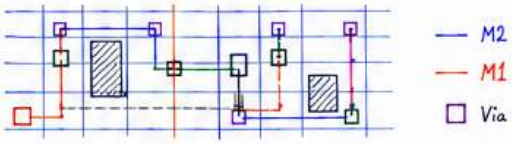
3. ROUTING STAGES

Routing is performed in two main stages.

A. Global Routing (GR)
→ Performed on abstract grid.
→ Estimates routing resources and finds possible paths.
→ Outputs routing guides / tracks for nets.
→ Helps in congestion analysis and layer assignment.



B. Detailed Routing (DR)
→ Uses routing guides from GR.
→ Exact wire shapes are created on actual metal layers.
→ Adds vias, meets DRC rules and antenna rules.
→ Output is DRC clean routed DEF.



KEY POINT

Routing converts routing guides into manufacturable interconnects using metal layers and vias while satisfying DRC, timing, SI and reliability constraints.

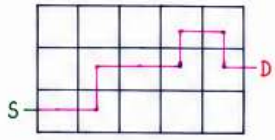
PRACTICAL OBSERVATION

- Good routing order reduces rework and congestion.
- Global routing quality strongly impacts final routability.
- Multi-layer routing is essential for timing and congestion.
- DRC clean routing avoids post-route violations.

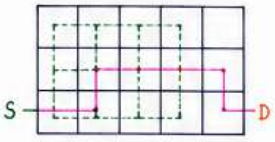
4. ROUTING ALGORITHMS

Different algorithms are used in GR and DR tools.

1. Maze Routing
- Simple and widely used.
- Explores grid in BFS manner.
- May produce non-optimal paths.



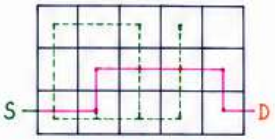
2. Lee's Algorithm (Improved Maze)
- Guarantees shortest path.
- Uses wave expansion.
- Higher memory usage.



3. Dijkstra's Algorithm
- Finds minimum cost path.
- Considers wire length, congestion cost.
- Used in advanced routers.



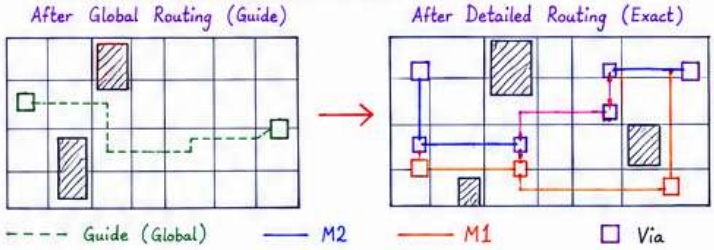
4. A* (A-Star) Algorithm
- Heuristic based.
- Fast and efficient.
- Popular in modern routers.



5. GLOBAL ROUTING vs DETAILED ROUTING (SUMMARY)

Parameter	Global Routing (GR)	Detailed Routing (DR)
Purpose	Estimate & plan routing	Create exact routes
Input	Placed DEF, Netlist, Tech files	GR guides, DEF, Tech files
Method	Coarse grid, fast congestion aware	Exact grid, DRC aware
Output	Routing guides, Track allocation	DRC clean routes, Final DEF
Main Concern	Congestion, Wirelength	DRC, SI, Antenna
Typical Issue	Over congestion	DRC violations, Antenna violation

6. ROUTING EXAMPLE (SAME NET)



7. WHY WE NEED ROUTING ?

☑ Creates physical interconnections between all cells, macros & I/Os.
☑ Ensures design connectivity as per netlist.
☑ Meets timing, power, SI and reliability requirements.
☑ Satisfies manufacturability and sign-off rules (DRC, Antenna, etc.).
☑ Enables final layout generation for tape-out.

INTERVIEW / VIVA POINT

Q. What is the main objective of routing ?
→ To create manufacturable interconnections satisfying DRC, timing, SI and reliability.

Q. What are the two main stages of routing ?
→ Global Routing and Detailed Routing.

Q. Why is routing order important ?
→ Proper order reduces congestion and improves timing and routability.

SUMMARY / REVISION POINTS

1. Routing connects all nets physically.
2. Routing order: Power → Clock → High speed → Signal → Low priority.
3. Two main types: Global & Detailed.
4. Global routing gives guides; detailed routing rreates exact wires.
5. Algorithms used: Maze, Lee's, Dijkstra's, A*.
6. Routing ensures DRC clean, reliable and timing closed design.

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PD_Routing

1. INTRODUCTION

☑ Metal stack (routing layers) is the set of metal layers provided in a technology file used to route signals and power inside the chip. Proper use of metal stack ensures low resistance, low capacitance, less congestion and good timing.

2. DEFINITION

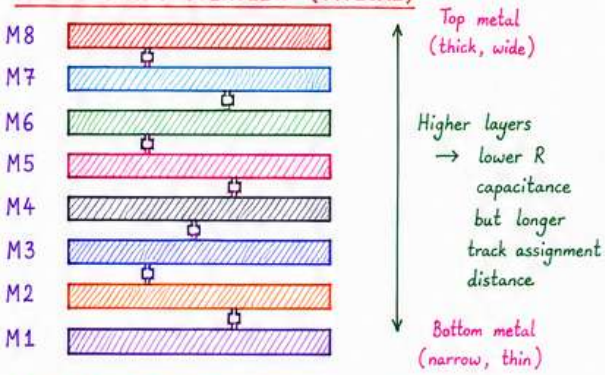
☑ Metal stack defines the available routing layers (M1..M_n) along with their width, spacing, preferred direction, pitch, thickness and via rules. These layers are used in global routing and detailed routing to form electrical interconnections.

3. GOALS OF METAL STACK

☑ Provide sufficient routing resources across the design.

- Minimize resistance (R) and capacitance (C) on critical nets.
- Support high performance and low power.
- Reduce congestion by using hierarchical layers.
- Follow preferred directions to avoid unnecessary vias.
- Meet DRC and reliability requirements.

4. METAL STACK OVERVIEW (TYPICAL)



5. PREFERRED ROUTING DIRECTION PER LAYER

Layer	Preferred Direction	Reason
M1	Horizontal (← →)	Better transistor gate alignment & diffusion rules
M2	Vertical (↑ ↓)	Orthogonal to M1, reduces jogs
M3	Horizontal	Longer wires, low
M4	Vertical	Crosses M3, reduces congestion
M5	Horizontal	Used for global nets
M6	Vertical	High performance nets
M7	Horizontal	Low distance buses
M8 / Top	Vertical	Power / clock / global

✳ Note : Alternate layers use orthogonal directions.

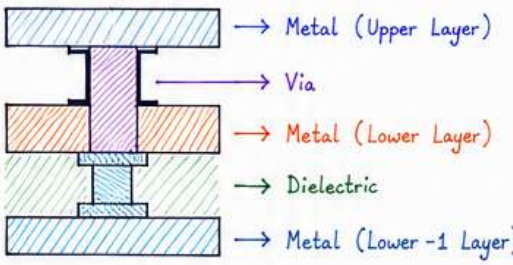
KEY POINT

Lower metal layers (M1–M2) are for local connections with high density. Higher metal layers (M5+) are for global routing with long wires and low resistance.

PRACTICAL OBSERVATIONS

- Using higher layers reduces resistance but increases layer assignment distance.
- Too many vias increase delay and area.
- Following preferred direction reduces congestion and DRC issues.
- Global nets (clock, power, reset) should use higher layers.

6. VIA STRUCTURE BETWEEN LAYERS



KEY POINT

Vias connect two adjacent metal layers.
More vias → more delay, more area.

7. METAL STACK PARAMETERS

- Width : Minimum wire width.
- Spacing : Minimum space between wires.
- Pitch : Repetition distance of wires.
- Thickness : Affects resistance.
- Resistance : Increases with length, decreases with width.
- Capacitance : Increases with area and proximity.

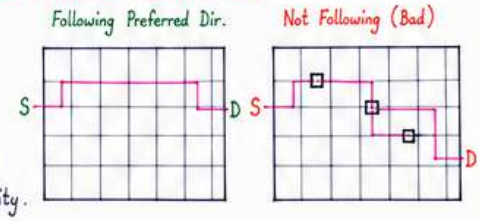
8. EXAMPLE METAL STACK TABLE

Layer	Material	Thickness (nm)	Min Width (nm)	Min Space (nm)	Preferred Direction
M1	Cu	30	100	100	Horizontal
M2	Cu	40	100	100	Vertical
M3	Cu	60	120	120	Horizontal
M4	Cu	80	140	140	Vertical
M5	Cu	120	160	160	Horizontal
M6	Cu	160	200	200	Vertical
M7	Cu	200	240	240	Horizontal
M8	Cu	250	280	280	Vertical

(Values are technology dependent and for illustration only)

9. WHY PREFERRED DIRECTION IS IMPORTANT ?

☑ Reduces number of vias.
☑ Leads to straighter wires.
☑ Minimizes congestion.
☑ Helps in DRC clean routing.
☑ Improves timing and reliability.



10. METAL STACK USAGE GUIDELINE

- Use M1–M2 for short, local, high density nets.
- Use M3–M4 for intermediate and medium length nets.
- Use M5 and above for long, global, critical nets (clock, power, resets).
- Avoid using lower layers for long wires.
- Respect layer assignment (from global router) during DR.

INTERVIEW / VIVA POINT

Q. What is metal stack ?
→ Set of metal layers with their design rules used for routing interconnections.

Q. Why do we have multiple metal layers ?
→ To reduce congestion, lower resistance, and to route long wires efficiently.

Q. Why alternate layers with orthogonal directions ?
→ To reduce vias, crossovers and DRC violations.

SUMMARY / REVISION POINTS

1. Metal stack consists of multiple routing layers (M1..Mn).
2. Lower layers → narrow, for local. Higher layers → wide, for global.
3. Alternate layers use orthogonal directions.
4. Higher layers have lower R, more width.
5. Proper metal stack usage reduces congestion and improves timing.
6. Metal stack parameters are defined in technology LEF / technology files.

Author : Konda Tharun Vivekananda

PD_Routing

1. INTRODUCTION

☑ Metal stack layers provide different routing resources with alternate preferred directions. This helps in reducing congestion, improving routability and timing.

2. DEFINITION

☑ Preferred routing direction of a metal layer is the direction in which wires are routed primarily to get maximum track availability and minimum vias.

Eg: M1 → Horizontal, M2 → Vertical, M3 → Horizontal ...

3. WHY ALTERNATE DIRECTIONS ?

☑ To increase routing efficiency & track utilization.

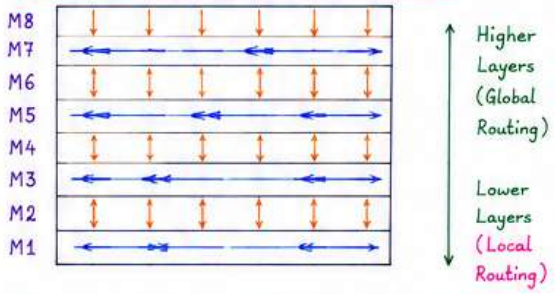
- To reduce congestion by orthogonal routing.
- To minimize number of vias and layer changes.
- To improve performance and reliability.
- To follow design rules and manufacturing guidelines.

4. PREFERRED ROUTING DIRECTION PER LAYER

Layer	Preferred Direction	Orientation
M1	Horizontal (← →)	
M2	Vertical (↑ ↓)	
M3	Horizontal (← →)	
M4	Vertical (↑ ↓)	
M5	Horizontal (← →)	
M6	Vertical (↑ ↓)	
M7	Horizontal (← →)	
M8	Vertical (↑ ↓)	

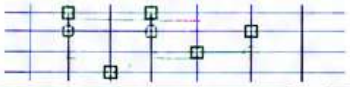
✱ Even layers → Vertical ✱ Odd layers → Horizontal

5. EXAMPLE - METAL STACK DIRECTION



6. PREFERRED vs NON-PREFERRED ROUTING STYLES

A. Preferred Routing Style (Follow Preferred Direction)

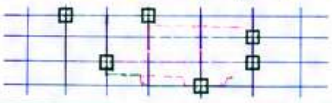


Eg: M1 → Horizontal
M2 → Vertical

Advantages
- Maximum track usage.
- Less congestion.
- Fewer vias required.
- Better timing & reliability.
- DRC clean routing.

Disadvantages
- Sometimes detours may increase wirelength slightly.

B. Non-Preferred Routing Style (Not Following Preferred Direction)



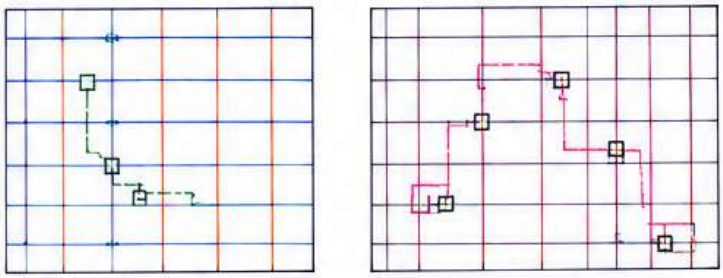
Eg: M1 → Vertical (Non-Preferred)

Advantages
- Useful when preferred tracks are congested.
- Shorter path in some cases.

Disadvantages
- Higher congestion.
- More vias (layer changes).
- Worse timing & RC.
- DRC & reliability issues.

7. EXAMPLE - PREFERRED vs NON-PREFERRED ROUTING

Preferred Routing (Good) | Non-Preferred Routing (Bad)



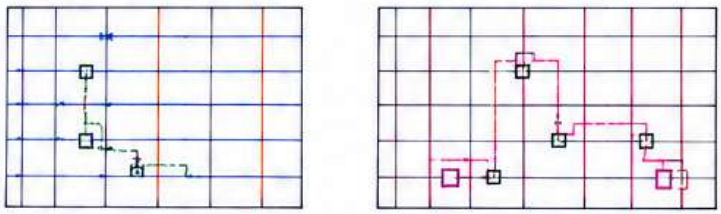
M1 (Horizontal) M2 (Vertical) ☐ Via

8. COMPARISON TABLE

Parameter	Preferred Routing	Non-Preferred Routing
Track Utilization	High	Low
Congestion	Low	High
Number of Vias	Less	More
Wirelength	Optimized	Usually higher
Timing	Better	Worse
DRC / Reliability	Easier to meet	Difficult
Power / EM	Lower stress	Higher stress
Routing Quality	Good	Poor

9. PRACTICAL EXAMPLE (SIGNAL ROUTE)

Preferred (M1-H, M2-V) | Non-Preferred (M1-V, M2-H)



M1 (Horizontal) M2 (Vertical) ☐ Via

10. KEY POINTS

☑ Metal stack has alternate preferred directions.
☑ Odd layers → Horizontal, Even layers → Vertical (General rule).
☑ Alternate directions improve routing efficiency.
☑ Following preferred direction reduces congestion & vias.
☑ Higher layers are used for long wires (global routing).
☑ Lower layers are used for short, local interconnects.

INTERVIEW / VIVA POINT

Q. What is preferred routing direction ?
→ The direction in which a metal layer is primarily used for routing.

Q. Why do we use alternate directions in metal stack ?
→ To avoid congestion, increase track utilization and reduce vias.

Q. What happens if we don't follow preferred direction ?
→ More congestion, more vias, poor timing and reliability issues.

SUMMARY / REVISION POINTS

1. Metal stack layers have alternate preferred directions.
2. Odd layers (M1, M3...) → Horizontal. Even layers (M2, M4...) → Vertical.
3. Alternate directions increase track usage and reduce congestion.
4. Following preferred direction minimizes vias and improves timing.
5. Higher layers are for global nets, lower layers for local nets.
6. Proper routing style ensures DRC clean and reliable design.

Author : Konda Tharun Vivekananda

PD _ Routing

1. INTRODUCTION

☑ Metal stack layers must follow strict DRC rules (width, spacing, enclosure, etc.). Correct layer usage and routing style directly affects DRC clean design.

2. DEFINITION

☑ Effect on DRC means how layer selection, routing directions, vias & spacing impact DRC violations. Good routing reduces rule breaks and helps sign-off.

3. HOW METAL STACK HELPS IN DRC

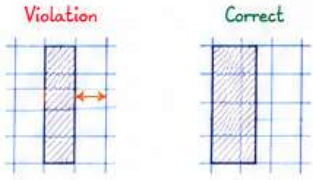
- Higher layers have more pitch & width → easy routing.
- Alternate directions → reduce parallel run violations.
- Vias with proper enclosure → avoid via DRC errors.
- Proper spacing → avoid min. spacing violations.
- Reduces congestion → less DRC and antenna issues.
- Follows design rules → easier verification.

4. EFFECT OF METAL STACK ON DRC RULES

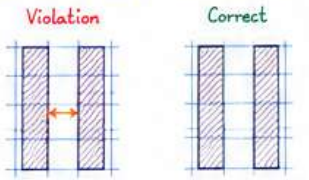
DRC Rule	Affected By	How Metal Stack Helps
Min Width	Metal Layer	Higher layers → larger min. width
Min Spacing	Adjacent Wires	Alternate directions increase spacing
Min Enclosure	Via to Metal	Proper vias and layer choice satisfy enclosure
Min Area	Metal Shape	Higher layers → more area
Min Step	Layer Transition	Following stack sequence avoids step errors
Antenna Rule	Long Nets	Higher layers reduce antenna ratio
Metal Density	Layer Usage	Balanced usage satisfies density

5. DRC EXAMPLES (VIOLATION vs CORRECT)

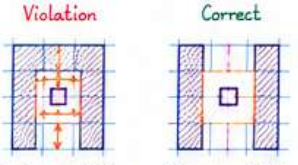
(i) Min Width Violation (M1)



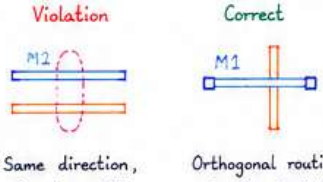
(ii) Min Spacing Violation (M2)



(iii) Via Enclosure Violation



(iv) Parallel Run Violation



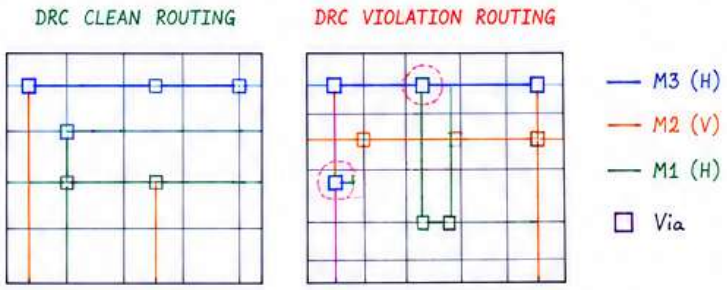
KEY POINT

Following proper metal stack usage, alternate directions, adequate spacing and via rules ensures DRC clean routing and reliable sign-off.

PRACTICAL OBSERVATIONS

- Wrong layer usage leads to width / spacing violations.
- Too many vias can cause enclosure and density violations.
- Parallel long wires increase crosstalk and DRC risk.
- Layer transition must follow the stack order.
- Always check DRC after each routing iteration.

6. DRC CLEAN vs DRC VIOLATION (EXAMPLE)



Violations: spacing, enclosure, parallel run, width etc.

7. ADVANTAGES

- Reduces number of DRC violations.
- Improves routing efficiency & track utilization.
- Better timing (less detours & wirelength).
- Less rework during sign-off.
- Helps in meeting metal density rules.
- Improves overall design reliability.

8. DISADVANTAGES

- Higher layers consume more area for vias.
- Too many layer changes increase delay and cap.
- Overuse of higher layers increases cost.
- Poor planning may cause congestion on upper layers.
- More complex to manage in large designs.

9. COMPARISON TABLE

Aspect	Good (DRC Friendly) Routing	Bad (DRC Violating) Routing
Layer Usage	All layers used as per stack	Wrong or limited layers used
Direction	Alternate (H & V)	Same direction on many layers
Spacing	Meets min. spacing	Spacing < rule
Via Usage	Proper enclosure & limited	Too many vias / poor enclosure
Congestion	Low	High
DRC Violations	Few or Zero	Many
Timing	Better	Worse
Sign-off	Easy	Difficult

10. INTERVIEW / VIVA POINT

Q. What is the effect of metal stack on DRC ?
→ Proper layer usage, direction, spacing and vias reduce DRC violations and help in sign-off.

Q. Why alternate directions are used in routing ?
→ To increase spacing, reduce parallel run violations and improve routing efficiency.

Q. What happens if we don't follow DRC rules ?
→ Violations occur → redesign → delay in tape-out and increased cost.

11. SUMMARY / REVISION POINTS

1. Metal stack must follow DRC rules.
2. Alternate directions reduce violations and congestion.
3. Higher layers have more width, spacing and less congestion.
4. Proper via enclosure and spacing ensure DRC clean design.
5. Good routing → fewer violations, better timing and easy sign-off.
6. DRC check must be run after each routing iteration.

12. DRC RULES CHECKLIST (During Routing)

✓ Min Width	✓ Min Step	✓ Overlap Check
✓ Min Spacing	✓ Parallel Run	✓ Edge Spacing
✓ Via Enclosure	✓ Antenna Rule	✓ Layer Transition Order
✓ Min Area	✓ Metal Density	

Author : Konda Tharun Vivekananda

PD_Routing

1. INTRODUCTION

☑ Routing is done in two main stages in physical design flow.
Global Routing (GR) → Detailed Routing (DR)
Both stages have different objectives, methods and outputs but together they create a complete, DRC clean layout.

2. GLOBAL ROUTING (GR)

☑ Global Routing estimates the routing resources and finds possible paths between source and sink pins on an abstract grid using all routing layers.
Outputs : Routing guides / tracks, congestion map.

3. DETAILED ROUTING (DR)

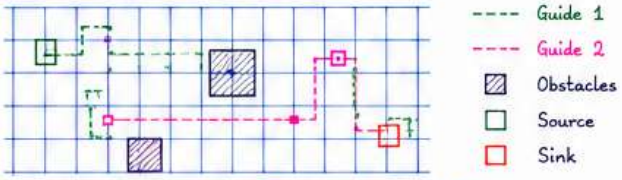
☑ Detailed Routing uses the global routing output and creates exact routes on actual metal layers, inserts vias and satisfies DRC design rules.
Outputs : DRC clean routed layout (DEF).

4. GLOBAL ROUTING vs DETAILED ROUTING (COMPARISON)

Parameter	Global Routing (GR)	Detailed Routing (DR)
Purpose	Estimate & plan routing resources	Create exact physical routes
Input	Placed DEF, Netlist, Tech files, Constraints	GR guides, Tech files, DRC rules
Level	Abstract (grid based)	Detailed (track based)
Layers Used	All routing layers	Actual routing layers
Method	Coarse grid, fast congestion aware	Exact grid, DRC aware
Output	Routing guides, Track allocation, Congestion map	DRC clean routes, Vias, Final DEF
Main Concern	Congestion, Wirelength estimate	DRC, SI, Antenna, Min spacing
Typical Issue	Over congestion, Over estimation	DRC violations, Antenna violations
Runtime	Faster	Slower
Iterative With	Placement, CTS, Power planning	DRC fixing, Optimization

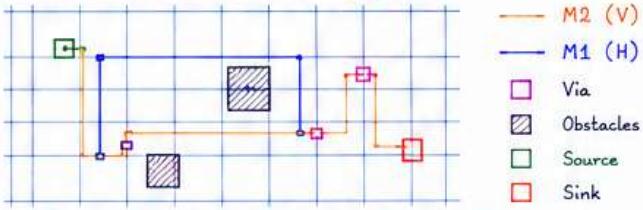
5. EXAMPLE DIAGRAMS

(i) GLOBAL ROUTING (Abstract Grid)



→ Only guides are found. Exact path is not fixed.

(ii) DETAILED ROUTING (On Metal Layers)

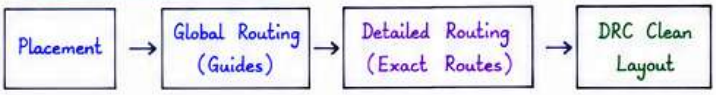


→ Exact route on tracks with vias satisfying DRC.

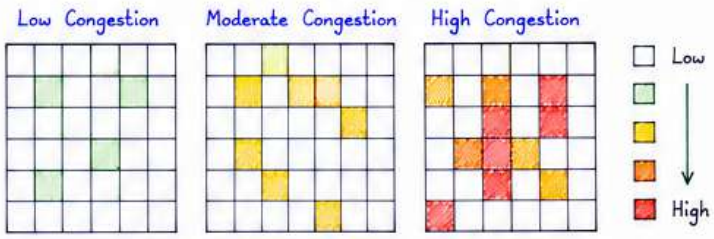
KEY POINT

Global Routing gives the big picture (where to route) while Detailed Routing gives the exact path (how to route). Both are essential for clean and manufacturable design.

6. FLOW RELATIONSHIP



7. CONGESTION VIEW (GLOBAL ROUTING OUTPUT)



High congestion → Detours → Longer wires → Timing degrade → Possible routing failure.

8. WHEN GR OR DR FAILS ?

- GR fails when congestion is too high.
  Solution → Re-do placement / adjust routing resources.
- DR fails when DRC / antenna / spacing rules are violated.
  Solution → Fix violations, use more vias or adjust tracks.

9. ADVANTAGES

Global Routing

- Fast and scalable.
- Good congestion estimation.
- Helps in early detection of routing hotspots.
- Provides wirelength and routing resource estimate.

Detailed Routing

- Creates exact manufacturable routes.
- Ensures DRC clean layout.
- Inserts vias automatically.
- Optimizes for timing, SI and antenna rules.

10. DISADVANTAGES

Global Routing

- Over-estimation of congestion.
- Does not consider exact DRC.
- Abstract model may not be accurate.

Detailed Routing

- Time consuming.
- May cause more layer changes.
- Sensitive to DRC rules.

11. PRACTICAL OBSERVATION

- Good global routing reduces detailed routing effort.
- Alternate directions in metal stack help global routing efficiency.
- Congestion seen in GR must be fixed before DR.
- DRC clean design needs multiple iterations of DR.

12. INTERVIEW / VIVA POINT

Q. What is difference between Global Routing and Detailed Routing ?
→ GR is abstract, estimates routes and generates guides.
DR is exact, creates final DRC clean routes on metal layers.
Q. Why is Global Routing done before Detailed Routing ?
→ To estimate resources, avoid congestion and plan routing.
Q. What is output of Global Routing ?
→ Routing guides, congestion map, track allocation.
Q. Why is Detailed Routing important ?
→ It creates exact routes and ensures DRC clean layout.

13. SUMMARY / REVISION POINTS

1. Routing has two stages – GR and DR.
2. GR works on abstract grid and estimates paths.
3. DR works on tracks and creates exact routes.
4. GR output is guides and congestion.
5. DR output is DRC clean routed DEF.
6. GR is fast, DR is accurate.
7. Both are required for timing, power and reliability.
8. Iterative process ensures good routing.

Author : Konda Tharun Vivekananda

PD_Routing

1. INTRODUCTION

☑ Global Routing (GR) is the first routing stage performed after placement. It is done on an abstract grid to estimate routing resources and find feasible paths between pins across all routing layers.
It does not create exact metal routes.

2. DEFINITION

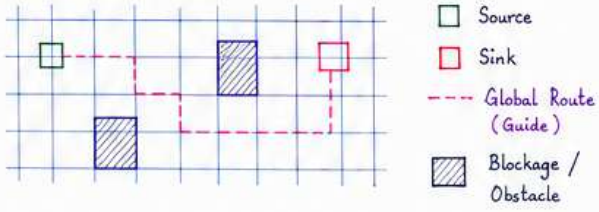
☑ Global Routing is the process of connecting nets between source and sink pins using an abstract model (grid based) considering layer capacities, congestion and design rules.

3. OBJECTIVES

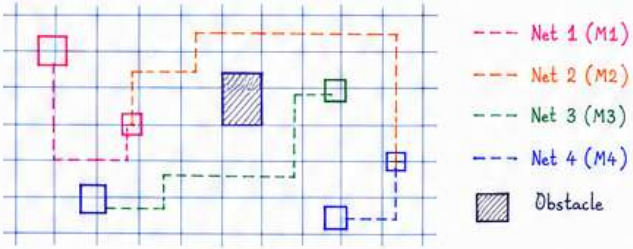
- Estimate routing resources and congestion.
- Generate routing guides / tracks for detailed routing.
- Reduce routing complexity and runtime of DR.
- Balance wirelength, congestion and layer usage.
- Support early detection of DRC, antenna and routability issues.

4. EXAMPLES (ABSTRACT GRID REPRESENTATION)

(i) Simple Net on Grid



(ii) Multiple Nets on Grid (Different Layers)



5. OUTPUTS (OF GLOBAL ROUTING)

- Routing Guides / Tracks on preferred layers.
- Congestion Map (overutilized hotspots).
- Layer assignment information.
- Estimated wirelength.
- DRC / Antenna / Design rule estimates.
- Input for Detailed Routing stage.

KEY POINT

Global Routing tells "WHERE" to route.
Detailed Routing tells "HOW" to route.

PRACTICAL OBSERVATION

- Good global routing reduces detours in detailed routing.
- High congestion areas must be fixed before DR.
- Layer assignment in GR affects timing, power and EM.
- Accurate models in GR improve overall PPA.

6. GLOBAL ROUTING FLOW

Netlist (Pins) → Abstract Grid Model → Routing Resource Estimation → Global Route (Guides) → Congestion Map

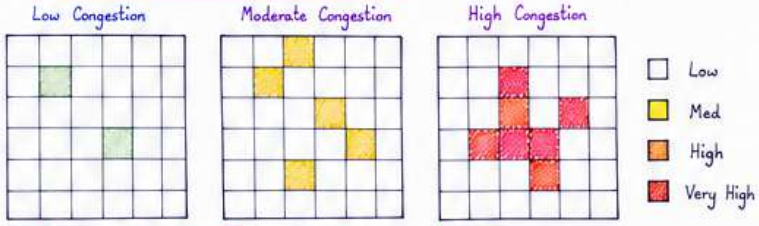
7. GLOBAL ROUTING ALGORITHMS (COMMON)

- Maze Routing : Simple grid based search.
- Line Search : Searches in one direction first.
- Steiner Minimum Tree (SMT) : For multi-pin nets.
- Path Finder / A* : Cost based search (wirelength + congestion).
- Negotiated Congestion : Iterative rip-up and re-route.

8. COMPARISON : GLOBAL ROUTING vs DETAILED ROUTING

Parameter	Global Routing (GR)	Detailed Routing (DR)
Stage	First routing stage	Final routing stage
Purpose	Estimate & plan routing	Create exact metal routes
Model	Abstract grid model	Actual metal layers
Accuracy	Approximate	Exact
Layer Info	Preferred layer assignment	Exact layer & via usage
Outputs	Routing guides, congestion map	DRC clean routes, DEF
Method	Coarse grid, fast	Track based, rule aware
DRC Check	Limited / early check	Complete DRC
Runtime	Faster	Slower
Typical Issue	Over/under estimation	Complexity, rule fixes
Dependency	Depends on placement	Depends on GR output

9. CONGESTION MAP (GR OUTPUT)



High congestion → Detours → More vias & wires
→ Longer delay → Possible routing failure.

10. ADVANTAGES

- Very fast and scalable for large designs.
- Helps early detection of routing problems.
- Provides congestion estimation and layer usage balance.
- Reduces runtime of detailed routing.
- Supports better PPA by early optimization.

11. LIMITATIONS

- Approximate, not exact.
- Depends on quality of models.
- May overestimate or underestimate congestion / wirelength.
- Cannot ensure full DRC cleanliness.

12. INTERVIEW / VIVA POINT

Q. What is Global Routing ?
→ It is first routing stage on abstract grid to estimate resources and generate routing guides.

Q. What is the output of Global Routing ?
→ Routing guides, congestion map, layer assignment and wirelength estimate.

Q. Why is Global Routing important ?
→ It detects congestion early and makes detailed routing easy and efficient.

13. SUMMARY / REVISION POINTS

1. Global Routing is the first routing stage.
2. Performed on abstract grid model.
3. Estimates routing resources and wirelength.
4. Generates routing guides for DR.
5. Outputs : Guides, congestion map, layer assignment.
6. Faster but approximate.
7. Helps in early DRC / congestion detection.
8. Good GR leads to better detailed routing and design closure.

14. GLOBAL ROUTING CHECKLIST

✓ All nets routed (no unrouted nets)	✓ Guide quality good	✓ Via count reasonable
✓ Congestion within limits	✓ Wirelength reasonable	✓ Hotspots identified
✓ Layer usage balanced	✓ DRC estimates clean	✓ Re-route if needed
✓ No excessive overflows	✓ Antenna estimates OK	

Author : Konda Tharun Vivekananda

PD _ Routing

1. GLOBAL ROUTING TERMS – 1 : CONGESTION

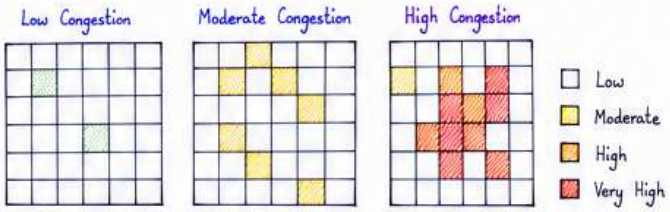
☑ 1.1 Definition

Congestion is the condition in which the routing demand (No. of nets crossing a region / tracks required) is greater than the routing capacity (tracks available).

☑ 1.2 Objectives

- Identify over-utilized regions early in the design.
- Guide placement or re-ordering to reduce congestion.
- Ensure routability and avoid routing failures in DR.
- Balance routing resources across all layers.

☑ 1.3 Example (Congestion Map)



High congestion → Detours → Longer wires → Timing degrade → Possible routing failure.

☑ 1.4 Outputs

- Congestion Map (per layer or overall).
- Hotspot List with utilization %.
- Guidance for placement / layer reassignment.
- Updated routing capacities for next stage.

1.5 Advantages

- Early detection of routing problems.
- Helps optimize placement.
- Reduces routing iterations.
- Improves routability and timing.

1.6 Limitations

- Depends on abstract model accuracy.
- May give false hotspots.
- Does not fix congestion itself.
- Cannot handle all DRC rules.

2. GLOBAL ROUTING TERMS – 2 : ROUTING GUIDE (GUIDE)

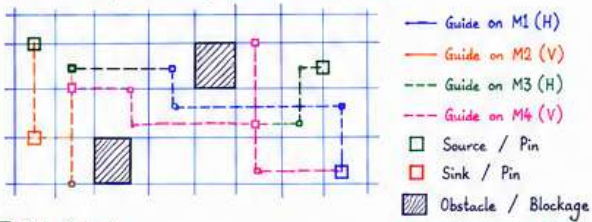
☑ 2.1 Definition

Routing Guide (or Guide / Track Guide) is a set of preferred on the abstract grid to generated by Global Routing on the abstract grid to indicate where nets should be routed in Detailed Routing.

☑ 2.2 Objectives

- Provide initial paths for all nets.
- Respect aitad directions and capacities.
- Minimize congestion and wirelength.
- Guide DR to complete exact metal routes.

☑ 2.3 Example (Routing Guide on Abstract Grid)



☑ 2.4 Outputs

- Routing Guides / Tracks for every net (per layer).
- Track assignment and preferred paths.
- Layer usage & capacity summary.
- Input for Detailed Routing.

2.5 Limitations

- Not exact routes (abstract only).
- May require modification in DR.
- Quality depends on congestion model.
- Cannot handle all DRC rules.

KEY POINT

Congestion tells "WHERE" the problem is.
Routing Guide tells "HOW" nets can be routed.
Both are essential outputs of Global Routing.

3. CONGESTION vs ROUTING GUIDE (OUTPUT COMPARISON)

Parameter	Congestion	Routing Guide
Nature	Analysis Output	Construction Output
Represents	Where routing is difficult	Where nets should go
Data Type	Utilization / Heat Map	Paths / Tracks
Granularity	Grid / Region based	Net / Track based
Purpose	Identify hotspots	Provide routing instructions
Used In	Placement / Optimization	Detailed Routing
Changes in DR	Re-computed if needed	Modified (rip-up / re-route)
Final Output	Heat map report	Exact metal routes (derived)

4. CONGESTION EXAMPLE (NUMERICAL)

Region (A)

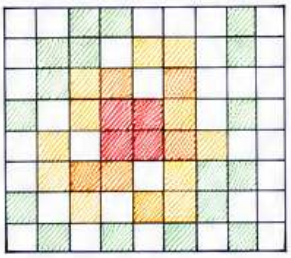
Capacity (tracks) = 100
Demand (tracks) = 150
Utilization = 150 / 100
= 1.5 (150%)
→ Over-utilized → Congested

Legend :

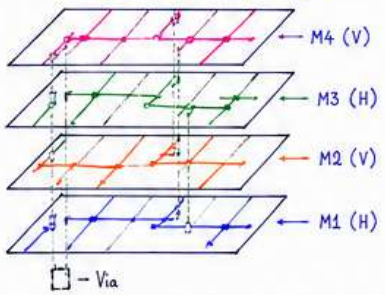
Demand > Capacity → Congested
Demand = Capacity → Balanced
Demand < Capacity → Under-utilized

5. TYPICAL GLOBAL ROUTING OUTPUT VIEW

(i) Congestion Map (Overall)



(ii) Routing Guide on Multiple Layers



6. ADVANTAGES SUMMARY

Congestion

- Early problem identification.
- Helps in placement & re-ordering.
- Avoids routing failures.
- Improves overall design clousire.

Routing Guide

- Provides clear path for DR.
- Respects layer directions.
- Reduces DR search space.
- Faster detailed routing.

7. PRACTICAL OBSERVATIONS

- Congestion must be minimized before going to Detailed Routing.
- Layer assignment in GR directly impacts timing and power.
- Good routing guides → fewer rip-ups in DR.
- Always re-check congestion after placement changes.

8. INTERVIEW / VIVA POINT

Q. What is congestion in global routing ?
→ Im is mode routing demand is more than available capacity in a region.

Q. What is routing guide ?
→ Set of preferred tracks / paths provided by GR to DR.

Q. Why is the output of global routing ?
→ Congestion map and routing guides.

9. SUMMARY / REVISION POINTS

1. Global Routing uses abstract grid model.
2. Congestion shows over-utilized regions.
3. Routing Guide gives preferred paths.
4. Both outputs are essential for DR.
5. Congestion drives optimization and placement adjustments.
6. Routing guide reduces DR search space and runtime.
7. Final DRC clean layout is obtained after Detailed Routing.

10. GLOBAL ROUTING TERMS CHECKLIST

✓ Congestion Map generated	✓ Layer usage balanced	✓ Via usage minimized
✓ Hotspots identified	✓ Capacity respected	✓ Holupus good (DEF / Tech)
✓ Routing Guides created	✓ Guide paths valid	✓ Ready for Detailed Routing

Author : Konda Tharun Vivekananda

PD _ Routing

1. INTRODUCTION

☑ Detailed Routing (DR) is the final routing stage in physical design. It converts global routing output (guides / tracks) into exact metal routes on actual metal layers while satisfying all DRC rules.
It inserts vias, connects pins and ensures full connectivity.

2. DEFINITION

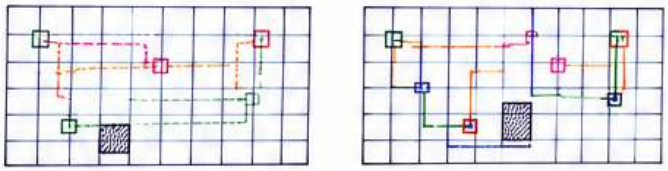
☑ Detailed Routing is the process of creating exact, manufacturable routes on specific metal layers using routing tracks and vias, following preferred directions and DRC rules.
Input : Global routing guides.
Output : DRC clean, exact routed layout (DEF / GDS).

3. OBJECTIVES

- Convert global routing guides into exact metal routes.
- Ensure connectivity between all source and sink pins.
- Follow design rules (width, spacing, enclosure, etc.).
- Minimize vias and wirelength while reducing congestion.
- Improve timing, IR drop and signal integrity.

4. EXAMPLES (DETAILED ROUTING ON METAL LAYERS)

(i) From Global Guide (M1-M3) (ii) After Detailed Routing



☐ Source / Pin ☐ Sink / Pin ▨ Blockage / Obstacle (Obstacle
--- Global Guide

5. OUTPUTS OF DETAILED ROUTING

- Exact wire shapes on each metal layer (M1, M2, ...).
- Via & cut at layer transitions.
- Updated DEF with exact connectivity.
- DRC clean routed layout.
- Enables sign-off and via count.
- Antenna / open / short free connections.

6. ADVANTAGES

- Produces manufacturable layout.
- Ensures full DRC compliance.
- Optimized wirelength and via usage.
- Improves timing and reliability.
- Enables sign-off and tape-out readiness.

7. LIMITATIONS

- Time consuming and complex.
- Depends on quality of global guides.
- Congestion may cause detours and timing impact.
- High computation for large designs.

8. CHALLENGES

- Congested regions with limited routing resources.
- Meeting all DRC rules (width, spacing, enclosure, step, etc.).
- Excessive vias can increase resistance and capacitance.
- Layer assignment conflicts.
- Antenna effect, electromigration and IR drop concerns.
- Maintaining timing while fixing DRC violations.

KEY POINT

Detailed Routing converts global routing guides into exact, DRC clean metal routes by inserting vias and following design rules. It is the final step before sign-off.

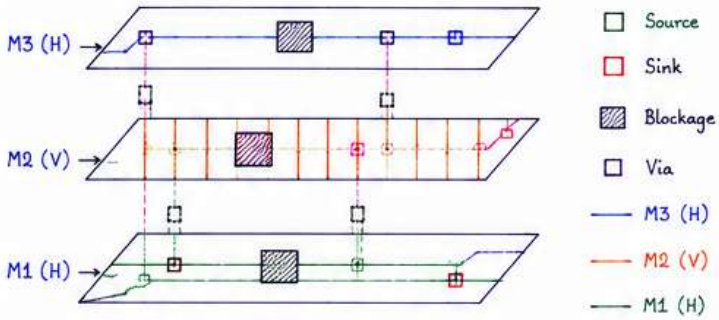
9. DETAILED ROUTING FLOW

Global Routing Output → Layer Assignment → Track Reservation → Detailed Routing → DRC Check → Routed DEF / GDS

Sub Steps :

- Read global routing guides.
- Assign layers to nets.
- Reserve tracks (blocking) to avoid conflicts.
- Route nets on tracks with preferred directions.
- Insert vias for layer changes.
- Perform rip-up and re-route if DRC violations occur.
- Final DRC clean routed layout.

10. DETAILED ROUTING DIAGRAM (ON MULTIPLE LAYERS)



Here, routes follow : M1 – Horizontal, M2 – Vertical, M3 – Horizontal

11. DETAILED ROUTING METRICS / REPORTS

- Wirelength (per net / total).
- Via count (per net / total).
- DRC violations count.
- Antenna violations count.
- Utilization % per layer.

12. COMPARISON : GR vs DR (Brief)

Aspect	Global Routing	Detailed Routing
Purpose	Find path & resources	Create exact routes
Model	Abstract grid	Actual metal layers
Accuracy	Approximate	Exact
Output	Guides / Tracks	DRC clean layout
DRC Check	Limited	Complete
Runtime	Fast	Slow
Changes	High level	Low level / Local

13. INTERVIEW / VIVA POINT

Q. What is Detailed Routing ?
→ Final routing stage that creates exact DRC clean metal routes on actual layers.

Q. What is the input of DR ?
→ Global routing output (guides / tracks).

Q. What is the output of DR ?
→ Exact routed layout (DEF / GDS) meeting all DRC rules.

Q. Why vias are inserted in DR ?
→ To connect between layers when route changes from one metal layer to another.

14. PRACTICAL OBSERVATIONS

- Good global routing reduces effort in DR.
- More congestion in GR increases detours in DR.
- Following preferred directions simplifies DR.
- DR is sensitive to grid resolution.
- Always run DRC after each iteration.

15. SUMMARY / REVISION POINTS

1. Detailed Routing is final routing stage using actual metal layers.
2. It converts global routing guides into exact routes.
3. Inserts vias, follows preferred directions and DRC rules.
4. Outputs are DRC clean layout, updated DEF and reports.
5. Advantages include manufacturability, reliability and better timing.
6. Limitations include runtime, quality dependence on GR and congestion.
7. Major challenges are congestion, DRC rules, vias and layer conflicts.
8. DR is essential for sign-off and tape-out.

16. DETAILED ROUTING CHECKLIST

✓ All nets routed	✓ DRC clean	✓ Min spacing satisfied	✓ Via count reasonable	✓ No opens / shorts	✓ Timing impact checked
✓ No unconnected pins	✓ Min width satisfied	✓ Enclosure rules OK	✓ Antenna clean	✓ Layer transitions valid	✓ Ready for sign-off

Author : Konda Tharun Vivekananda

PD _ Routing

1. INTRODUCTION

☑ Routing Tracks are the predefined paths on a metal layer along which wires can be routed.
They are generated during routing preparation and used by global and detailed routing.

2. DEFINITION

☑ A routing track is a line on a specific metal layer centered at regular pitch in a given direction (horizontal or vertical).
Tracks act as preferred channels for routing wires.

3. WHY DO WE NEED TRACKS ?

- Simplify routing by limiting wire positions.
- Speed up global and detailed routing.
- Help estimate routing capacity and congestion.
- Ensure uniform spacing and design rule compliance.
- Reduce wirelength and avoid unnecessary detours.

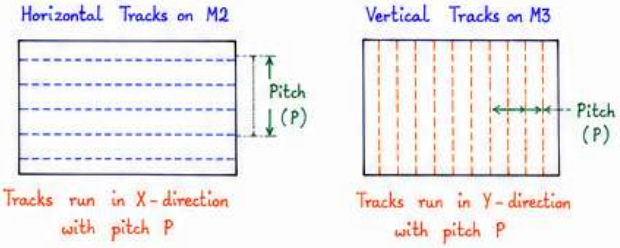
4. EXAMPLE OF TRACKS GENERATED ?

- Based on technology LEF / tech files.
- Using routing pitch (preferred spacing) and offset.
- For each metal layer, direction is decided first (H or V).
- Tracks are created across the core area at regular pitch.

Track Generation Formula

$$\text{No. of Tracks} = \frac{(\text{Core_Width} + 2 \times \text{Offset})}{\text{Routing_Pitch}} + 1$$

5. EXAMPLE OF TRACKS GENERATION

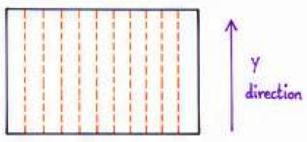


6. TYPES OF TRACKS

(i) Horizontal Tracks
- Run along X-direction.
- Used on layers with horizontal preferred direction.
- Example : M1, M3 ...



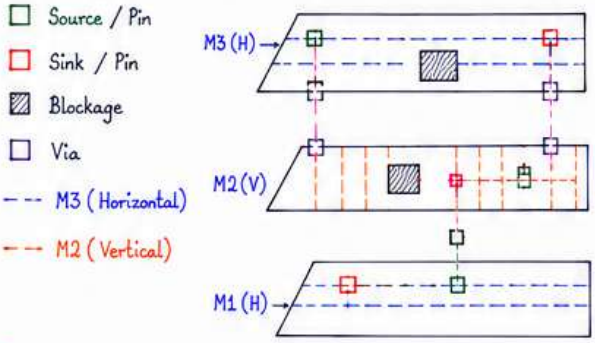
(ii) Vertical Tracks
- Run along Y-direction.
- Used on layers with vertical preferred direction.
- Example : M2, M4 ...



KEY POINT

Routing tracks define WHERE we can route.
Good track generation ensures less congestion, lower wirelength and DRC clean routing.

7. TRACK EXAMPLE WITH VIAS AND CONNECTIONS



Vias connect tracks between adjacent layers to complete routing.

8. OUTPUTS OF ROUTING TRACKS

- Track Grid File (per layer).
- Track map used by global & detailed router.
- Capacity information per track.
- Helps in congestion estimation and path planning.

9. COMPARISON : HORIZONTAL vs VERTICAL TRACKS

Parameter	Horizontal Tracks	Vertical Tracks
Direction	Along X-axis	Along Y-axis
Used in Layers	M1, M3, M5 ...	M2, M4, M6 ...
Routing Style	Easier for wide buses running horizontally	Easier for wide buses running vertically
Via Connections	Needs Vias to vertical layers	Needs Vias to horizontal layers
Congestion	May increase if too many horizontal nets	May increase if too many vertical nets
Wirelength Impact	Good for horizontal dominant paths	Good for vertical dominant paths

10. ADVANTAGES

- Organized routing structure.
- Faster routing convergence.
- Easy congestion analysis.
- Reduced design rule violations.
- Better routability and QoR.

11. LIMITATIONS

- Fixed pitch may waste area.
- Over-constraint can increase wirelength.
- Incorrect track density may cause congestion.
- Depends on quality of tech file and routing rules.

12. CHALLENGES

- Choosing correct pitch and offset for each layer.
- Maintaining balance between capacity and area.
- Avoiding track over-utilization.
- Handling layer transitions and via blockage.
- Meeting timing, DRC and antenna rules.

13. INTERVIEW / VIVA POINTS

Q. What is a routing track ?
→ It is a preferred line on a metal layer along which wires are routed.
Q. Why do we need routing tracks ?
→ To guide routing, estimate congestion, and ensure DRC clean layout.
Q. How are tracks generated ?
→ Using routing pitch, offset and layer direction from tech/LEF files.
Q. Difference between horizontal and vertical tracks ?
→ To guide routing since on congestion located along metal layers.
Q. Ea all tracks aun routing congestlity.
→ Inconte contacts n dainength. and corchoity with metalility routing.

14. REVISION POINTS

- ✓ Tracks are generated per layer with fixed pitch.
- ✓ They guide routing and help in congestion analysis.
- ✓ Horizontal tracks run in X-direction.
- ✓ Vertical tracks run in Y-direction.
- ✓ Tracks are used in both Global and Detailed Routing.
- ✓ Proper track generation improves routability.
- ✓ Tracks must follow tech rules and design constraints.
- ✓ Good tracks → Less congestion, better timing.

Author : Konda Tharun Vivekananda

PD_Routing

1. INTRODUCTION

☑ Routing tracks are the parallel, equally spaced lines on a metal layer along which wires are placed during routing.
The spacing between adjacent tracks is called Track Pitch.

2. DEFINITION

☑ Track Pitch is the center-to-center distance between two adjacent routing tracks.
It defines the routing grid density on a metal layer.

Formula

Track Pitch (P) = Center to Center distance between two adjacent tracks.

3. WHY DO WE NEED TRACK PITCH ?

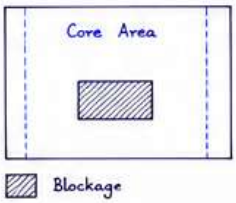
- Determines routing capacity of a layer.
- Controls congestion and wire density.
- Affects timing (RC), IR drop and EM reliability.
- Ensures manufacturability and DRC compliance.
- Helps balance area utilization and performance.

4. HOW ARE TRACKS GENERATED ?

- Based on Technology LEF / Tech files (routing layer width, spacing, pitch, direction).
- For each metal layer, routing direction is fixed (H or V).
- Router creates equally spaced tracks across the core area with given pitch.
- Offset may be applied to align tracks with blockages or boundaries.
- Tracks are stored in a grid and used by global and detailed router.

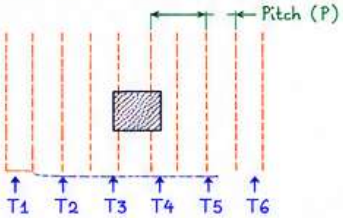
Inputs for track generation

- Layer Name (M1, M2 ...)
- Direction (H or V)
- Routing Pitch (P)
- Offset
- Blockages / Keep-outs



5. TRACK PITCH EXAMPLE

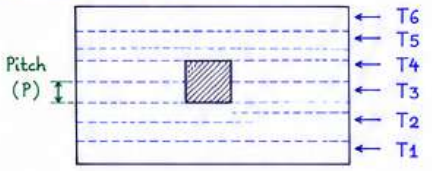
Consider Metal 2 (Vertical)



If P = 0.4 μm
Tracks are at
0, 0.4, 0.8, 1.2,
1.6, 2.0 μm ...
(center to center)

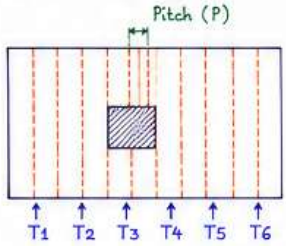
6. TYPES OF TRACKS

(i) HORIZONTAL TRACKS (H)



- Run along X-axis
- Used on layers with horizontal preferred direction.
- Example : M1, M3, M5 ...

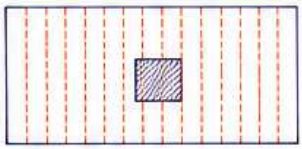
(ii) VERTICAL TRACKS (V)



- Run along Y-axis
- Used on layers with vertical preferred direction.
- Example : M2, M4, M6 ...

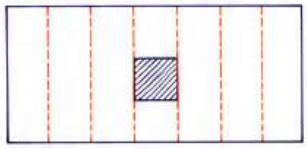
7. TRACK PITCH VISUAL COMPARISON

Small Pitch (High Density)



- More tracks
- High routing capacity.
- More congestion possible.

Large Pitch (Low Density)



- Fewer tracks
- Low routing capacity.
- Less congestion.

8. TRACK PITCH EFFECTS

Parameter	Small Pitch	Large Pitch
Routing Capacity	High	Low
Congestion	High (possible)	Low
Wirelength	Low	High (detours)
RC Delay	Low	High
IR Drop / EM	More risk (dense)	Less risk
Area	High (more wires)	Low
Manufacturability	Tight DRC, harder	Easy DRC
Runtime	Higher	Lower

9. ADVANTAGES

- Provides uniform routing grid.
- Improves routability and design closure.
- Helps in congestion control.
- Ensures DRC and design rule compliance.
- Supports better timing and reliability.

10. LIMITATIONS

- Incorrect pitch may cause congestion or under-utilization.
- Smaller pitch increases area and runtime.
- Depends heavily on tech file accuracy and design rules.
- Layer assignment conflicts.

11. CHALLENGES

- Choosing correct pitch for each layer.
- Balancing capacity vs area and timing.
- Handling blockages and keep-outs.
- Track alignment across multiple layers.
- Meeting all DRC, antenna and EM rules.

12. PRACTICAL OBSERVATION

- Pitch is usually decided based on target congestion and timing.
- Industrial flows auto-generate tracks from tech LEF files.
- Always check congestion after global routing and adjust pitch if required.

KEY POINT

Track Pitch controls the routing grid density.
Smaller pitch → More tracks → Higher capacity but higher area and congestion.
Larger pitch → Fewer tracks → Lower capacity but less congestion and area.

13. INTERVIEW / VIVA POINTS

Q. What is a track ?
→ A line on a metal layer along which wires are routed.
Q. What is track pitch ?
→ Center to center distance between two adjacent tracks.
Q. Why do we need track pitch ?
→ To control routing capacity, congestion, timing and DRC.
Q. How a are tracks generated ?
Q. Drfrm tech LEF files using pitch, direction and offset.
→ Horizontal runs in X-direction, vertical in Y-direction

14. SUMMARY / REVISION POINTS

1. Routing tracks are predefined parallel lines on a metal layer.
2. Track Pitch is the center-to-center spacing.
3. Tracks are generated using tech LEF parameters.
4. Smaller pitch → more capacity, more congestion.
5. Larger pitch → less capacity, less congestion.
6. Tracks guide global and detailed routing.
7. Correct pitch selection improves PPA and closure.
8. Horizontal tracks → X direction, Vertical → Y direction.

15. TRACK PITCH CHECKLIST

✓ Pitch values follow tech LEF	✓ Uniform track distribution	✓ Congestion within limits	✓ Antenna / EM rules satisfied
✓ Correct direction per layer	✓ No overlaps / DRC safe	✓ Via count reasonable	✓ Ready for routing stages

Author : Konda Tharun Vivekananda

PD _ Routing

1. INTRODUCTION

☑ Routing Grid is a 2-D array of routing tracks (H and V) defined on each metal layer of the chip.
It forms the basic framework on which global routing guides and detailed routing are performed.

2. DEFINITION

☑ A Routing Grid is a set of intersecting horizontal and vertical routing tracks spaced at regular intervals (pitch) covering the core area.
It defines legal routing locations for wires and vias.

3. WHY DO WE NEED ROUTING GRID ?

- Provides structured environment for routing.
- Simplifies routing by restricting wires to tracks.
- Ensures uniform spacing and DRC compliance.
- Helps congestion estimation and resource planning.
- Speeds up routing convergence and reduces runtime.
- Maintains design rule constraints (width, spacing, enclosure).

4. HOW IS ROUTING GRID GENERATED ?

- Based on Technology LEF / Tech file parameters.
- For each metal layer, router reads :
  - Routing direction (H or V)
  - Track pitch (P
  - Wire
  - Spacing rules

Input from Tech File
- Layer name
- Direction
- Pitch (P)
- Width
- Spacing

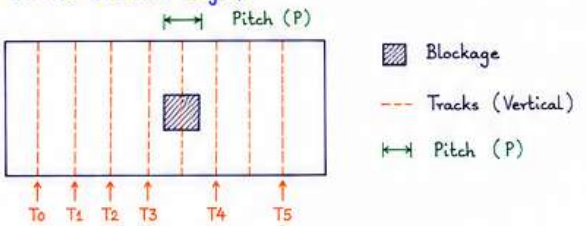
- Tracks are generated across the core area at pitch P starting from an offset (O).
- Offset ensures alignment with pin grid / blockages.
- Track index is assigned to each track.

Track Position Formula

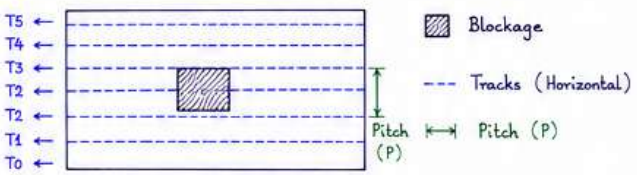
$$Position_i = Offset + i \times Pitch \quad (i = 0, 1, 2 \ldots N)$$

5. EXAMPLE : ROUTING GRID ON METAL LAYER

(i) On M2 (Vertical Layer)



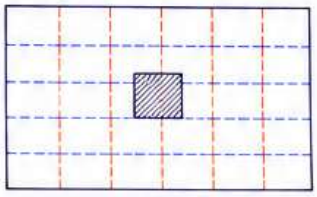
(ii) On M3 (Horizontal Layer)



KEY POINT
- Routing grid defines WHERE routing is allowed.
- Tracks on horizontal layers run in X-direction.
- Tracks on vertical layers run in Y-direction.
- Grid spacing (Pitch) controls routing density and congestion

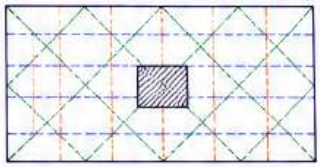
6. TYPES OF ROUTING GRID

(i) Orthogonal Grid (Most Common)



- Tracks are only horizontal or vertical.
- Used in most ASIC technologies.
- Simple, fast and DRC friendly.

(ii) Non-Orthogonal / Diagonal Grid (Rare)

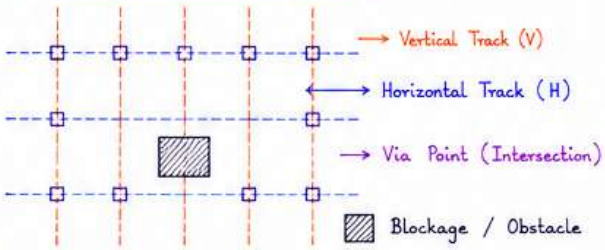


- Includes 45° or other angles.
- Used in high performance / analog designs.
- More complex and slower.

7. ROUTING GRID LAYERS EXAMPLE (4 METAL LAYERS)

Layer	Direction	Pitch (μm)	Track Type	Typical Use
M1	Horizontal (H)	0.20	H-Tracks	Local routing
M2	Vertical (V)	0.20	V-Tracks	Local / std cell
M3	Horizontal (H)	0.40	H-Tracks	Medium range
M4	Vertical (V)	0.40	V-Tracks	Long range / global

8. ROUTING GRID COMPONENTS



Vias are placed at track intersections to connect different layers.

9. ADVANTAGES

- Structured routing environment.
- Ensures uniform spacing.
- Easy congestion estimation.
- DRC clean routing.
- Faster global and detailed routing.
- Improves QoR and reliability.

10. LIMITATIONS

- Restricts route flexibility.
- May cause detours in congested regions.
- Depends on accurate tech file.
- Large designs need fine pitch which increases runtime.

11. PRACTICAL OBSERVATIONS

- Smaller pitch → more tracks → higher capacity → more congestion.
- Larger pitch → fewer tracks → lower capacity → longer wires.
- Offset selection affects pin access and congestion.
- Always check grid alignment across all layers.

12. INTERVIEW / VIVA QUESTIONS

Q. What is a routing grid ?
→ A set of horizontal and vertical tracks on each metal layer.
Q. Why do we need routing grid ?
→ To provide structured routing, control congestion and follow DRC
Q. What is track pitch ?
→ Center to center distance between adjacent tracks.
Q. How are tracks generated ?
→ Using tech file parameters (direction, pitch, width, spacing).
Q. What are the types of routing grid ?
→ Orthogonal grid (H/V) and Non-orthogonal grid.

13. SUMMARY / REVISION POINTS

1. Routing grid is base for routing.
2. Tracks are spaced at regular pitch.
3. Tracks run H on H-layers, V on V-layers.
4. Grid generated from tech LEF.
5. Controls capacity, congestion and QoR.
6. Vias placed at intersections.
7. Smaller pitch → more capacity.
8. Larger pitch → less capacity

14. ROUTING GRID CHECKLIST

✓ Correct layer directions	✓ Uniform track distribution	✓ No overlaps / DRC clean grid	✓ Congestion within limits	✓ Grid validated for all layers
✓ Correct track pitch	✓ Proper offset applied	✓ Via locations aligned	✓ Tech file rules satisfied	✓ Ready for global routing

Author : Konda Tharun Vivekananda

PD _ Routing

1. INTRODUCTION

☑ Routing resources are the available physical resources on each metal layer that can be used to create routed connections.
These resources are defined by the routing grid (tracks and vias) generated during physical design.

2. DEFINITION

☑ Routing resources = set of routing tracks (H/V) and vias on a metal layer defined by the routing grid, that can be used to route nets while meeting design rules.
They are limited by track pitch, layer width, blockages and design rules.

3. WHY DO WE NEED ROUTING RESOURCES ?

- To provide legal paths for nets on actual layers.
- To enforce spacing and DRC rules.
- To estimate routing capacity and congestion.
- To balance resources across layers.
- To ensure successful routing and timing closure.

4. TYPES OF ROUTING RESOURCES

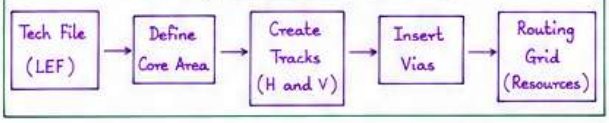
(i) 1-D Resources :
Only tracks are considered on a specific layer (Horizontal or Vertical).

(ii) 2-D Resources :
Tracks on multiple layers + vias (inter-layer connections). Used in global routing.

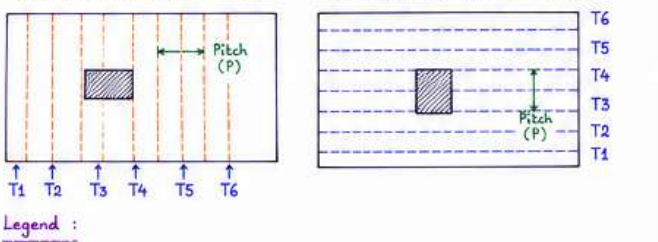
5. HOW ROUTING RESOURCES ARE GENERATED ?

- Read Technology LEF / Tech file.
- Get layer width, preferred direction, pitch and spacing.
- Define core area and placement/blockages.
- Generate routing grid layer by layer.
- Create tracks along preferred directions with pitch P.
- Insert legal vias at track intersections.
- Respect design rules (width, spacing, enclosure, ...).

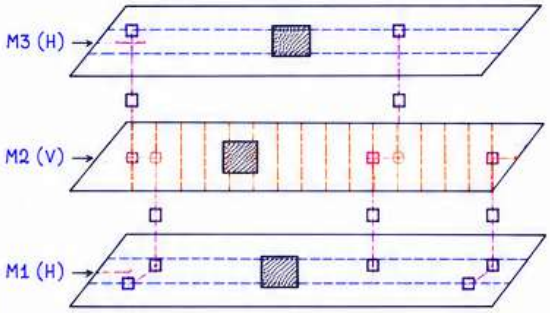
Routing Grid Generation Flow



6. EXAMPLE : ROUTING RESOURCES ON M2 LAYER



7. ROUTING RESOURCES OVERVIEW (ON MULTIPLE LAYERS)



- M1, M3 → Horizontal resources
- M2 → Vertical resources
- Vias connect resources across layers.

8. TRACK UTILIZATION

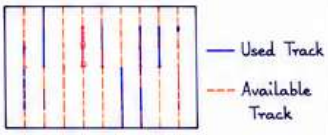
8.1 Definition

Track Utilization is the percentage of routing tracks that are used by nets.

8.2 Formula

$$\text{Track Utilization (\%)} = \left(\frac{\text{Used Tracks}}{\text{Total Available Tracks}}\right) \times 100$$

8.3 Example



Total Tracks = 10, Used Tracks = 6

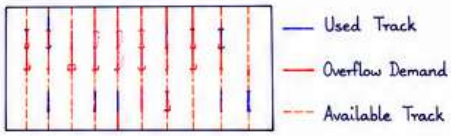
Utilization = $\frac{6}{10}$ x 100 = 60%

9. TRACK OVERFLOW

9.1 Definition

Track Overflow occurs when the routing demand (required tracks) exceeds the available routing capacity.

9.2 Example



Total Tracks = 8, Demand = 11

Overflow = 11 - 8 = 3 tracks

9.3 Consequences

- Increased congestion
- Long detours
- Timing degradation
- IR drop increase
- Routing failure
- DRC violations

10. ROUTING RESOURCES SUMMARY

Parameter	Description	Impact
Tracks (H/V)	Parallel lines on a layer	Provide routing paths
Vias	Connections between layers	Enable multi-layer routing
Track Pitch	Spacing between tracks	Controls capacity
Available Tracks	Total legal tracks	Defines capacity
Used Tracks	Tracks occupied by nets	Affect utilization
Overflow Tracks	Demand > Capacity	Causes congestion

11. ADVANTAGES

- Organized and legal routing paths.
- Easy congestion estimation.
- Better utilization of resources.
- DRC clean routing.
- Improved timing and reliability.

12. LIMITATIONS

- Fixed resources, may become limited.
- Depends on track pitch and layer width.
- Blockages reduce available resources.
- Overflow leads to performance loss.
- Requires accurate tech data.

KEY POINT

Routing resources define HOW and WHERE the nets can be routed on each metal layer.
They are limited, shared and must be managed carefully to avoid congestion and routing failures.

13. INTERVIEW / VIVA POINTS

Q. What are routing resources ?
→ They are the available tracks and vias on metal layers used for routing nets.
Q. Why are routing resources required ?
To provide legal paths, control congestion and
Q. What is track utilization ?
Percentage of tracks used = (Used / Total) x 100.

14. SUMMARY / REVISION POINTS

1. Routing resources are tracks and vias defined by routing grid.
2. They provide legal paths for routing and must follow DRC rules.
3. Resources are limited and shared among all nets.
4. Track utilization shows how much capacity is used.
5. Track overflow causes congestion, detours and timing problems.
6. Resources are generated from tech file using pitch and rules.
7. Proper resource management is essential for successful routing.

15. ROUTING RESOURCES CHECKLIST

✓ Tech file verified	✓ Track generation ?	✓ Track generation Vias	✓ Generation considered	✓ Utilization within limits	✓ Blockages considered
✓ Track pitch correct	✓ Track generation done	✓ DRC generation done	✓ No overlaps / shorts	✓ No track overflow	✓ Ready for routing

PD _ Routing

1. INTRODUCTION

☑ Routing requires physical resources on metal layers in order to connect sources and sinks.
These resources are in the form of routing tracks and vias defined by the routing grid.
Proper usage of routing resources is critical for achieving timing, reliability and DRC clean layout.

2. DEFINITION

☑ Routing Resources are the available routing tracks (H & V) and vias on a metal layer which are used to establish electrical connections between nets while satisfying design rules.

3. WHY DO WE NEED ROUTING RESOURCES ?

- To provide legal physical paths for nets.
- To enforce design rules (width, spacing, enclosure).
- To estimate routing capacity and congestion early.
- To balance resources across the design area.
- To ensure timing closure and signal integrity.

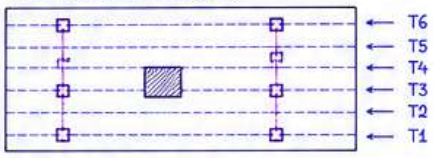
4. ROUTING RESOURCES COMPONENTS

- Routing Tracks
  - Horizontal Tracks (H-Tracks)
  - Vertical Tracks (V-Tracks)
- Vias
  - Via cuts for layer connections
- Track Pitch
  - Spacing between adjacent tracks
- Blockages / Keep-outs
  - Reduced or blocked resources

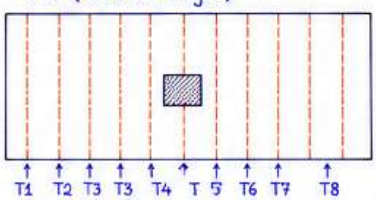
Legend
---- H - Track
---- V - Track
□ Via
▨ Blockage

5. ROUTING RESOURCES EXAMPLE (ON M2 & M3)

M3 (Horizontal Layer)



M2 (Vertical Layer)



- Vias connect tracks between M2 & M3.
- Blockage reduces the available resources.

6. ROUTING RESOURCES OVERVIEW

Routing Grid (Tracks + Vias) → Available Resources → Track Utilization → Congestion Check → Detailed

7. TRACK UTILIZATION

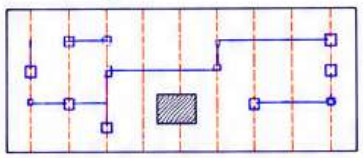
7.1 Definition

Track Utilization is the percentage of available tracks used by routed nets in a region or on a layer.

7.2 Formula

$$\text{Track Utilization (\%)} = \frac{\text{Used Tracks}}{\text{Total Available Tracks}} \times 100$$

7.3 Example



Available Tracks = 12, Used Tracks = 7

Utilization = $\frac{7}{12} \times 100 = 58.3\%$

Good Utilization → 50% to 70%

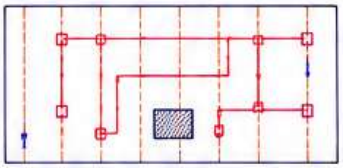
High Utilization → > 80% (Risk of congestion)

8. TRACK OVERFLOW

8.1 Definition

Track Overflow occurs when the routing demand (required tracks) exceeds the available tracks in a region or layer.

8.2 Example



Available Tracks = 8, Demand = 12

Overflow = 12 − 8 = 4 tracks (Overflow)

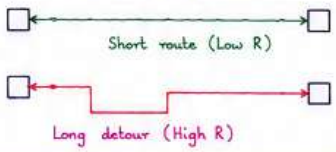
8.3 Consequences

- High congestion
- Long detours
- Increased resistance
- Increased capacitance
- Higher delay
- IR drop increase
- DRC violations
- Routing failure

9. EFFECTS OF CONGESTION ON RC AND DELAY

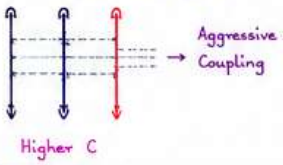
9.1 Increased Resistance (R)

Longer routes and narrow detours increase wire length and R.



9.2 Increased Capacitance (C)

More neighboring wires + longer length → higher coupling C.



9.3 Increased Delay (T)

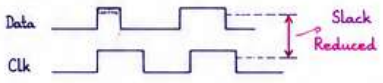
Delay ≈ 0.69 × R × C

High R and High C together cause large delay.



9.4 Timing Impact

Positive Slack reduces, may cause setup violations.



10. HOW CONGESTION WAS RESOLVED ? (SOLUTIONS)

- Increase routing resources (more tracks).
- Optimize placement to reduce congestion.
- Use buffering / repeater insertion.
- Layer change / via jumping.
- Use shielded / wider wires for critical nets.
- Re-route nets for better distribution.
- Nice / global routing congestion-driven adjustments.

11. PRACTICAL OBSERVATIONS

- Congestion should be checked after placement and CTS.
- Uniform resource distribution improves routability.
- Always monitor track utilization and overflow maps.
- Early fixes save iterations and improve QoR.

12. INTERVIEW / VIVA POINTS

Q. What are routing resources ?
→ Tracks and vias on metal layers defined by routing grid.
Q. What is track utilization ?
→ % of available tracks used by routed nets.
Q. What is track overflow ?
→ When required tracks exceed available tracks.
Q. What are effects of congestion ?
→ High R, high C, high delay, timing violations, IR drop.
Q. How do you fix congestion ?
→ Increase resources, placement optimization, buffering, layer change.

13. SUMMARY / REVISION POINTS

1. Routing resources are tracks and vias.
2. Tracks are generated using routing grid.
3. Utilization measures how many tracks are used.
4. Overflow occurs when demand > capacity.
5. Congestion increases R, C and Delay.
6. High congestion causes timing failures.
7. Solutions: add resources, optimize placement, buffering, layer change.
8. Proper management ensures routability and timing closure.

14. ROUTING RESOURCES CHECKLIST

✓ Routing grid defined	✓ Tracks generated	✓ Utilization checked	✓ No track overflow	✓ Congestion within limits	✓ Blockages considered	✓ DRC clean resources
✓ Vias placed properly	✓ Layer assignment valid	✓ Buffering done if needed	✓ Detours minimized	✓ Timing met	✓ Ready for detailed routing	

Author : Konda Tharun Vivekananda

PD _ Routing

1. VIA TYPES - INTRODUCTION

☑ Vias are used to connect one metal layer to another adjacent layer. They allow vertical current flow between layers.
Different via types are used based on which layers are connected.

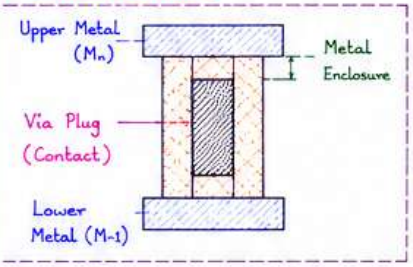
2. DEFINITION

☑ A Via is a small conductive plug (contact) that connects two adjacent metal layers in the routing stack.
It is surrounded by via enclosure and satisfies design rules.

3. WHY DO WE NEED VIAS ?

- To route nets vertically between metal layers.
- To cross over other nets without using long detours.
- To utilize multi-layer routing resources efficiently.
- To reduce congestion on single layers.
- To implement compact and timing optimized routes.

4. VIA STRUCTURE



Structure Parts :

- Upper Metal
- Lower Metal
- Via Plug (Conductive)
- Via Enclosure
- Cut Size
- Spacing to other vias

5. VIA TYPES (BASED ON LAYERS CONNECTED)

Via Type	Connects	Example	Used Between
VIA1	M1 ↔ M2	VIA1	Lowest metal layers
VIA2	M2 ↔ M3	VIA2	Lower to middle layers
VIA3	M3 ↔ M4	VIA3	Middle layers
VIA4	M4 ↔ M5	VIA4	Upper middle layers
VIA5	M5 ↔ M6	VIA5	Upper layers
...	.. ...	...	... ...

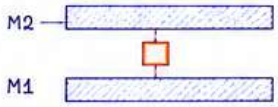
6. VIA RESISTANCE (R)

☑ Via resistance depends on the number of cuts (vias) used in parallel. More cuts → Lower resistance.

$$R_{via} = \frac{\rho_{via} \times h_{via}}{N_{cut} \times A_{cut}}$$

ρ_{via} = Resistivity of via material
h_{via} = Via height (layer thickness)
A_{cut} = Area of one via cut
N_{cut} = Number of via cuts in parallel

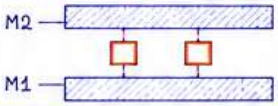
6.1 Single Via (1 Cuts)



$$R_{single} = \frac{\rho h}{1 \times A_{cut}} = \frac{R_o}{2}$$

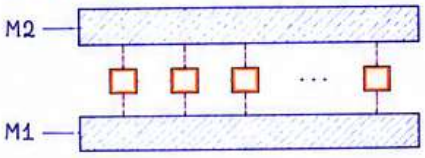
- N_{cut} = 1
- Highest resistance.
- Used for low current or non-critical nets.

6.2 Double Via (2 Cuts)



- N_{cut} = 2
- Resistance reduces to half of single via.
- Used for medium critical nets.

6.3 Multi-Cut Via (N Cuts)



$$R_{multi} = \frac{\rho h}{N \times A_{cut}} = \frac{R_o}{N}$$

- N_{cut} = N
- Resistance reduces inversely with number of cuts.
- Used for high current / timing critical nets.

Observation
More cuts in parallel → Lower Via Resistance
Lower R → Lower IR drop → Better timing

7. VIA RESISTANCE COMPARISON

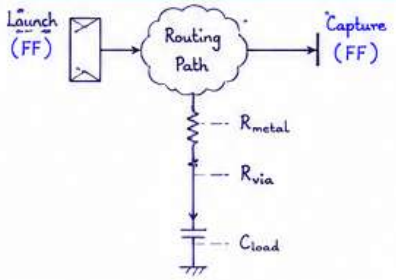
Via Type	No. of Cuts	Relative Resistance	Usage
Single Via	1	R_o	Low current nets
Double Via	2	$R_o/2$	Medium nets
4-Cut Via	4	$R_o/4$	High current nets
N-Cut Via	N	R_o/N	Critical timing nets

8. EFFECTS OF MORE VIA CUTS

- Lower resistance → Lower IR drop
- Lower delay → Better timing
- Better current handling capacity
- Improved reliability
- But → More area, More DRC checks

Trade-off
Too many cuts increase area and routing blockage.
Need optimal number of via cuts.

9. HOW VIA RESISTANCE AFFECTS DELAY

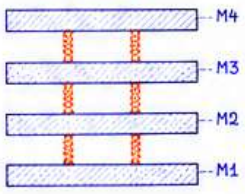


Delay approx.,
$t_d \approx 0.69 \times (R_{metal} + R_{via}) \times C_{load}$
Higher R_{via} → Higher Delay
More cuts → Lower R_{via} → Lower Delay

10. SOLUTIONS TO MANAGE VIA RESISTANCE

1. Use multiple via cuts in parallel for critical nets.
2. Follow technology recommended cut arrays.
3. Optimize via placement to reduce detours.
4. Use higher metal layers for long routes.
5. Use via ladders / stacks when needed.
6. Bufferring / repeater insertion for long high R paths.

Via Ladder



11. PRACTICAL OBSERVATIONS

- Single vias are used in non-critical or local connections.
- Double or multi-cut vias are used on power, clock and timing critical nets.
- Always verify via rules : cut size, spacing, enclosure.
- Excessive via cuts may increase congestion and area.

12. ADVANTAGES

- Enables multi-layer routing
- Reduces wirelength
- Improves timing and reliability
- Less congestion
- Better IR drop control

13. INTERVIEW / VIVA POINTS

Q. What is a via ?
A. A via is a conductive plug that connects two adjacent metal layers.
Q. Why do we need vias ?
A. To route nets between layers, cross over other nets and reduce congestion.
Q. How does via resistance depend on cuts ?
A. More via cuts in parallel → Lower resistance.
Q. Multi-cuts vias with low resistances ?

14. SUMMARY / REVISION POINTS

1. Vias connect adjacent metal layers.
2. Via resistance depends on number of cuts.
3. $R_{via} = \frac{\rho h}{N_{cut} \times A_{cut}}$
4. More cuts → Lower R → Lower delay.
5. Use optimal via cuts based on net criticality.
6. Proper via usage ensures timing, reliability and DRC clean design.

15. VIA CHECKLIST

- ✓ Correct via type (VIA1, VIA2 ...)
- ✓ Sufficient enclosure on both metals
- ✓ Correct cut size and spacing
- ✓ Use minimum required cuts for critical nets
- ✓ No via DRC / antenna violations
- ✓ Optimize location to avoid congestion

Author : Konda Tharun Vivekananda

PD_Routing

1. VIA ARRAYS - INTRODUCTION

☑ Via arrays are multiple vias placed in parallel between two metal layers to carry large current and reduce via resistance.
They provide reliable connection and improve electrical performance in routing.

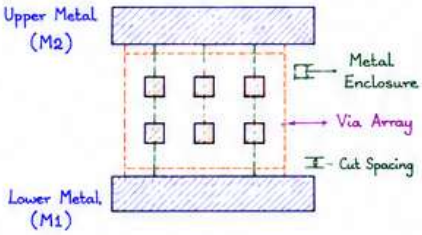
2. DEFINITION

☑ A via array is a group (N) of vias connected in parallel between two adjacent metal layers inside a defined enclosure.
It is used when a single via cannot meet current, timing and reliability requirements.

3. WHY DO WE NEED VIA ARRAYS ?

- To reduce via resistance and IR drop.
- To increase current carrying capability.
- To improve reliability and electromigration margin.
- To reduce delay on vertical connections.
- To avoid via failure and increase routing robustness.
- To meet DRC rules for wide / high current nets.

4. VIA ARRAY STRUCTURE

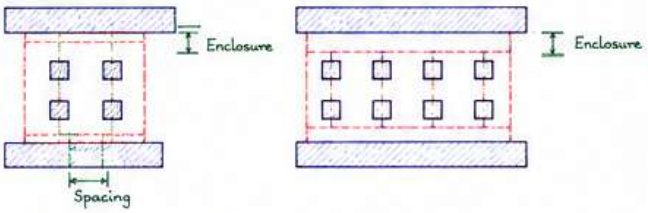


Structure Parts :

- Upper Metal
- Lower Metal
- Via (Conductive Plug)
- Enclosure
- Cut Size
- Cut Spacing
- Array Rows & Columns

5. VIA ARRAY EXAMPLE

(i) 2 x 2 Via Array (ii) 2 x 4 Via Array



Metal ☐ Via (Cut) ---- Enclosure

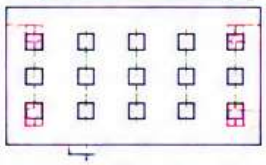
6. REDUNDANT VIA

☑ Redundant via means inserting more vias than the minimum required to provide extra margin.
It helps to avoid open via / bad via / DRC failures.

Types :

- Extra redundant cuts in array.
- Dummy vias at array edges.
- Staggered redundant vias.

Redundant Via Example



☐ Redundant Via
☐ Required Via

KEY POINT

Via arrays reduce resistance and improve reliability.
More cuts in parallel → Lower via resistance.
Redundant vias → Extra safety and margin.
Proper spacing and enclosure → DRC clean & robust routing.

7. VIA RESISTANCE (R_{via})

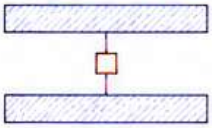
Via resistance is inversely proportional to the number of cuts in the array.
More cuts → Lower resistance.

Formula :

$$R_{via_array} = \frac{\rho_{via}}{N \times A_{cut}}$$

ρ_{via} → Resistivity of via material
A_{cut} → Area of single via cut
N → Number of vias in parallel

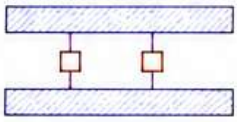
7.1 Single Via (1 Cuts)



$N = 1$
$R_{single} = R_o$ (Highest)

- Highest resistance
- Used for low current or small nets.
- Less reliable.

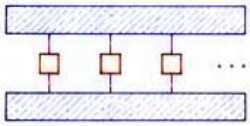
7.2 Double Via (2 Cuts)



$N = 2$
$R_{double} = \frac{R_o}{2}$

- Resistance reduces to half,
- Used for medium current nets.

7.3 Multi-Cut Via (N Cuts)



$N = N$
$R_{multi} = \frac{R_o}{N}$

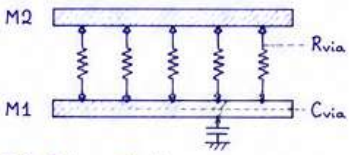
- More cuts → Lower resistance.
- Used for high current or critical nets.
- Better reliability.

8. VIA ARRAY IMPACTS ON ROUTING

Aspect	Impact
Resistance (R)	More cuts → Lower resistance → Lower IR drop
Capacitance (C)	More metal area → Slightly higher capacitance
Delay (t_d)	Lower via R → Lower RC delay → Better timing
Routing Capacity	High current nets can be routed safely
Congestion	Large arrays occupy more area → Possible congestion
DRC Complexity	More rules (enclosure, spacing, cut limits) to satisfy

9. EFFECT OF VIA ARRAY ON RC AND DELAY

(i) RC Model of Via Array



- R_{via} decreases with more cuts.
- C_{via} increases slightly.
- Overall RC delay improves.

(ii) Delay Trend

Cuts (N)	1	2	4	8
Delay (t_d)	High	Medium	Low	Very Low

10. HOW VIA ARRAY HELPS TO RESOLVE ISSUES

1. Increase number of cuts in congested / high current regions.
2. Use redundant vias to avoid open via and improve reliability.
3. Follow recommended cut arrays from technology file.
4. Optimize via placement to balance routing resources.
5. Use buffering / repeater insertion if delay still high.

Result

- Low R and IR drop
- Better timing
- Reliable routing
- DRC clean layout

11. PRACTICAL OBSERVATIONS

- Never use single via on high current nets.
- Always prefer arrays with enough redundant cuts.
- Staggered redundant vias improve reliability.
- Via arrays must satisfy enclosure and spacing rules.
- Overuse of large arrays can increase area and congestion.

12. ADVANTAGES

- Low resistance
- High current handling
- Better reliability
- Lower delay
- Robust routing

13. INTERVIEW / VIVA POINTS

Q. What is a via array ?
Q. Why do we use via arrays instead of single via ?
Q. How via resistance changes with number of cuts ?
Q. What is redundant via and why it is used ?
Q. What is impact of via array on routing ?
Q. What are enclosure rules for via array ?

14. SUMMARY / REVISION POINTS

1. Via arrays are multiple vias in parallel.
2. They reduce resistance and IR drop.
3. Improve current handling and reliability.
4. Redundant via adds extra safety margin.
5. Proper spacing, enclosure and cut size are very important.
6. Via array design directly impacts timing and power.

Author : Konda Tharun Vivekananda

PD_Routing

1. DRC RULES - INTRODUCTION

☑ Design Rule Check (DRC) rules are a set of geometric constraints defined by the technology to ensure that the physical layout can be manufactured reliably. All routing shapes (wires, vias, cuts, etc.) must satisfy these rules.

2. DEFINITION

☑ A DRC rule is the minimum requirement for dimensions, spacing and enclosure of routing features to avoid manufacturing defects and to ensure device reliability.

3. WHY DO WE NEED DRC RULES ?

- Ensure manufacturability and yield.
- Avoid shorts / opens due to lithography variations.
- Prevent reliability issues (EM, IR drop, stress).
- Ensure safety margins for process variations.
- Maintain electrical performance and timing.

4. ALL 6 MAJOR DRC RULES

S.No.	Rule	Purpose	Parameters
1	Minimum Width	Minimum width of line	Wmin
2	Minimum Spacing	Minimum spacing between two shapes	Smin
3	End of Line	Keep distance at line ends	EOLmin
4	Cut Spacing	Spacing between two via cuts	SCUTmin
5	Via Enclosure	Minimum metal overlap around via	Encmin
6	Parallel Run Length Rule	Limit length of parallel coupled wires	PRLmax

5. DRC RULES - EXPLANATIONS & DIAGRAMS

5.1 Minimum Width (Wmin)

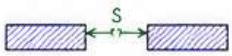
- The width of a routing wire must be greater than or equal to Wmin.



$W \geq W_{min}$

5.2 Minimum Spacing (Smin)

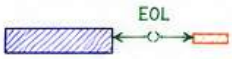
- Minimum spacing between two parallel wires or shapes.



$S \geq S_{min}$

5.3 End of Line Rule (EOLmin)

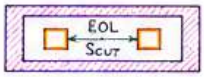
- Distance from the end of a wire to the nearest another wire.



$EOL \geq EOL_{min}$

5.4 Cut Spacing (SCUTmin)

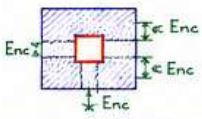
- Minimum spacing between two via cuts on the same layer.



$S_{CUT} \geq S_{CUTmin}$

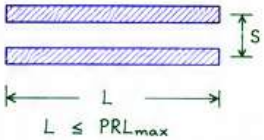
5.5 Via Enclosure (Encmin)

- Minimum overlap (enclosure) of metal around the via cut.



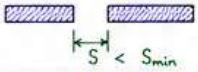
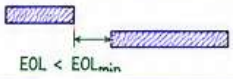
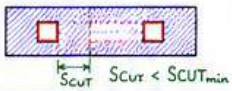
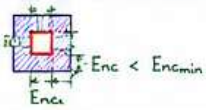
5.6 Parallel Run Length Rule (PRLmax)

- Two parallel wires running close to each other should not exceed maximum length PRLmax.



$L \leq PRL_{max}$

6. COMMON ROUTING VIOLATIONS - EXAMPLES

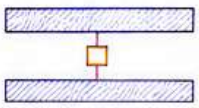
Violation	Example (Top View)	Description
1. Width Violation		Wire width is less than minimum allowed.
2. Spacing Violation		Spacing between wires is less than rule.
3. End of Line Violation		Line end is too close to another object.
4. Cut Spacing Violation		Via cuts are placed too close to each other.
5. Via Enclosure Violation		Via enclosure on one or more sides is less than minimum.
6. Parallel Run Length Violation		Parallel wires run longer than maximum allowed length.

7. FIX TECHNIQUES (RESOLUTION METHODS)

Violation Type	Fix Technique
Width Violation	Increase wire width to ≥ Wmin
Spacing Violation	Increase spacing or reduce wire width if possible
End of Line Violation	Extend the line end or move the other object
Cut Spacing Violation	Increase spacing between via cuts
Via Enclosure Violation	Increase metal overlap (enclosure) around via
Parallel Run Length Violation	Insert shields / change layer / jog one wire or reduce parallel length

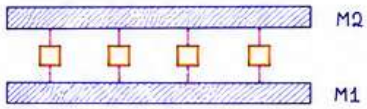
8. VIA ARRAY VS SINGLE VIA (BENEFITS)

Single Via



- Higher resistance
- Less current capacity
- Less reliable

Via Array (Multi Cuts)



- Lower resistance
- Higher current capacity
- Better reliability and EM margin

9. SUMMARY OF DRC RULES

- ✓ Minimum Width (Wmin) → Avoids open / thin lines.
- ✓ Minimum Spacing (Smin) → Avoids shorts between shapes.
- ✓ End of Line (EOLmin) → Protects line ends from bridging.
- ✓ Cut Spacing (SCUTmin) → Ensures via pattern safety.
- ✓ Via Enclosure (Encmin) → Strong via connection & reliability.
- ✓ Parallel Run Length (PRLmax) → Controls coupling & crosstalk.

10. PRACTICAL OBSERVATIONS

- DRC must be run at every stage after routing.
- Technology files contain all rule values.
- Violations increase with congestion.
- Fix early → saves iterations.
- Use via arrays on power / clock for better reliability.

11. INTERVIEW / VIVA QUESTIONS

Q. What is DRC in routing ?
→ Checking layout against design rules.

Q. Why minimum spacing is important ?
→ To avoid shorts due to process variations.

Q. What is via enclosure ?
→ Metal overlap around via cut.

Q. Why via array is used ?
→ To reduce resistance and improve reliability.

KEY POINT

Following DRC rules ensures manufacturable, reliable and high performance routing without shorts, opens or timing issues.

Author : Konaa Tharun Vivekananda

PD_Routing

1. ROUTING ALGORITHMS - INTRODUCTION

☑ Routing algorithms are mathematical methods used by auto routers to find a feasible path to connect two points (source and sink) while satisfying design rules and optimizing certain cost (length, congestion, etc).

2. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Maze Routing	Search based, explores grid step by step	Lee, DFS, BFS
Graph Based	Represents routing area as graph nodes and edges	Dijkstra, A*
Negotiation Based	Multiple nets negotiate for resources	PathFinder, NTHU-Route
Steiner Tree Based	Used for multi-pin nets, builds Steiner tree	Steiner minimal tree, FLUTE
Optimization Based	Improves routing iteratively using cost functions	Simulated Annealing, ILP based

3. ADVANTAGES OF ROUTING ALGORITHMS

- Automates routing for large designs.
- Finds shortest / near shortest paths.
- Reduces routing congestion.
- Ensures DRC compliance.
- Improves timing and performance.

4. APPLICATIONS

- Global routing and detailed routing in VLSI physical design.
- Inter connect synthesis in ASIC / FPGA / SoC.
- PCB routing.
- On-chip RF routing.
- Multi-layer routing for complex systems.

5. LEE ALGORITHM

5.1 DEFINITION

☑ Lee algorithm is a maze routing technique (BFS based) that guarantees the shortest path in terms of number of grid steps on a uniform cost grid.

5.2 IDEA

☑ Start from source and expand wavefront layer by layer in all 4 directions (U, D, L, R). Each cell is assigned a distance (level). When sink is reached, backtrace to find the shortest path.

5.3 ALGORITHM (Conceptual Steps)

1. Initialize grid with 0 at source; others as UNVISITED.
2. Put source in queue.
3. Pop cell from queue and examine its 4 neighbors.
4. If neighbor is free and unvisited → assign distance = current distance + 1 and push to queue.
5. Repeat until sink is reached or queue is empty.
6. Backtrace from sink to source following decreasing distance to obtain shortest path.

LEGEND

[S] Source [T] Target / Sink

[#] Blockage / Obstacle

. Free Cell

n Distance Label

KEY POINT

Lee algorithm always finds shortest path in terms of number of grid steps.

5.4 PROCEDURE - EXAMPLE (STEP BY STEP)

Consider a 7 × 7 grid

Step 0 : Initial Grid

	1	2	3	4	5	6	7
1	S	.	.	.	.	.	.
2	.	.	.	#	.	.	.
3	.	.	.	#	.	.	.
4	.	.	.	#	.	.	.
5	.	.	.	#	.	.	.
7	.	.	.	.	.	T	.

S (1,1)

T (7,7)

→ Blockage

Step 1 : After Expanding from S

	1	2	3	4	5	6	7
1	S	1	.	.	.	.	.
2	1	1	.	#	.	.	.
3	.	1	.	#	.	.	.
4	.	.	1	#	.	.	.
5	.	.	.	#	.	.	.
6	.	.	.	.	.	.	.
7	.	.	.	.	.	.	T

Step 2 : Next Expansion

	1	2	3	4	5	6	7
1	S	1	2	.	.	.	.
2	1	1	2	#	.	.	.
3	2	1	2	#	.	.	.
4	.	.	2	#	.	.	.
5	.	.	.	#	2	.	.
6	.	.	.	.	.	.	.
7	.	.	.	.	.	.	T

Step 3 : Wavefront Expansion

	1	2	3	4	5	6	7
1	S	1	2	3	.	.	.
2	1	1	2	#	3	.	.
3	2	1	2	#	3	.	.
4	3	.	2	#	3	.	.
5	.	3	.	2	3	.	.
6	.	.	3	.	.	.	.
7	.	.	.	.	.	.	T

Step 4 : Continue until Sink Reached

	1	2	3	4	5	6	7
1	S	1	2	3	4	5	6
2	1	1	2	#	4	5	6
3	2	1	2	#	4	5	6
4	3	2	2	#	4	5	6
5	4	3	3	#	5	6	6
6	5	4	3	5	6	6	6
7	6	5	5	5	5	5	T

Step 5 : Backtrace Shortest Path

	1	2	3	4	5	6	7
1	S	1	2	3	4	5	6
2	1	1	2	#	4	5	6
3	2	1	2	#	4	5	6
4	3	2	2	#	4	5	6
5	4	3	3	#	5	6	6
6	5	4	3	5	6	6	6
7	6	5	5	5	5	5	T

Path Length (in steps) = 12

Backtrace Rule :

From T, move to neighbor with value = current - 1 until S is reached.

This gives the shortest path.

5.5. ADVANTAGES OF LEE ALGORITHM

- Guarantees shortest path on a uniform grid.
- Simple and easy to implement.
- Completeness → always finds a path if exists.
- Suitable for grid based routing.

5.6. DISADVANTAGES

- High memory requirement.
- Slow for large grids.
- Does not handle congestion well.
- Not suitable for multi-pin nets.

5.7. APPLICATIONS

- Used in detailed routing for two terminal nets.
- Used in academic and industrial routers.
- Good for small / medium sized designs.
- Basis for advanced algorithms (e.g., multi-source BFS).

SUMMARY OF LEE ALGORITHM

Initialize → Expand → Reach Sink
↓
Backtrace → Shortest Path

6. COMMON ROUTING VIOLATIONS AND FIX TECHNIQUES

S.No.	Violation Type	Description	Fix Techniques / Resolution
1	Width Violation	Wire width < minimum required.	Increase wire width to ≥ rule width or use higher metal layer.
2	Spacing Violation	Spacing between wires < minimum.	Increase spacing or insert dummy fillers / shift wires.
3	End of Line (EOL) Violation	Wire end is too close to another object.	Extend wire end or add EOL extension to meet rule.
4	Cut Spacing Violation	Via cuts too close to each other.	Increase cut spacing or reduce number of cuts.
5	Via Enclosure Violation	Metal overlap around via < minimum.	Increase metal enclosure on one or more sides.
6	Parallel Run Length Violation	Parallel wires run longer than limit.	Reduce parallel length, add spacing, or insert shield wires / change layer.

IMPORTANT NOTE

Always run DRC after routing and fix all violations to ensure manufacturable and reliable layout.

KEY BENEFITS OF DRC CLEAN ROUTING

- ✓ Reliable & manufacturable chips
- ✓ Better yield and performance
- ✓ Avoids re-spins and delays
- ✓ Meets foundry requirements

7. INTERVIEW / VIVA QUESTIONS

Q. What are routing algorithms?
Q. What is Lee algorithm?
Q. Why Lee algorithm is called BFS based?
Q. What are the advantages of Lee algorithm?
Q. What are common DRC rules in routing?
Q. Why cut spacing rule is important?
Q. What is via enclosure rule?

8. SUMMARY / REVISION POINTS

1. Routing algorithms are used to find paths.
2. Lee algorithm guarantees shortest path.
3. It expands wavefront layer by layer.
4. Backtrace gives the shortest route.
5. DRC rules must be satisfied in routing.
6. Fix violations using width, spacing, enclosure, extensions and layer changes.

9. ROUTING ALGORITHMS QUICK RECAP

Maze Based → Lee, DFS, BFS
Graph Based → Dijkstra, A*
Negotiation Based → PathFinder, NTHU
Steiner Tree Based → FLUTE, MST
Optimization Based → SA, ILP

10. DRC RULES QUICK RECAP (6 Major Rules)

- ☐ Minimum Width (Wmin)
- ☐ Minimum Spacing (Smin)
- ☐ End of Line (EOLmin)
- ☐ Cut Spacing (SCUTmin)
- ☐ Via Enclosure (Encmin)
- ☐ Parallel Run Length (PRLmax)

Author : Konda Tharun Vivekananda

B) MAZE ALGORITHM

1. DEFINITION

☑ Maze routing algorithm is a search based routing technique that explores a grid cell by cell to find a feasible path between source (S) and sink (T) avoiding obstacles. It does not guarantee the shortest path but is simple and easy to implement.

2. IDEA

☑ Start from the source and explore neighboring cells in a fixed order (U, R, D, L). Mark visited cells. If dead end occurs, backtrack and try another direction until the sink is reached.

3. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Exact Algorithms	Guarantee optimal (shortest) path	Lee, Dijkstra, Heng-Pan
Heuristic Algorithms	Find good path, not necessarily optimal	Maze, A*, BFS
Negotiation Based	Multiple nets negotiate for resources	PathFinder, NTHU-Muzo
Steiner Tree Based	Minimize number of pins using Steiner tree	Steiner minimal tree, FLUTE
Optimization Based	Improve routing iteratively using cost functions	Simulated Annealing, ILP based

4. ADVANTAGES OF MAZE ALGORITHM

- Simple and easy to implement.
- Less memory requirement.
- Works well for small and medium sized designs.
- Can handle irregular obstacles.
- Good for interactive routing and ECO fixes.

5. DISADVANTAGES OF MAZE ALGORITHM

- Does not guarantee shortest path.
- May produce longer routes.
- High runtime for large designs.
- Explores many unnecessary cells.
- Not suitable for congestion driven global routing.

6. APPLICATIONS

- Detailed routing in VLSI layout.
- Interactive router for manual assistance.
- ECO routing and fix metal opens.
- Small block routing, std cell level routing.
- Academic research and teaching purposes.

7. ALGORITHM (Steps)

1. Initialize all cells as UNVISITED (-).
2. Put source cell (S) in stack.
3. Mark the cell as VISITED.
4. Pop a cell from stack and check its neighbors in order U, R, D, L.
5. If a neighbor is free and UNVISITED → mark it visited and push to stack.
6. If sink (T) is reached → stop.
7. If no unvisited neighbor → backtrack (pop) and try next path.
8. Repeat until stack is empty (no path) or sink is reached.

Legend

- S — Source
- T — Target / Sink
- # — Obstacle / Blockage
- * — Free Cell
- - — Unvisited Cell
- ✱ — Visited Cell

Neighbor Order

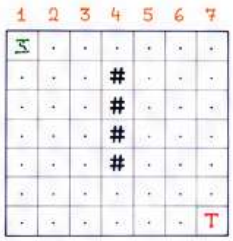
Always check in order:

U (Up) → R (Right)

D (Down) → L (Left)

8. STEP BY STEP MOVING EXPLANATION

Step 0 : Initial Grid



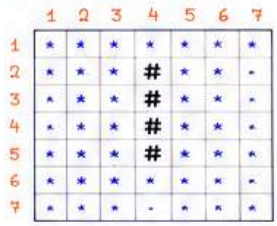
Step 1 : From S explore U,R,D,L



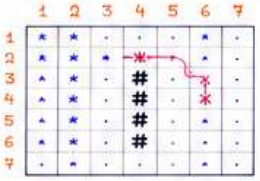
Step 2 : Continue exploration



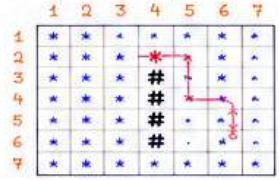
Step 3 : More expansion



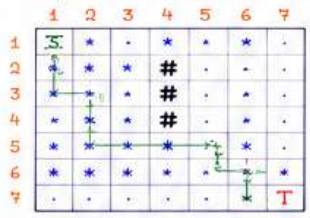
Step 4 : Dead end → backtrack



Step 5 : Another path tried



Step 6 : Sink Reached (Path Found)



Observation

Maze algorithm finds a feasible path but not necessarily the shortest path.

9. PROCEDURE EXAMPLE (TRACE TABLE)

Step	Stack Top (Cell)	Action	Neighbors Checked (U,R,D,L)	Next Cell Pushed	Status
1	(1,1)	Start from Source	U(-) R(2,1) D(1,2) L(-)	(2,1)	Visited
2	(2,1)	Expand	U(-) R(2,2) D(2,2) L(-)	(2,2)	Visited
3	(2,2)	Expand	U(1,2) R(2,3) D(3,2) L(2,1)	(1,2)	Visited
4	(1,2)	Expand	U(-) R(1,3) D(2,2) L(-)	(1,3)	Visited
5	(1,3)	Expand	U(-) R(1,4) D(2,3) L(1,2)	(1,4)	Visited
...	...				
N	(7,7)	Sink Reached	- - -	-	Success

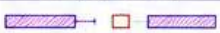
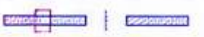
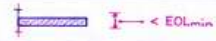
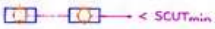
10. COMPLEXITY OF MAZE ALGORITHM

- Worst case time complexity : O(M × N) where M = rows, N = columns.
- Space complexity : O(M × N) (for stack / visited matrix).
- Does more backtracking than Lee.
- hence longer runtime.

11. COMPARISON : LEE vs MAZE

Aspect	Lee Algorithm	Maze Algorithm
Optimal Path	Yes (Shortest)	No (Not guaranteed)
Search Method	Wavefront (BFS)	Backtracking (DFS like)
Exploration	Level by level	Cell by cell
Memory	Higher	Lower
Runtime	Less	More
Usage	Global / detailed	Detailed / ECO

12. COMMON ROUTING VIOLATIONS (MAZE) AND FIX TECHNIQUES

Violation Type	Description	Example (Top View)	Fix Technique
Open / Unconnected	Path not connected to source or sink		Extend wire or add via to connect properly.
Wrong Layer Jump	Via placed without proper enclosure		Move via inside enclosure or increase enclosure area.
Short Circuit	Two nets touch or overlap		Increase spacing or change layer.
Min Width Violation	Wire width less than minimum		Increase wire width to ≥ Wmin.
Spacing Violation	Spacing between wires less than minimum		Increase spacing between wires.
End of Line Violation	Line end too close to another object		Extend line end or move object.
Cut Spacing Violation	Vias placed too close to each other		Increase spacing between vias.
Antenna Violation	Long metal connected to gate without diode		Add antenna diode or break metal.

13. KEY POINT

Maze algorithm explores the grid using backtracking. It is simple but slower than Lee algorithm and produces non-optimal paths.

14. PRACTICAL OBSERVATIONS

- Use maze for small blocks, ECO routing.
- Lee is preferred for global routing.
- Maze is useful when memory is limited.
- Proper heuristics (neighbor order) improves efficiency.

15. INTERVIEW / VIVA QUESTIONS

Q. What is Maze routing algorithm?
Q. How is it different from Lee algorithm?
Q. What is backtracking in maze routing?
Q. What are advantages of maze algorithm?
Q. Why maze is not used for global routing?
Q. When do we prefer maze algorithm?

16. SUMMARY / REVISION POINTS

✓ Maze uses backtracking search technique. ✓ Explores cells in order U, R, D, L. ✓ Does not guarantee shortest path. ✓ Simple but may take more time. ✓ Suitable for small designs / ECO.

17. QUICK RECAP

Initialize → Push Source → Explore Neighbors → Backtrack if Dead End → Repeat → Reach Sink or Fail

Author : Konda Tharun Vivekananda

PD_Routing

C) LINE SEARCH ALGORITHM

1. DEFINITION

☑ Line Search algorithm is an improvement over maze algorithm. It searches in straight lines (horizontal or vertical) instead of exploring all four directions at every step. It finds a feasible path by extending lines until an obstacle is hit, then tries other directions.

2. IDEA

☑ From the current cell, extend in one direction as far as possible in a straight line until blocked. At that point, turn and continue searching. This reduces number of visited cells and runtime.

3. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Exact Algorithms	Guarantee optimal (shortest) path.	Lee, Dijkstra, Heng-Pan
Heuristic Algorithms	Find good path, not necessarily optimal.	Maze, A*, BFS
Negotiation Based	Multiple nets negotiate for resources.	PathFinder, NTHU-route
Steiner Tree Based	Minimize number of pins using Steiner tree.	Steiner minimal tree, FLUTE
Optimization Based	Improve routing iteratively using cost functions.	Simulated Annealing, ILP based

4. ADVANTAGES OF LINE SEARCH ALGORITHM

- Faster than maze algorithm.
- Fewer cells are visited → low memory & runtime.
- Good for large grids and dense routing.
- Produces near optimal paths.
- Works well in global routing and detailed routing.

5. DISADVANTAGES

- May miss optimal path in some cases.
- Performance depends on search order of directions.
- Can get trapped in local dead ends.
- Not suitable for extremely irregular obstacles.

6. APPLICATIONS

- Global routing in VLSI physical design.
- Detailed routing with track based grids.
- Used in academic and industrial routers.
- Especially useful when performance is important.
- PCB routing and FPGA routing.

LEGEND

- S → Source
- T → Target / Sink
- # → Obstacle / Blockage
- . → Free Cell
- * → Visited / Expanded Cell
- → → Routed Path (Result)

KEY POINT

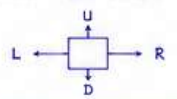
Line Search algorithm searches in straight lines reducing number of explored cells compared to Maze algorithm.

7. PROCEDURE (ALGORITHM)

1. Initialize all cells as UNVISITED.
2. Put source cell (S) in stack.
3. Mark source as VISITED.
4. From current cell, try to extend in one direction (U, R, D, or L).
5. Move in straight line marking each cell as VISITED until an obstacle (#) or boundary is hit.
6. If target cell (T) is found → stop.
7. If blocked, try next direction.
8. If all directions are blocked and stack not empty → backtrack.
9. Repeat until target found or stack becomes empty (no path).

Neighbor Order (Search Priority)

Default : U → R → D → L



Observations

- Straight line extension makes search fast.
- Backtracking ensures we try alternative routes.
- Path length may not be strictly minimum in some cases.

8. STEP BY STEP EXPLANATION (EXAMPLE)

Example Grid : 7 x 7

S at (1,1) T at (7,7) # → Blockage

Step 0 : Initial Grid

	1	2	3	4	5	6	7
1	S	.	.	.	.	.	.
2	.	.	.	#	.	.	.
3	.	.	.	#	.	.	.
4	.	.	.	#	.	.	.
5	.	.	.	#	.	.	.
6	.	.	.	.	.	.	.
7	.	.	.	.	.	.	T

Step 1 : Extend Right

	1	2	3	4	5	6	7
1	S	*	*	*	*	.	
2	.	.	.	#	.	.	
3	.	.	.	#	.	.	
4	.	.	.	#	.	.	
5	.	.	.	#	.	.	
6	.	.	.	.	.	.	
7	.	.	.	.	.	T	

Move right till blocked at (1,6)

Step 2 : Extend Down

	1	2	3	4	5	6	7
1	S	*	*	*	*	.	
2	.	.	.	#	.	.	
3	.	.	.	#	.	.	
4	.	.	.	#	.	.	
5	.	.	.	#	.	.	
6	*	*	*	*	*	.	
7	.	.	.	.	.	T	

Move down till blocked at (5,5)

Step 3 : Extend Left

	1	2	3	4	5	6	7
1	S	*	*	.	.	.	
2	.	.	.	#	.	.	
3	.	.	.	#	.	.	
4	.	.	.	#	.	.	
5	*	*	*	#	.	.	
6	.	.	.	.	.	.	
7	.	.	.	.	.	T	

Move left till blocked at (5,2)

Step 4 : Extend Up

	1	2	3	4	5	6	7
1	S	*	.	.	.	.	
2	.	*	.	#	.	.	
3	.	*	.	#	.	.	
4	.	*	.	#	.	.	
5	.	*	.	#	.	.	
6	.	*	.	.	.	.	
7	.	.	.	.	.	T	

Move up till blocked at (1,2)

Step 5 : Backtrack & Extend Down

	1	2	3	4	5	6	7
1	S	*	.	.	.	.	
2	.	*	.	#	.	.	
3	.	*	.	#	.	.	
4	.	*	.	#	.	.	
5	.	*	*	#	*	*	
6	.	.	.	.	.	.	
7	.	.	.	.	.	T	

Backtrack to (5,2) → Try Down

Step 6 : Extend Right

	1	2	3	4	5	6	7
1	S	*	.	.	.	.	
2	.	*	.	#	.	.	
3	.	.	.	#	.	.	
4	.	*	.	#	.	.	
5	.	*	*	#	*	*	
6	.	.	.	.	.	.	
7	.	.	.	.	.	T	

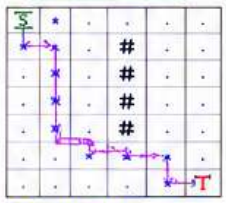
Move right till boundary

Step 7 : Extend Down to Target

	1	2	3	4	5	6	7
1	S	*	.	.	.	*	
2	.	.	.	#	.	*	
3	.	.	.	#	.	*	
4	.	.	.	#	.	*	
5	.	.	.	#	.	*	
6	.	.	.	.	.	*	
7	.	.	.	.	.	T	

Move down → Target reached

Final Routed Path



Path taken (sequence)

(1,1) → (1,6)
↓
(5,6)
←
(5,2)
↓
(7,2)
/
(7,7)

Total steps = 18

9. PROCEDURE EXAMPLE (TRACE TABLE)

Step	Current Cell	Direction Tried	Action	Next Cell	Status
1	(1,1)	Right	Extend till blocked	(1,6)	Visited
2	(1,6)	Down	Extend till blocked	(5,6)	Visited
3	(5,6)	Left	Extend till blocked	(5,2)	Visited
4	(5,2)	Up	Extend till blocked	(1,2)	Visited
5	(1,2)	Backtrack	Return to (5,2)	(5,2)	Backtrack
6	(5,2)	Down	Extend till boundary	(7,2)	Visited
7	(7,2)	Right	Extend till boundary	(7,7)	Visited
8	(7,7)	-	Target reached	-	Success

10. COMPLEXITY

- Time Complexity : O(N x M) in worst case (N = rows, M = columns)
- Space Complexity : O(N x M) (for visited matrix + stack)
- Fewer cells explored than maze algorithm.

11. COMPARISON - MAZE vs LINE SEARCH

Aspect	Maze Algorithm	Line Search Algorithm
Search Style	Step by step in all 4 dirn	Straight line extension
Cells Visited	More	Less
Runtime	Higher	Lower
Memory	Higher	Lower
Optimality	Always optimal	Near optimal
Best Used For	Small / Irregular grids	Large / Track based grids

13. PRACTICAL OBSERVATIONS

- Line Search is simple but effective.
- Direction priority affects path quality.
- Good for channel routing and track based routers.
- Backtracking is essential for success.
- Helps reduce congestion compared to maze.

14. VIVA QUESTIONS

Q. What is Line Search algorithm?
Q. How is it different from Maze algorithm?
Q. What is the idea behind Line Search?
Q. Why backtracking is needed?
Q. When do we prefer Line Search algorithm?

12. COMMON VIOLATIONS WITH LINE SEARCH & FIXES

Violation Type	Description	Example (Top View)	Fix Technique
Blocked Path	No possible path between source and target.	S # # # T	Modify blockage, use Via or reroute around obstruction.
Dead End Trap	Algorithm gets stuck in dead end.	S #	Enable backtracking to previous cell and try new direction.
Long Detour	Path is longer than necessary.	S T	Change direction priority or use cost based optimization.
Wrong Direction	Wrong turn taken early.	S #	Try alternative direction when
Orthogonality Violation	Too many bends in route.	S T	Prefer extension in same direction, reduce turns

15. SUMMARY / REVISION POINTS

- ✓ Line Search searches in straight lines.
- ✓ Fewer cells visited → faster routing.
- ✓ Backtracking ensures feasible path.
- ✓ Near optimal but not always shortest.
- ✓ Used widely in global and detailed routing.

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PD _ Routing

D) MIKAMI TABUCHI ALGORITHM

1. DEFINITION

☑ Mikami Tabuchi algorithm is an improvement over Line Search algorithm. It restricts the search to a certain number of preferred directions (two or three) and uses a history to modify the cost when backtracking. It reduces the chance of getting trapped in local dead ends and produces near optimal paths.

2. IDEA

☑ Search is performed along a set of preferred directions (usually Right and Up). If the path is blocked, the algorithm backtracks and tries other allowed directions (Left or Down) using a history counter to avoid repeating same wrong turns.

3. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Exact Algorithms	Guarantee optimal (shortest) path.	Lee, Dijkstra, Heng-Jan
Heuristic Algorithms	Find good path, not necessarily optimal.	Maze, A*, BFS
Negotiation Based	Multiple nets negotiate for resources.	PathFinder, NTHU-route
Steiner Tree Based	Minimize number of pins using Steiner tree.	Steiner minimal tree, FLUTE
Optimization Based	Improve routing iteratively using cost functions.	Simulated Annealing, ILP based

4. ADVANTAGES

- Better than Line Search in avoiding local traps.
- Produces shorter paths with less exploration.
- Works well for medium / large designs.
- Good trade-off between runtime and path quality.

5. DISADVANTAGES

- More complex than Line Search.
- Still not guaranteed to find shortest path.
- Requires memory for history / cost update.
- Runtime increases with more preferred directions.

6. APPLICATIONS

- Detailed routing in ASIC / FPGA.
- Used in academic routers and advanced tools.
- Global routing refinement.
- ECO routing and IP block routing.

LEGEND

- [S] → Source
- [T] → Target / Sink
- [#] → Obstacle / Blockage
- * → Visited Cell
- * → Backtracked Cell
- → → Routed Path

KEY POINT

Mikami Tabuchi algorithm uses preferred directions with history based cost update to reduce back-tracking and finds near optimal path efficiently.

7. PROCEDURE (ALGORITHM)

1. Initialize all cells as UNVISITED.
2. Put source cell (S) in stack / list.
3. Define preferred directions = {Right (R), Up (U)}; Backtrack directions = {Left (L), Down (D)}.
4. Start exploring in preferred directions in order.
5. If next cell is free and UNVISITED → move there.
6. If all preferred directions are blocked, backtrack using backtrack list and try other directions.
7. Update history / cost for backtracked cells.
8. Repeat until target (T) is reached or no more cells to explore.

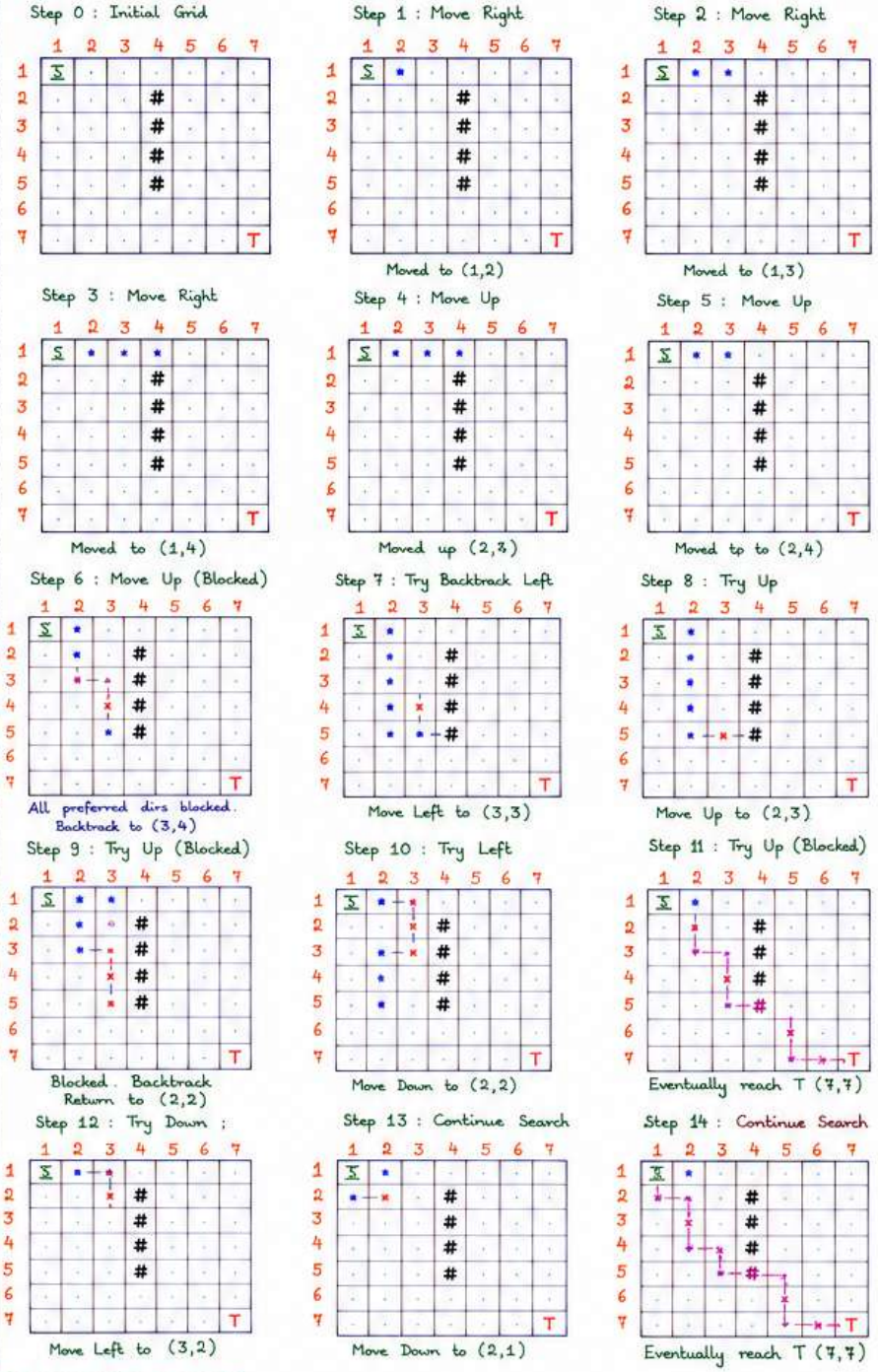
8. STEP BY STEP EXPLANATION WITH DIAGRAMS

Example Grid : 7 x 7

S at (1,1) T at (7,7)

Preferred direction order : R → U : Backtrack : L, D

→ Blockage



9. PROCEDURE EXAMPLE (TRACE TABLE)

Step	Current Cell (r,c)	Attempt Dir.	Next Cell (r,c)	Status	Action
1	(1,1)	R	(1,2)	Move	
2	(1,2)	R	(1,3)	Move	
3	(1,3)	R	(1,4)	Move	
4	(1,4)	U	(2,4)	Move	
5	(2,4)	U	(3,4)	Move	
6	(3,4)	U	(4,4)	Blocked	
7	(3,4)	L	(3,3)	Move	
8	(3,3)	U	(2,3)	Move	
9	(2,3)	U	(1,3)	Blocked	
10	(2,3)	L	(2,2)	Move	
11	(2,2)	D	(3,2)	Move	
...	...	...	...	...	...
N	(7,6)	R	(7,7)	Target	Reached

10. COMPLEXITY

- Time Complexity : O(N x M) where N = rows, M = columns.
- More efficient than Line Search. slightly higher than Maze.
- Space Complexity : O(N x M) (for visited, cost and stack)

11. COMPARISON

Aspect	Lee	Line Search	Mikami Tabuchi
Optimal Path	Yes	No	Near Optimal
Search Space	Large	Limited	Reduced
Backtracking	No	Yes	Yes with History
Runtime	High	Low	Medium
Memory	High	Low	Medium
Use Case	Small grids	Simple nets	Complex nets

12. COMMON ROUTING VIOLATIONS AND FIX (Relevant)

Violation	Description	Fix Technique
Open	Path not connected	Extend wires / add via
Wrong Layer Jump	Via placed on wrong layer	Use correct via layer
Min Width	Wire width < minimum	Increase width
Spacing	Spacing < minimum.	Increase spacing
End of Line	Line end too close object	Extend / add EOL
Cut Spacing	Vias too close	Increase via spacing

13. PRACTICAL OBSERVATIONS

- History based cost update reduces repeating wrong turns.
- Preferred directions should be chosen wisely for better performance.
- Good for dense routing with many obstacles.
- Works well in prefer Mayer routing.

14. INTERVIEW / VIVA QUESTIONS

Q. What is Mikami Tabuchi algorithm ?
Q. How is it differred from Line directions ?
Q. What are preferred and backtrack directions ?
Q. Why history is used in this algorithm ?

15. SUMMARY / REVISION POINTS

- ✓ Mikami Tabuchi improves Line Search.
- ✓ Uses preferred directions and backtrack.
- ✓ Uses history / cost to avoid dead ends.
- ✓ Finds near optimal path efficiently.
- ✓ Suitable for real time VLSI routing tools.

Author : Konda Tharun Vivekananda

E) HANDLOCK ALGORITHM

1. DEFINITION

☑ The Handlock algorithm is a channel routing / rip-up and re-route technique. It routes nets sequentially in a channel using a set of preferred directions (horizontal or vertical). If a conflict (handlock) occurs, the algorithm rips-up one of the nets and tries alternate directions until all nets are routed without conflicts.

2. IDEA

☑ Route nets one by one in a channel using preferred direction order. When a conflict (overlap) occurs, rip-up the conflicting net and re-route it in the next preferred direction. Continue until the net is successfully placed or no more directions left.

3. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Exact Algorithms	Guarantee optimal (shortest) path.	Lee, Dijkstra, Heng-Jan
Heuristic Algorithms	Find good path, not necessarily optimal.	Maze, Line Search, Handlock, Mikami-Tabuchi
Negotiation Based	Multiple nets negotiate for resources.	PathFinder, NTHU-route
Steiner Tree Based	Minimize number of pins using Steiner tree.	Steiner minimal tree, FLUTE
Optimization Based	Improve routing iteratively using cost functions.	Simulated Annealing, ILP based

4. ADVANTAGES

- Simple and easy to implement.
- Good for channel / global routing.
- Handles congestion by rip-up and re-route.
- Provides good results with preferred directions.
- Fast and memory efficient.

5. DISADVANTAGES

- May take more iterations for dense designs.
- Not guaranteed to find optimal path.
- Performance depends on direction order.
- Rip-up can increase runtime in congested areas.

6. APPLICATIONS

- Channel routing in ASICs and FPGAs.
- Global routing during early physical design.
- Used in academic tools and industrial routers.
- ECO routing and congestion handling.

LEGEND

- [S] → Source
- [T] → Target / Sink
- [#] → Obstacle / Blockage
- • → Free Cell
- * → Visited / Routed Cell
- → → Routed Path

KEY POINT

Handlock algorithm routes nets using preferred directions and resolves conflicts by rip-up and re-route until a valid path is obtained.

7. PROCEDURE (ALGORITHM)

1. Initialize all cells as UNVISITED.
2. Put source cell (S) in stack / list.
3. Define preferred directions (e.g., Right (R), Up (U), Left (L), Down (D)).
4. Route the net in the first preferred direction.
5. If conflict occurs (handlock), rip-up the conflicting net (with lower priority or longer length).
6. Push the rip-up net back to list with next preferred direction.
7. Continue routing other nets in the channel.
8. Repeat until all nets are routed or all directions are exhausted.

Preferred Direction Order

1st : → (Right)
2nd : ↑ (Up)
3rd : ← (Left)
4th : ↓ (Down)

Observation

- Rip-up net is usually the one with lower priority.
- History (cost) helps to avoid oscillation.
- Handlock is local deadlock while routing in channel.

8. STEP BY STEP EXPLANATION WITH DIAGRAMS

Example Channel : 7 x 3 grid (routing channel)
Two nets to route : Net-1 : S_2 (1,1) → T_1 (6,1)
Net-2 : S_2 (1,3) → T_2 (6,3)
Preferred direction order : R → U → L → D
→ Blockage

Step 0 : Initial Grid

	1	2	3	4	5	6	7
3	S_2	.	.	#	.	.	T_2
2	.	.	.	.	.	.	.
1	S_1	.	.	#	.	.	T_1

Row numbers shown on left.

Step 1 : Route Net-1 (Right)

	1	2	3	4	5	6	7
3	S_2	.	.	#	.	.	T_2
2	.	.	.	.	.	.	.
1	S_1	*	*	#	*	*	T_1

Net-1 routed from (1,1) → (6,1)

Step 2 : Route Net-2 (Right) → Conflict

	1	2	3	4	5	6	7
3	S_2	*	*	#	*	*	T_2
2	.	.	.	.	.	.	.
1	S_1	*	*	#	*	*	T_1

Conflict at cells (2,1)-2) with Net-1.
Handlock occurs.

Step 3 : Rip-up Net-1 (Lower Priority)

	1	2	3	4	5	6	7
3	S_2	*	*	#	*	*	T_2
2	.	.	.	.	.	.	.
1	S_1	.	.	#	.	.	T_1

Net-1 is removed to resolve conflict.

Step 4 : Route Net-2 with Successfully

	1	2	3	4	5	6	7
3	S_2	*	*	#	*	*	*
2	S_1	↕	↕	↑ ↓	.	.	.
1	.	.	.	#	.	.	.

Net-2 routed successfully

Step 5 : Re-route Net-1 with 2nd Pref (Up)

	1	2	3	4	5	6	7
3	S_2	*	*	#	*	*	T_2
2	↕	↕	↕	↕	.	.	.
1	.	.	.	#	.	.	T_1

Net-1 moved from (1,1) → (1,2) → (1,3)

Step 6 : Continue Net-1 to Right at Row 3

	1	2	3	4	5	6	7
3	S_1	*	*	#	*	*	T_1
2	S_2	*	*	#	*	*	T_2
1	.	.	.	.	.	.	.

Net-1 now moves right on row-3.

Step 7 : Final Routing (Both Nets Routed)

	1	2	3	4	5	6	7
3	S_1	*	*	#	*	*	T_1
2	S_2	*	*	#	*	*	T_2
1	.	.	.	#	.	.	T_1

All nets successfully routed.

9. PROCEDURE EXAMPLE (TRACE TABLE)

Step	Current Net	Action	Preferred Dir Tried	Result	Status
1	Net-1	Route	R	–	Routed
2	Net-2	Route	R	Conflict with Net-1	Handlock
3	Net-1	Rip-up	–	–	–
4	Net-2	Route	R	–	Routed
5	Net-1	Route	U	Partial (to row-3)	–
6	Net-1	Route	R	Success	Routed
Final	–	–	–	Both nets routed	Done

10. COMPLEXITY

- Time Complexity : O (N × D × I) where N = nets, D = directions, I = iterations (rip-ups).
- Space Complexity : O (N × M) (for grid storage and history).

11. COMPARISON

Aspect	Line Search	Maze	Handlock
Approach	Greedy straight ext.	Explore all directions	Preferred dirs + rip-up
Backtracking	No	Yes	Yes (selective)
Optimality	Less	Less	Less
Runtime	Low	High	Medium
Congestion Handling	Poor	Good	Good

13. PRACTICAL OBSERVATIONS

- Handlock is a simple yet effective negotiation method.
- Proper priority and history improve convergence.
- Works well in global routing before detailed routing.
- Used in many academic and commercial routers.

14. INTERVIEW / VIVA QUESTIONS

Q. What is Handlock algorithm ?
Q. How does Handlock resolve conflicts ?
Q. What is rip-up and re-route ?
Q. When is Handlock preferred over Line Search ?
Q. What is the role of preferred directions ?

15. COMMARY VIOLATIONS & POINTS

- ✓ Handlock routes nets using preferred directions.
- ✓ Conflicts are solved by rip-up and re-route.
- ✓ Uses history and priority to avoid deadlocks.
- ✓ Good for channel and global routing.
- ✓ Not guaranteed optimal but efficient.
- ✓ Improves routing quality with iterations.

16. QUICK RECAP FLOW

Route Net → Conflict → Rip-up → Try Next Direction → Success or All Directions Tried → Fail

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PD _ Routing

F) HIGH TOWER ALGORITHM

1. DEFINITION

☑ High Tower algorithm is a cost based channel routing algorithm. It assigns higher cost to congested / blocked cells and lower cost to free cells and selects the minimum cost path in the channel. The path is built gradually like a 'tower' by always choosing the lowest cost cell while rip-up and re-route happen when conflicts occur.

2. IDEA

☑ Instead of exploring all paths, the algorithm uses a cost map (tower values) to guide the search. Nets grow from source towards target by picking minimum cost neighbor. Conflicts are resolved by increasing cost (tower) of congested cells and re-routing.

3. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Exact Algorithms	Guarantee optimal (shortest) path.	Lee, Dijkstra, Hung-Jan
Heuristic Algorithms	Find good path, not necessarily optimal.	Maze, Line Search, High Tower, M.T.
Negotiation Based	Multiple nets negotiate for resources.	PathFinder, NTHU-route
Steiner Tree Based	Minimize number of pins using Steiner tree.	Steiner minimal tree, FLUTE
Optimization Based	Improve routing iteratively using cost functions.	Simulated Annealing, ILP based

4. ADVANTAGES

- Handles congestion effectively using cost values.
- Faster convergence compared to simple maze.
- Good quality routing with fewer rip-ups.
- Scalable for large designs.
- Uses historical information (tower map).

5. DISADVANTAGES

- Cost update and re-routing increase runtime.
- Not guaranteed to find global optimum.
- Needs good cost initialization.
- More memory than simple path search.

6. APPLICATIONS

- Channel routing in ASIC / FPGA.
- Global routing in early physical design.
- Widely used in commercial routers.
- Useful for designs with high congestion.

LEGEND

- [S] → Source
- [T] → Target / Sink
- [#] → Obstacle / Blockage
- • → Free Cell
- C(x) → Tower Cost (integer)
- → → Routed Path

KEY POINT

High Tower algorithm assigns high cost to congested cells and grows the net towards target by always selecting the cell with minimum tower cost, thereby reducing congestion and improving routing quality.

7. PROCEDURE (ALGORITHM)

1. Initialize all cells as FREE and assign base cost = 1.
2. Put source cell (S) in active list.
3. For each neighboring cell, compute tentative tower cost.
4. Select the neighbor with minimum tower cost and add to path.
5. Mark routed cell as PERMANENT.
6. Repeat steps 3-5 until target (T) is reached.
7. If conflict occurs, increase cost of conflicting cells and rip-up.
8. Re-route using updated tower map.

Preferred Direction Order

1st : → (Right)
2nd : ↑ (Up)
3rd : ← (Left)
4th : ↓ (Down)

Observation

Higher tower value means higher congestion. Algorithm avoids high tower cells and chooses lower tower path.

8. STEP BY STEP EXPLANATION WITH DIAGRAMS

Example Channel : 7 x 3 grid (routing channel)

Net : S at (1,1) → T at (7,3)

Preferred direction order : R → U → L → D # → Blockage

Step 0 : Initial Grid (Base cost = 1)

	1	2	3	4	6	7
3	1	1	1	1	1	1
2	:	:	1	#	:	:
1	S	1	1	1	1	T

C( )

Step 1 : Initialize from Source (1,1)

	1	2	3	4	5	6	7
3	1	1	1	1	1	1	1
2	:	:	1	#	1	1	1
1	S(0)	1	1	1	1	1	T

C(0)

Step 2 : Expand Right (min cost neighbor)

	1	2	3	4	5	6	7
3	1	1	1	1	1	1	1
2	1	1	1	#	:	1	1
1	S	1(=1)	2	2	2	2	T

C(1)

Step 3 : Continue Right

	1	2	3	4	5	6	7
3	1	1	1	1	1	1	1
2	1	1	1	#	1	1	1
1	S	1	2	3	3	3	T

C(2)

Step 4 : Move Up at (4,1)

	1	2	3	4	5	6	7
3	1	1	1	1	1	1	1
2	:	1	1	#	↑	4	1
1	S	1	2	3	3	3	7

C(3)

Step 5 : Move Up again to (4,2)

	1	2	3	4	5	6	7
3	1	1	1	5	5	5	1
2	1	1	1	#	4	4	1
1	S	1	2	3	3	3	T

C(4)

Step 6 : Continue Right on row-2.

	1	2	3	4	5	6	7
3	1	1	1	5	6	6	6
2	1	1	1	#	4	4	4
1	S	1	2	3	3	3	T

C(5)

Step 7 : Reach Target (7,3)

	1	2	3	4	5	6	7
3	1	1	1	5	6	7	T(7)
2	1	1	1	#	4	4	4
1	S	1	2	3	3	3	3

C(6)

Final Routed Path : (1,1) → (2,1) → (3,1) → (4,1) → (4,2) → (5,2) → (6,2) → (7,2) → (7,3)

9. PROCEDURE EXAMPLE (TRACE TABLE)

Step	Current Cell	Selected Dir.	Next Cell	Tower Cost	Action	Status
1	(1,1)	R	(2,1)	1	Expand	OK
2	(2,1)	R	(3,1)	2	Expand	OK
3	(3,1)	R	(4,1)	3	Expand	OK
4	(4,1)	U	(4,2)	4	Expand	OK
5	(4,2)	R	(5,2)	5	Expand	OK
6	(5,2)	R	(6,2)	6	Expand	OK
7	(6,2)	R	(7,2)	6	Expand	OK
8	(7,2)	U	(7,3)	7	Reach Target	Success

10. COMPLEXITY

- Time Complexity : O(N x D x I) where N = cells, D = directions (4), I = iterations (rip-ups).
- Space Complexity : O(N x M) (for cost map storage).
- In practice, efficient routing with fewer rip-ups.

11. COMPARISON TABLE

Aspect	Maze	Line Search	High Tower
Approach	Explore all dirs	Limited dirs	Cost guided
Cost Function	No	No	Yes (tower)
Congestion Handling	Poor	Medium	Good
Rip-up freq.	High	Medium	Low
Runtime	High	Medium-High	Medium
Optimality	No Opt.	Near optimal	Good quality

13. PRACTICAL OBSERVATIONS

- Tower values should be updated after every rip-up to reflect congestion.
- Good initial cost map reduces iterations.
- Preferred directions help in faster convergence.
- Works well in global routing before detailed routing stage.
- Widely used in industrial routers.

14. INTERVIEW / VIVA QUESTIONS

Q. What is High Tower algorithm ?
Q. How are tower costs assigned ?
Q. How does it handle conflicts ?
Q. What is the role of preferred directions ?
Q. Advantages of High Tower over Maze ?
Q. Where is it used in physical design flow ?

15. SUMMARY / REVISION & PINTS

✓ High Tower uses cost (tower) values to guide routing.
✓ Higher cost for congested cells, lower for free cells.
✓ Net grows from source to target using minimum cost neighbor.
✓ Conflicts are resolved by rip-up and cost update.
✓ Faster convergence than simple maze.
✓ Good quality routing with fewer rip-ups.
✓ Widely used in global routing.
✓ History of congestion improves routing.

16. QUICK RECAP FLOW

Initialize Costs → Put Source in List →
Expand (Min Tower Neighbor) → Add to Path →
Reach Target ? → Yes : Done
No Conflict : Continue Expansion
Conflict : Raise Cost → Rip-up → Re-route

Author : Konda Tharun Vivekananda

G) STENIER ALGORITHM

1. DEFINITION

☑ Steiner algorithm is a tree based routing algorithm used to interconnect a set of terminals (pins) using minimum total wire length. It may introduce additional points called Steiner points to reduce overall length. It is commonly used in global routing for multi-terminal nets.

2. IDEA

☑ Group the terminals pairwise and build minimum length connection (Steiner tree) gradually. At each step, combine the two groups with least connection cost until a single tree connects all terminals.

3. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Exact Algorithms	Guarantee optimal (shortest) path.	Lee, Dijkstra, Steiner tree
Heuristic Algorithms	Find good path, not necessarily optimal.	Maze, Line Search, High Tower, Steiner
Negotiation Based	Multiple nets negotiate for resources.	PathFinder, NTHU-route
Steiner Tree Based	Minimize number of pins using Steiner tree.	Steiner minimal tree, FLUTE
Optimization Based	Improve routing iteratively using cost functions.	Simulated Annealing, ILP based

4. ADVANTAGES

- Produces minimum length interconnections.
- Reduces overall wire length significantly.
- Handles multi-terminal nets efficiently.
- Well suited for global routing.
- Better routability in congested regions.

5. DISADVANTAGES

- Computationally expensive for large nets.
- Added Steiner points increase routing complexity.
- Not always accurate for detailed routing.
- Requires more memory for large designs.

6. APPLICATIONS

- Global routing of large multi-terminal nets.
- Used in ASIC and FPGA physical design.
- Interconnect planning in SoC and IP blocks.
- Floorplan driven early wirelength estimation.

LEGEND

- [S] → Source / Terminal
- [T] → Target / Terminal
- ✱ → Steiner Point
- # → Obstacle / Blockage
- → → Routed Wire
- .... → Possible Connection

KEY POINT

Steiner algorithm creates a minimum length tree by allowing additional Steiner points to connect all given terminals with least total wirelength.

7. PROCEDURE (ALGORITHM)

1. Identify all terminals (pins) to be connected.
2. Compute pairwise shortest path cost between all terminal pairs.
3. Select two groups with minimum connection cost.
4. Merge the two groups and add a Steiner point (if needed) to connect them.
5. Update cost table for the new group.
6. Repeat steps 3 to 5 until only one group remains.
7. The final tree connects all terminals.

Preferred Direction Order

1st : → (Right)
2nd : ↑ (Up)
3rd : ← (Left)
4th : ↓ (Down)

8. STEP BY STEP EXPLANATION WITH DIAGRAMS

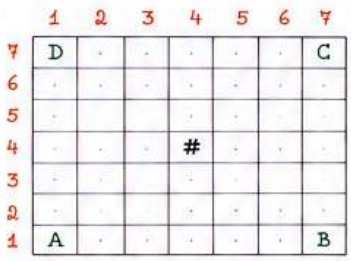
Example Channel : 7 x 7 grid (routing channel)

Net : Connect terminals A(1,1), B(1,7), C(7,1), D(7,7)

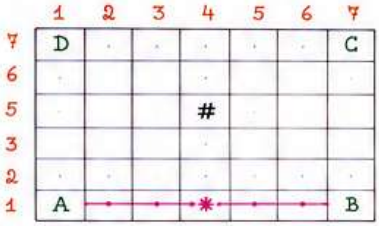
Preferred direction order : R → U → L → D

→ Blockage

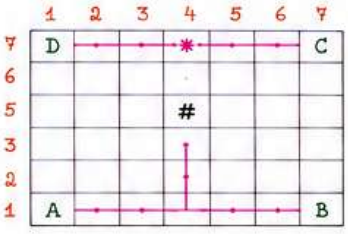
Step 0 : Initial Grid with Terminals



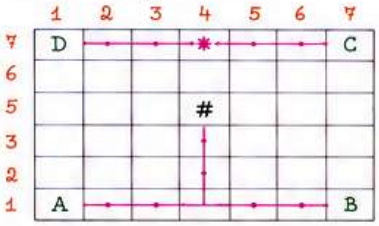
Step 1 : Connect A(1,1) and B(1,7)



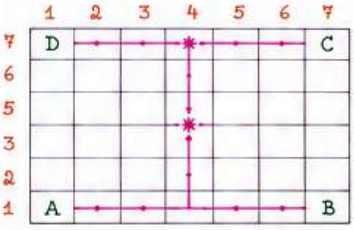
Step 2 : Connect C(7,1) and D(7,7)



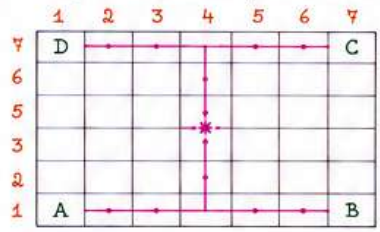
Step 3 : Tangier Steines



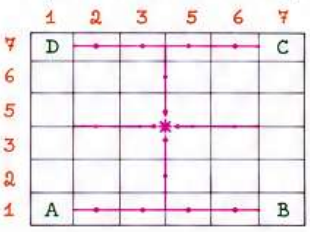
Step 4 : Merge (A,B) group with (C,D) group



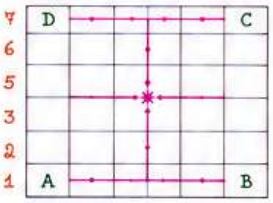
Step 5 : Add Steiner point at Steiner point



Step 6 : Extend Corniner at (4,4)



Step 7 : Final Steiner Tree



Final Routed Path Sequence

A(1,1) → (1,4) → (4,4)
B(1,7) → (1,4) → (4,4)
C(7,1) → (4,1) → (4,4)
D(7,7) → (4,7) → (4,4)
Steiner Point at : (4,4)

9. PROCEDURE EXAMPLE (TRACE TABLE)

Step	Groups	Cost Metric (wirelength)	Selected to Merge	New Steiner Pt	Total Cost	Status
0	{A} {B} (C) (D)	–	–	–	0	Start
1	(A,B) (C) (D)	6	S	–	6	Merged
2	(A,B) (C,D)	6	6	–	12	Merged
3	(A,B) (C,D)	8	Both	–	20	Merged
4	Single Tree	–	–	–	20	Done

10. COMPLEXITY

- Time Complexity : $O(n^3)$ (for n terminals)
- Space Complexity : $O(n^2)$ n = number of terminals.

11. COMPARISON TABLE

Aspect	Maze	Steiner	High Tower
Optimality	No	Approx.	Approx.
Extra Points	No	Yes (Y)	Yes
Wirelength	Higher	Lower	Lower
Congestion	High	Lower	Lower
Use Case	Detailed	Global	Global

12. PRACTICAL OBSERVATIONS

- Steiner points reduce total wirelength.
- Cost metric must be updated after each merge.
- Better for global routing of multi-terminal nets.
- Quality depends on choice of cost and merge strategy.
- Widely used in industrial global routers.

13. INTERVIEW / VIVA QUESTIONS

Q. What is Steiner algorithm in routing ?
Q. Why are Steiner points used ?
Q. What is time complexity of Steiner algorithm?
Q. Difference between Maze and Steiner algorithm?
Q. Where is Steiner algorithm used in PD flow?

14. SUMMARY / REVISION POINTS

✓ Steiner algorithm builds minimum length tree using Steiner points.
✓ Works on cost metric (tower values) not equal direction exploration.
✓ Reduces wirelength and improves routability.
✓ Used in global routing for multi-terminal nets.
✓ Adds Steiner points to reduce congestion and length.
✓ Higher computational cost than simple maze.
✓ Not used directly for detailed routing.
✓ Important for early routing estimation and floorplanning.

15. QUICK RECAP FLOW

Identify Terminals → Build Cost Table → Merge Min Cost Groups → Add Steiner Pt → Update Cost → Repeat → Final Tree

Author : Konda Tharun Vivekananda

PD _ Routing

H) A* ALGORITHM

1. DEFINITION

☑ A* (A-star) algorithm is an informed search algorithm used to find the minimum cost path from a source to a target in a graph. It uses both the actual cost from start to current node ($g(n)$) and an estimated cost from current node to goal ($h(n)$). It always expands the node with the least total cost $f(n) = g(n) + h(n)$ which leads to optimal routing in VLSI grids.

2. IDEA

☑ Guide the search using a heuristic estimate to reach the target faster by avoiding unnecessary expansions. The heuristic must never overestimate the remaining cost for optimality.

3. CLASSIFICATION OF ROUTING ALGORITHMS

Type	Description	Examples
Exact Algorithms	Guarantee optimal (shortest) path.	Lee, Dijkstra, Steiner tree
Heuristic Algorithms	Find good path, not necessarily optimal.	Maze, Line Search, High Tower, A*
Negotiation Based	Multiple nets negotiate for resources.	PathFinder, NTHU-route
Steiner Tree Based	Minimize number of pins using Steiner tree.	Steiner minimal tree, FLUTE
Optimization Based	Improve routing iteratively using cost functions.	Simulated Annealing, ILP based

4. ADVANTAGES

- Finds optimal shortest path.
- Uses heuristic to reduce exploration.
- Efficient for large routing grids.
- Well suited for multi-terminal nets.
- Balances exploration and exploitation.

5. DISADVANTAGES

- Heuristic function design is critical.
- Memory requirement can be high (open list).
- Performance degrades if heuristic is weak.
- Not ideal when heuristic overestimates.
- Higher computational cost than simple maze.

6. APPLICATIONS

- Global routing in ASIC / FPGA.
- Detailed routing with complex obstacles.
- SoC interconnect planning.
- VLSI floorplan driven routing.
- Power/Clock grid routing in advanced nodes.

LEGEND

- S → Source / Start
- T → Target / Goal
- # → Obstacle / Blockage
- • → Free Cell
- $g(n)$ → Cost from Start
- $h(n)$ → Heuristic to Goal
- $f(n)$ → Total Cost = g+h
- → → Expanded Path

KEY POINT

A* algorithm evaluates a node using $f(n) = g(n)+h(n)$. It expands the node with minimum $f(n)$ which guarantees the optimal route when $h(n)$ is admissible (never overestimates).

7. PROCEDURE (ALGORITHM)

1. Initialize OPEN list with source node (S).
2. Set $g(S) = 0$, compute $h(S)$, $f(S) = g + h$.
3. Repeat until OPEN list is empty :
   a) Pick node n in OPEN with minimum $f(n)$.
   b) If n is goal (T), reconstruct path and stop.
   c) Move n from OPEN to CLOSED list.
   d) For each neighbor m of n :
      - If m is obstacle → skip.
      - Compute $g(m) = g(n) + cost(n,m)$
      - Compute $h(m)$, $f(m) = g(m) + h(m)$
      - If m in CLOSED with lower g → skip.
      - Else add/update m in OPEN.
4. If OPEN is empty, no path exists.

Preferred Direction Order

1st : → (Right)
2nd : ↑ (Up)
3rd : ← (Left)
4th : ↓ (Down)

Heuristic Examples

- Manhattan Distance : $h = |x_1 - x_2| + |y_1 - y_2|$
- Euclidean Distance : $h = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
- Diagonal Distance : $h = \max(dx, dy)$ where $dx = |x_1 - x_2|$, $dy = |y_1 - y_2|$

8. STEP BY STEP EXPLANATION WITH DIAGRAMS

Example Channel : 5 x 5 grid (routing channel)
Net : S (1,1) → T (5,5)
Heuristic : Manhattan Distance

→ Blockage

Step 0 : Initial Grid

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	.
3	.	#	.	.	#
2	.	.	.	.	.
1	S	.	.	.	.

Step 1 : Expand Start (1,1)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	.
3	.	#	.	.	#
2	1	.	.	.	.
1	S	.	.	.	.

g(2,1) = 1
h = 7
f = 8

Step 2 : Expand (2,1)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	.
3	.	#	.	.	#
2	1	2	.	.	.
1	S	.	.	.	.

(3,1)
g = 2
h = 6
f = 8

Step 3 : Expand (3,1)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	.
3	.	#	.	.	#
2	1	2	3	.	.
1	S	.	.	.	.

(4,1)
g = 2
h = 5
f = 8

Step 4 : Expand (4,1)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	.
3	.	#	.	.	#
2	1	2	3	4	.
1	.	.	.	.	.

(4,2)
g = 4
h = 4
f = 8

Step 5 : Expand (4,2)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	.
3	.	#	.	.	#
2	1	2	3	4	.
1	.	.	.	.	.

(5,2)
g = 5
h = 3
f = 8

Step 6 : Expand (4,2)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	.
3	.	#	.	.	#
2	1	2	3	4	6
1	.	.	.	.	.

(5,3)
g = 6
h = 2
f = 8

Step 7 : Expand (5,2)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	7
3	.	#	.	.	#
2	1	2	3	4	6
1	.	.	.	.	.

(5,4)
g = 7
h = 1
f = 8

Step 8 : Expand (5,4)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	8
3	.	#	.	.	#
2	.	.	.	.	.
1	.	.	.	.	.

(5,5)
g = 8
h = 0

Step 9 : Goal Reached (5,5)

	1	2	3	4	5
5	.	.	.	.	T
4	.	#	.	.	8
3	.	#	.	.	7
2	1	2	3	4	6
1	S	.	.	.	.

Optimal path cost = 8

Path :
(1,1) → (2,1) → (3,1) → (4,1) → (4,2) → (5,2) → (5,3) → (5,4) → (5,5)

9. PROCEDURE EXAMPLE (TRACE TABLE)

Step	Current Node	g(n)	h(n)	f(n)	OPEN List (f,node)	CLOSED List
0	(1,1)	0	8	8	{(1,1)}	{ }
1	(2,1)	1	7	8	{(2,1)}	{(1,1)}
2	(3,1)	2	6	8	{(3,1)}	{(1,1),(2,1)}
3	(4,1)	3	5	8	{(4,1)}	{(1,1),(2,1),(3,1)}
4	(4,2)	4	4	8	{(4,2)}	{(1,1),(2,1),(3,1),(4,1)}
5	(5,2)	5	3	8	{(5,2)}	{(1,1),(2,1),(3,1),(4,1),(4,2)}
6	(5,3)	6	2	8	{(5,3)}	{..., (5,2)}
7	(5,4)	7	1	8	{(5,4)}	{..., (5,3)}
8	(5,5)	8	0	8	{ }	{..., (5,4)} Goal

10. COMPLEXITY

- Time Complexity : $O(E \log V)$ (using priority queue)
- Space Complexity : $O(V)$ where V = number of cells/nodes, E = number of edges

11. COMPARISON TABLE

Aspect	Lee	Dijkstra	A* (A-star)
Heuristic Use	No	No	Yes
Optimal Path	Yes	Yes	Yes
Exploration	High	Medium	Low
Speed	Slow	Faster	Fastest
Memory	Low	High	High
Use Case	Small grids	Weighted grids	Large grids

12. PRACTICAL OBSERVATIONS

- Good heuristic reduces routing runtime significantly.
- Manhattan distance works well in orthogonal grids.
- A* explores fewer nodes than Dijkstra.
- Widely used in global and detailed routing engines.
- Path quality depends on cost model and heuristic.

13. INTERVIEW / VIVA QUESTIONS

Q. What is A* algorithm used for in routing ?
Q. What is $f(n)$ in A* algorithm ?
Q. Why heuristic must be admissible ?
Q. Advantage of A* over Dijkstra ?
Q. How does A* handle obstacles ?

14. SUMMARY / REVISION POINTS

✓ A* algorithm is an informed search routing algorithm.
✓ Uses $f(n) = g(n) + h(n)$ to guide the search.
✓ Expands node with minimum total cost.
✓ Guarantees optimal path with admissible heuristic.
✓ Less node expansion → faster routing.
✓ Suitable for large VLSI routing grids.
✓ Widely used in industrial routing tools.
✓ Combines advantages of speed and optimality.

15. QUICK RECAP FLOW

Start → Initialize OPEN → Evaluate f(n) → Expand Min f Node → Update Neighbors → Goal ? → Yes : Reconstruct Path → Stop
→ No : Repeat

Author : Konda Tharun Vivekananda

PD – Routing

CONGESTION ANALYSIS

1. INTRODUCTION

☑ Congestion occurs when routing demand in a region exceeds the available routing capacity. It is one of the major reasons for routing detours, timing degradation and even routing failures.

2. DEFINITION

☑ Congestion is the condition where the routing resources required to connect all nets in a region are more than the routing resources available in that region.

3. WHY IT HAPPENS ?

- Poor floorplan / macro placement
- Too many nets passing through a small region
- Uneven cell / net distribution
- Insufficient routing tracks due to technology or blockage
- High connectivity (high fanout, buses)
- Lack of early congestion driven optimization

4. ROUTING DEMAND

Routing demand is the total number of routing tracks required by all nets to pass through a routing region (tile) in a specific direction.

Example : In tile, if 12 nets need to pass horizontally, demand = 12 tracks.

5. ROUTING CAPACITY

Routing capacity is the total number of routing tracks available in a region in a specific direction considering blockages and reserved resources.

Example : In same tile if only 8 horizontal tracks are available, capacity = 8 tracks.

6. CONGESTION FORMULAE

Congestion (C)	Overflow	Normalized Overflow
$C = \frac{\text{Demand}}{\text{Capacity}}$	$O = \text{Demand} - \text{Capacity}$	$NO = \frac{\text{Demand} - \text{Capacity}}{\text{Capacity}}$
(per tile / per direction)	(tracks)	(when Demand > Capacity)

7. WHAT IS OVERFLOW ?

☑ Overflow is the number of extra tracks required beyond the available capacity to route all nets.

Overflow = max (0, Demand − Capacity)

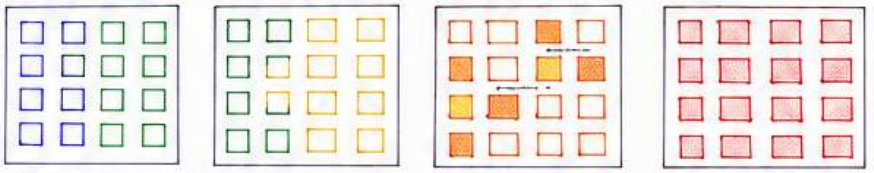
If Overflow > 0 ⇒ Congested (C > 1)
If Overflow = 0 ⇒ No congestion (C ≤ 1)

8. TYPES OF CONGESTION

- Global Congestion : When large areas of the design have high demand.
- Local Congestion : Small localized regions exceed capacity.
- Pin Congestion : Too many connections to a pin or a small pin region.
- Via Congestion : Excessive via usage between metal layers.
- Track Congestion : More nets compete for the same routing tracks.

9. CONGESTION MAPS

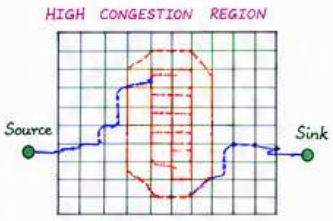
☑ Congestion map is a color coded heat map showing congestion level across the chip.



0 – 0.5 (No issue) | 0.5 – 0.8 (Low) | 0.8 – 1.2 (Moderate) | 1.2 – 1.5 (High) | > 1.5 (Severe)

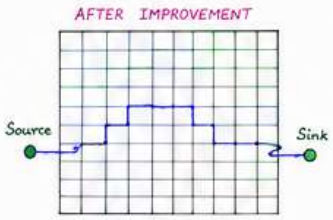
10. CONGESTION – IMPACTS

- Increases routing wirelength (detours)
- Causes timing violations (longer delays)
- More vias → more RC → more delay
- Higher power due to long interconnects
- Possible routing failures (unroutable regions)
- Increases design closure time



11. CONGESTION FIX TECHNIQUES

- Improves floorplanning and macro placement
- Spread high fanout nets using buffering / replication
- Adjust placement to spread cells evenly
- Increase routing resources (wider channels, more layers)
- Use congestion driven placement (CDP)
- Reroute congestion nets with higher cost for congested tiles
- Add routing blockages / guides to steer nets
- Optimize pin access (better pin placement, more access)



12. CONGESTION FLOW IN PD

Placement → Global Routing → Congestion Analysis → Fixes (CDP) → Reroute → Detailed Routing

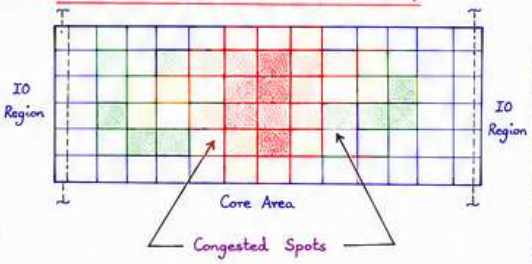
13. EXAMPLE – CONGESTION CALCULATION

Tile ID	Direction	Demand (Tracks)	Capacity (Tracks)	Congestion C = D/C	Overflow O = D − C	Normalized Overflow C = D/C
T1	Horizontal	12	8	1.50	4	0.50
T1	Vertical	6	8	0.75	0	0
T2	Horizontal	9	9	1.00	0	0
T2	Vertical	15	10	1.50	5	0.50

Interpretation

- C ≤ 1 → No congestion
- 1 < C ≤ 1.2 → Moderate
- 1.2 < C ≤ 1.5 → High
- C > 1.5 → Severe

14. CONGESTION MAP EXAMPLE (CONCEPTUAL)



15. PRACTICAL OBSERVATIONS

- Early congestion analysis saves rework.
- Congestion usually appears after global routing and during placement.
- Fixing congestion late increases wirelength and timing risk.
- Balanced utilization of routing resources is key to good PPA.

16. KEY POINT

Congestion = Demand > Capacity

High congestion leads to detours, timing issues and routing failures.

Continuous analysis and optimization are essential for good routing closure.

17. APPLICATIONS

- ASIC physical design
- SoC routing optimization
- Congestion driven placement
- Floorplanning evaluation
- High performance designs

18. INTERVIEW / VIVA QUESTIONS

Q1. What is congestion in routing ?
A: When routing demand exceeds available routing capacity in a region.
Q2. How do you measure congestion ?
A: By congestion factor C = Demand / Capacity.
Q3. What are the impacts of congestion ?
A: Longer wirelength, timing violations, power increase and routing failures.
Q4: How can congestion be reduced ?
A: Better floorplan, placement, buffering, increasing resources and CDP.

19. SUMMARY / REVISION POINTS

- ✓ Congestion occurs when demand > capacity.
- ✓ Congestion factor C = Demand / Capacity.
- ✓ Overflow = max(0, Demand − Capacity).
- ✓ Types : global, local, pin, via, track.
- ✓ Congestion map shows hotspots.
- ✓ High congestion → detours & timing issues.
- ✓ Fix using CDP, placement spread, buffering, and more routing resources.

Author : Konda Tharun Vivekananda

SIGNAL INTEGRITY DURING ROUTING

1. INTRODUCTION

☑ As interconnects become faster and denser, signal integrity (SI) issues become critical during routing. Unwanted interaction between nearby nets causes delays, noise and logic failures.

2. DEFINITION

☑ Signal Integrity is the ability of a signal to travel from source to destination without degradation of its intended functionality due to noise, interference or distortion.

3. WHY IT HAPPENS ?

- Close proximity of parallel interconnects.
- Capacitive & inductive coupling between nets.
- High switching activity on nearby nets.
- Long wires and multiple metal layers.
- Aggressor switching causes noise on victim.

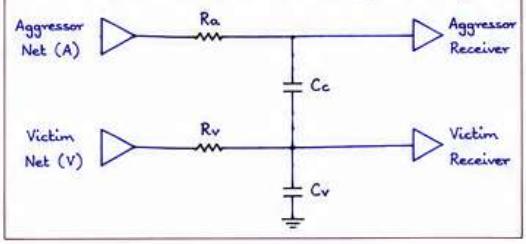
4. CROSSTALK – COUPLING CAPACITANCE

Crosstalk is unwanted noise coupled from an aggressor net to a victim net due to mutual capacitance (Cc) between them.

Types of Coupling

- Capacitive (dominant)
- Inductive (at high frequency)

COUPLING CAPACITANCE MODEL (CAPACITIVE)



5. AGGRESSOR NET

- Net which switches and causes noise.
- Source of disturbance.
- Denoted as Net A.

6. VICTIM NET

- Net which is affected by the aggressor.
- Carries the original signal.
- Denoted as Net V.

7. CROSSTALK DELAY

☑ Coupled noise can change the voltage on victim net and modify the effective propagation delay.

a) Positive Crosstalk (Delay Increase)

- Aggressor causes extra charge on victim.
- Slows down rising edge → increases delay.
- Critical for setup timing.

b) Negative Crosstalk (Delay Decrease)

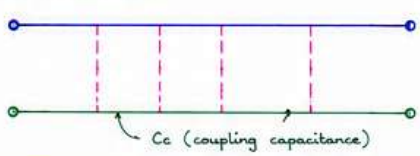
- Aggressor removes charge from victim.
- Speeds up rising edge → decreases delay.
- Critical for hold timing.

KEY POINT

✓ SI issues must be considered during routing.
✓ Keep sufficient spacing between parallel nets.
✓ Shielding and layer assignment help reduce noise.
✓ Balanced crosstalk avoids setup & hold failures.

8. CROSSTALK DIAGRAMS & EXPLANATION

(a) Two Parallel Nets

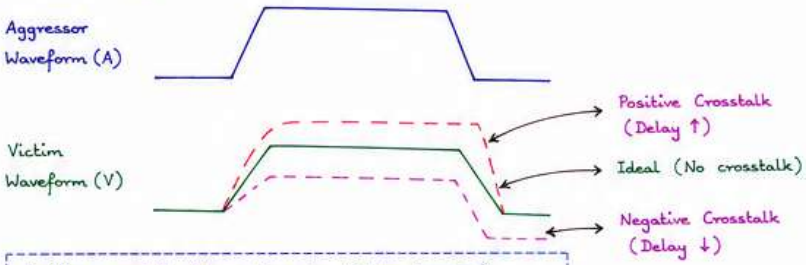


LEGEND

- Aggressor net
- Victim net
- Coupling Capacitance
- Shield / Ground

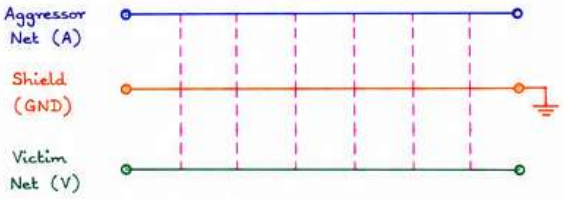
Explanation :
When aggressor (A) switches, coupling capacitance (Cc) injects noise current into victim (V) causing unwanted voltage change.

(b) Crosstalk Effect on Victim Waveform



Positive crosstalk delays the edge (delay increase).
Negative crosstalk speeds up the edge (delay decrease).

(c) Shielding to Reduce Crosstalk



Shield layer (connected to GND) absorbs coupling and protects victim from noise.

9. CROSSTALK DELAY EQUATIONS

Noise voltage induced on victim due to aggressor switching :

$$\Delta V = \frac{Cc}{Cv + Cpar} \times \Delta V_A$$

where,
Cc : Coupling capacitance
Cv : Total capacitance of victim net
Cpar : Parasitic capacitance to ground
ΔV_A : Voltage change on aggressor

Approx. delay change on victim :

$$\Delta t \approx \frac{\Delta V}{(dV/dt)_v}$$

$\left(\frac{dV}{dt}\right)_v$: Slope of victim signal

Delay Interpretation :

- Positive ΔV → Delay increase → Setup risk
- Negative ΔV → Delay decrease → Hold risk

10. CROSSTALK SUMMARY TABLE

Parameter	Description	Effect on Victim
Aggressor Net	Net that switches	Creates noise
Victim Net	Net that is affected	Receives noise
Coupling Capacitance (Cc)	Capacitance between parallel nets	Couples voltage
Positive Crosstalk	Same polarity noise	Delay ↑ (Setup risk)
Negative Crosstalk	Opposite polarity noise	Delay ↓ (Hold risk)
Shielding	Insert grounded shield	Reduces noise

11. CROSSTALK IMPACTS

- Timing violations (setup / hold failures)
- Increase in bit error rate (BER)
- Glitches and functional failures
- Reduced noise margin
- Higher dynamic power

12. CROSSTALK MITIGATION TECHNIQUES

- Increase spacing between nets
- Use shielding (GND wires)
- Proper layer assignment
- Reduce parallel run length
- Control aggressor switching (balance activity)
- Use wide / low resistance wires

13. PRACTICAL OBSERVATIONS

- Crosstalk is more critical in advanced nodes.
- Parallel bus routing requires careful spacing.
- Shielding is essential for high speed clocks.
- Always verify SI in sign-off using tools.

14. INTERVIEW / VIVA QUESTIONS

Q1. What is crosstalk in VLSI routing ?
A : Unwanted noise coupled from an aggressor net to a victim net due to mutual capacitance / inductance.
Q2. Difference between positive & negative crosstalk ?
A : Positive delays the signal, negative advances the signal.
Q3. How to reduce crosstalk during routing ?
A : Increase spacing, shielding, layer assignment, reduce parallel length and balance switching activity.

Author : Konda Tharun Vivekananda

ANTENNA VIOLATIONS & FIXING

1. INTRODUCTION

☑ During fabrication, plasma exists in the etch / deposition chamber. Long metal interconnects behave like antenna and collect charge from the plasma. This charge can damage the thin gate oxide of MOS devices connected to it.

2. DEFINITION

☑ Antenna violation occurs when the ratio of metal area connected to a gate (antenna area) to the gate oxide area exceeds the technology specific limit.

$$\text{Antenna Ratio (AR)} = \frac{\text{Antenna Area (Aa)}}{\text{Gate Oxide Area (Ag)}}$$

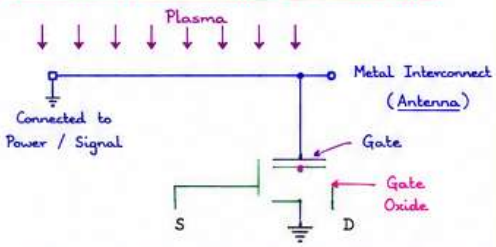
If $AR > AR_{limit}$ → Antenna Violation

(AR_{limit} is given in tech file)

3. WHY THIS IS DANGEROUS ?

- Excess charge at the gate may puncture the gate oxide.
- Leads to increase in gate leakage current.
- Can cause threshold voltage shift (ΔV_{th}).
- May result in device failure or early lifetime degradation.

4. ANTENNA PHENOMENON (EXPLANATION)



- Plasma induces charge on exposed metal.
- Charge flows through the interconnect to the gate.
- Gate oxide is very thin → may get damaged.

5. ANTENNA SIGNOFF

☑ Antenna check is performed as part of signoff DRC.

- Tools report Antenna Ratio for each net / pin.
- All violations must be fixed before tape-out.

Parameter	Description
Antenna Ratio (AR)	Aa / Ag
Antenna Limit	Technology specific limit
Antenna Area (Aa)	Total metal area connected to the gate
Gate Area (Ag)	Gate oxide area of the device
Status	Pass if AR ≤ Limit, else Violation

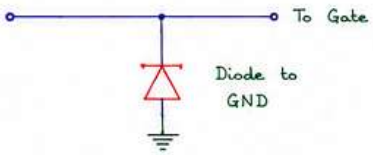
KEY POINT

- ✓ Long metal without proper discharge → Antenna effect.
- ✓ Keep Antenna Ratio below technology limit.
- ✓ Excess charge can damage gate.
- ✓ Antenna signoff is mandatory before tape-out.

6. FIX TECHNIQUES

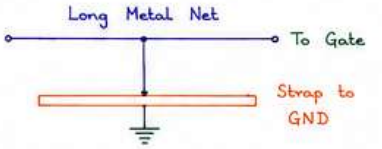
① Insert Diode (Antenna Diode)

- Connect a diode between the net and ground / VDD.
- Provides a path to discharge the collected charge.



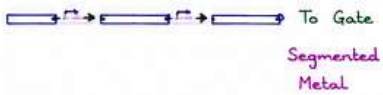
② Add Antenna Strap / Jumper

- Connect the long net to a nearby metal layer tied to GND or VDD.
- Distributes and discharges the charge.



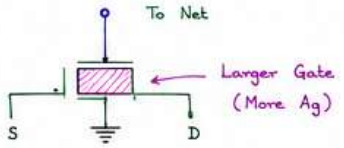
③ Reduce Metal Area

- Break long metal into smaller segments using additional vias / layers.
- Reduces effective antenna area.



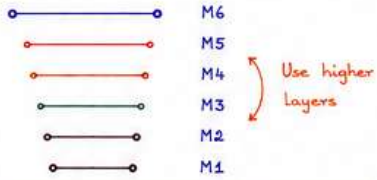
④ Increase Gate Oxide Area

- Use larger gate devices if functionally allowed.
- Increases Ag and reduces Antenna Ratio.



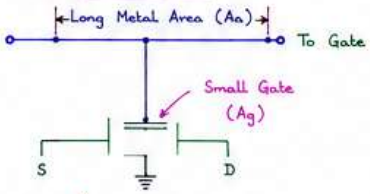
⑤ Use Higher Metal Layers

- Higher layers are less exposed to plasma.
- Metal density and fill help in charge dissipation.



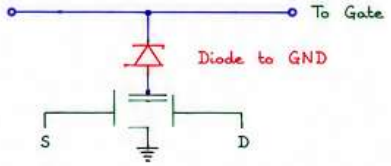
7. ANTENNA VIOLATION EXAMPLE

(a) VIOLATION CASE



$$AR = \frac{Aa}{Ag} = \text{High} > \text{Limit}$$

(b) AFTER FIX (DIODE INSERTION)



Charge discharges → No violation

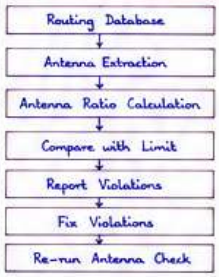
8. ANTENNA RATIO GUIDELINE (EXAMPLE)

Technology Node	Antenna Limit (AR_{limit})	Typical Fix
90 nm	~ 400	Diode / Strap
65 nm	~ 300	Diode / Strap / Segmentation
45 nm	~ 200	Diode / Strap / Higher Layers
28 nm	~ 100	Multiple techniques

Factors Affecting Antenna

- Metal Length
- Metal Width
- Number of Metal Layers
- Metal Density
- Plasma Exposure Time

9. ANTENNA CHECK FLOW



10. PRACTICAL OBSERVATIONS

- Antenna problems are more in early routing with long wires.
- Fix using diode is most common and reliable.
- Adding straps also helps IR drop and charge discharge.
- Always check antenna after major routing ECOs.

11. INTERVIEW / VIVA QUESTIONS

- Q1. What is antenna effect in VLSI ?
- Q2. Define antenna ratio.
- Q3. Why antenna violations are dangerous ?
- Q4. How do you fix antenna violations ?
- Q5. What is an antenna diode ?
- Q6. Is antenna check part of signoff ?
- Q7. What factors affect antenna ?

12. SUMMARY / REVISION POINTS

- ✓ Long metal wires act as antennas in plasma.
- ✓ Antenna Ratio (Aa/Ag) must be below limit.
- ✓ Excess charge can damage gate oxide.
- ✓ Fix using diode, straps, segmentation, higher layers.
- ✓ Antenna check is mandatory signoff rule.
- ✓ Fix early to avoid late stage rework.
- ✓ Proper metal density and layers help reduce antenna.
- ✓ Protects device reliability and ensures tape-out success.

Author : Konda Tharun Vivekananda

NDR ROUTING FOR CLOCK NETS

1. INTRODUCTION

☑ Clock nets are critical for synchronous design. They must be distributed with low skew, low latency and high reliability. Default routing may not meet these requirements. Hence Non-Default Rule (NDR) routing is used for clock nets.

2. WHY NDR FOR CLOCK NETS ?

☑ Reduce clock skew and insertion delay.

- Lower resistance and IR drop.
- Lower capacitance → less delay.
- Improve signal integrity and reduce jitter.
- Avoid crosstalk and maintain robustness.
- Meet timing closure and CTS requirements.

3. DEFAULT RULE VS NDR

Parameter	Default Rule	NDR for Clock
Wire Width	Standard width	Wider than default
Spacing	Standard spacing	Increased spacing
Metal Layers	Any allowed layer	Prefer higher metal layers
Resistance (R)	Higher	Lower
Capacitance (C)	Higher	Lower
Delay	Higher	Lower
IR Drop	Higher	Lower
Skew	Higher	Reduced

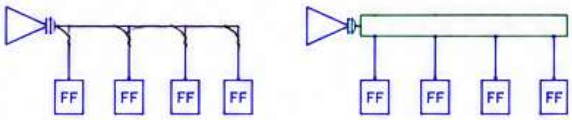
4. EXAMPLE – WITHOUT NDR Vs WITH NDR

Without NDR (Default)

- Narrow wire
- High resistance
- High capacitance
- Higher delay & skew

With NDR

- Wide wire (low R)
- Low capacitance
- Lower delay & skew

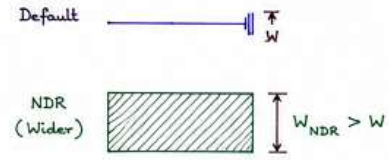


Result : NDR routing ensures faster, balanced and reliable clock distribution.

5. TYPES OF NDR (3 TYPES)

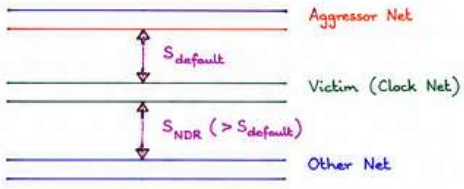
① WIDER WIRE NDR

- Use wider wires on higher metal layers.
- Reduces resistance.
- Used for clock trunks and spines.



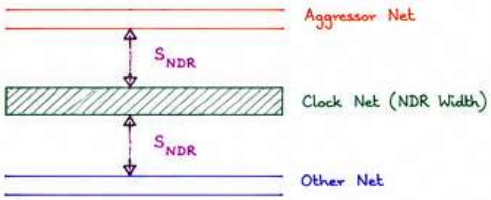
② SPACING NDR

- Increase spacing between clock nets and other signals.
- Reduces crosstalk.
- Provides better signal integrity.



③ COMBINED WIDTH + SPACING NDR

- Uses wider wires along with increased spacing.
- Best for critical clock nets with long distance and high load.

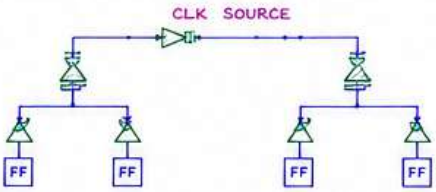


6. NDR RULE PARAMETERS (EXAMPLE)

Parameter	Default Rule	NDR for Clock (Example)	Effect
Metal Layer	M2 – M6	M6 – M8	Lower R, C
Wire Width	0.20 µm	0.60 µm	Lower R
Wire Spacing	0.20 µm	0.60 µm	Less crosstalk
Via Enclosure	Standard	Increased	More reliability
Cut Spacing	Standard	Increased	Reduce EM risk

7. NDR IN CLOCK TREE SYNTHESIS (CTS)

- CTS tool assigns NDR rule to clock trees.
- Clock trunks and spines use wide & high metal.
- Leaves may use lighter rules but still follow NDR spacing.
- Ensures balanced insertion delay and low skew.



NDR Applied On :

- Clock trunk
- Clock spine
- Clock branches
- Critical leaves

8. HOW NDR WORKS ?

① Define NDR rule in technology / LEF / tool.
② Create NDR (Width / Spacing / Layer preference).
③ Assign NDR rule to clock nets.
④ CTS builds clock tree using this rule.
⑤ Routing uses wider wires and spacing as per NDR.
⑥ Results in low delay, low skew & reliable clock.

Define NDR → Assign to Clock Nets → CTS Builds Tree → Routing with NDR Rule → Low Skew & Reliable Clock

KEY POINT

✓ NDR is essential for high performance clock distribution.
✓ Wider width reduces resistance and delay.
✓ Increased spacing reduces crosstalk.
✓ Higher metal layers improve performance.
✓ Proper NDR setup ensures low skew clock network.

9. NDR SIGNOFF

- Antenna, DRC, LVS, ERC must pass with NDR rules.
- Check min width, spacing, via rules.
- Verify IR Drop & Electromigration.
- STA → Check skew, insertion delay, latency.
- Final GDS must follow NDR rule.

10. PRACTICAL OBSERVATIONS

- Use M6 and above for clock trunks.
- Keep clock nets away from aggressor signals.
- Always verify skew after routing.
- Overuse of wide wires increases area and coupling → optimize carefully.

11. COMPARISON TABLE

Aspect	Default Routing	NDR Clock Routing
Wire Width	Normal	Wider
Spacing	Normal	Increased
Metal Layer	Any	Higher Preferred
Resistance	High	Low
Capacitance	High	Low
Skew	High	Low
Signal Integrity	Moderate	Improved
Usage	General Nets	Clock Nets

12. INTERVIEW / VIVA QUESTIONS

Q1. Why NDR is used for clock nets ?
A: To reduce skew, delay, crosstalk and improve reliability.

Q2. What are the types of NDR ?
A: (1) Wider wire NDR (2) Spacing NDR (3) Combined width + spacing NDR.

Q3. Which metal layers are preferred for clock NDR ?
A: Higher metal layers like M6, M7, M8.

Q4. How does NDR affect clock performance ?
A: It lowers resistance & capacitance → reduces delay and skew.

13. SUMMARY / REVISION POINTS

✓ NDR routing is used for clock nets to meet performance & reliability.
✓ Three types : Width NDR, Spacing NDR, Combined NDR.
✓ NDR reduces R, C → lowers delay & skew.
✓ Higher metal layers and spacing improve SI.
✓ Always perform signoff with NDR rules.

Author : Konda Tharun Vivekananda

ROUTING OPTIMIZATION

1. INTRODUCTION

☑ Routing optimization improves the quality of routed design to meet timing, congestion, DRC, SI and power goals. It is performed after initial detailed routing or during rip-up and re-route iterations.

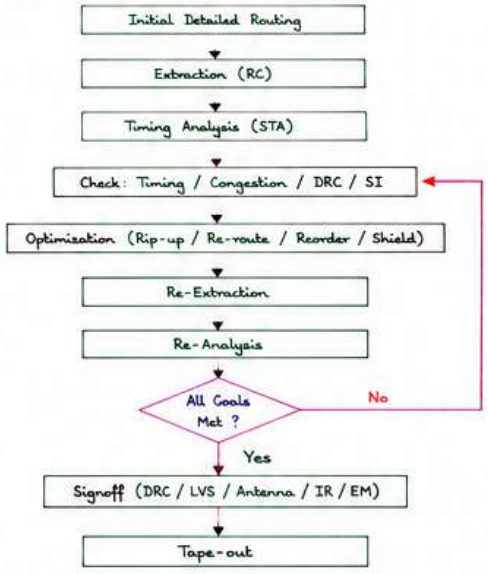
2. DEFINITION

☑ Routing Optimization is the process of modifying existing routed nets (or performing limited rip-up and re-route) to improve design metrics such as timing, congestion, wirelength, and reliability without changing the logical connectivity.

3. OBJECTIVES

- Meet timing (reduce setup/hold violations).
- Reduce congestion and overflow.
- Minimize wirelength and total capacitance.
- Improve signal integrity (reduce crosstalk).
- Minimize IR drop and EM (electromigration) issues.
- Meet DRC and antenna rules.
- Optimize for power and reliability.

4. OVERALL ROUTING OPTIMIZATION FLOW



5. METRICS IMPROVED BY OPTIMIZATION

Metric	How Optimized
Timing	Reduce delay, fix violations, balance paths
Congestion	Redistribute routing, remove hotspots
Wirelength	Shorter routes, better detours
SI (Crosstalk)	Increase spacing, shielding
Power	Reduce cap, avoid long unnecessary routes
Reliability	Meet EM limits, reduce IR drop

KEY POINT

✓ Optimization is an iterative process.
✓ Balance timing, congestion and reliability.
✓ Use report driven optimization.

6. TIMING DRIVEN ROUTING

a) Introduction

Timing driven routing (TDR) considers timing information during routing to reduce delay and fix violations.

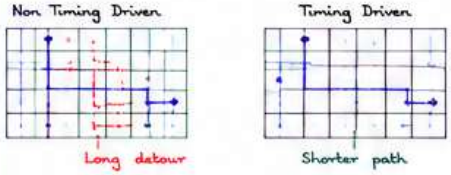
b) How it Works

- Uses critical path information from STA.
- Routes critical nets first.
- Uses preferred directions and layers.
- Minimizes delay by reducing R and C.
- Applies buffering / layer promotion where required.

c) Timing Driven Techniques

- Critical net first routing.
- Layer promotion for critical nets.
- Use wider wires for critical nets.
- Avoid long detours on critical paths.
- Buffer insertion during routing

EXAMPLE - CRITICAL NET ROUTING



Result: Reduced delay, improved slack.

7. CONGESTION OPTIMIZATION

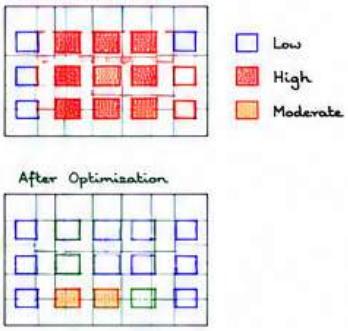
a) Introduction

High congestion leads to routing failures, detours and timing degradation. Congestion optimization reduces overflow and distributes routing demand.

b) How it Works

- Identify congested regions (overflow).
- Rip-up congested nets partially or fully.
- Reroute using alternate paths / layers.
- Spread routing demand more evenly.
- Use wire spreading and blockages adjustment.

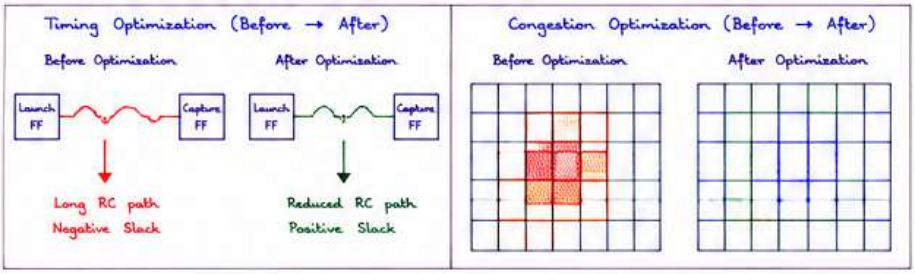
CONGESTION EXAMPLE



8. COMMON OPTIMIZATION TECHNIQUES

Technique	Purpose	Description / Effect
Rip-up and Re-route	Fix congestion / timing	Remove and reroute nets to better paths
Layer Promotion	Reduce resistance & delay	Move net to higher metal layer
Wire Widening	Reduce delay / IR drop	Increase width to reduce R & improve EM
Buffer Insertion	Fix timing	Insert buffers to reduce delay / balance
Wire Spreading	Reduce crosstalk	Increase spacing between aggressor / victim
Shielding	Reduce crosstalk	Insert shield nets or use ground shields
Via Optimization	Reduce delay	Optimize via count & placement
Blockage Adjustment	Reduce congestion	Add / remove blockages to guide routing

9. EXAMPLE - TIMING & CONGESTION OPTIMIZATION



10. OPTIMIZATION CONSIDERATIONS

- Over optimization may increase wirelength.
- Maintain DRC and antenna compliance.
- Avoid excessive layer promotions.
- Validate after each iteration using accurate extraction.
- Use ECO friendly optimization steps.

11. PRACTICAL OBSERVATIONS

- Always start with congestion optimization before timing.
- Critical clock nets need both timing and reliability focus.
- Layer promotion on critical nets gives major timing benefit.
- Shielding helps in high speed and long routes.
- Optimization is an iterative and trade-off process.

12. INTERVIEW / VIVA QUESTIONS

Q1. What is routing optimization ?
Ans: Process to improve routed design quality (timing, congestion, SI, power, reliability).
Q2. Why timing driven routing is important ?
Ans: It reduces delay and fixes timing violations.
Q3. What causes congestion ?
Ans: High routing demand, narrow routing channels, poor placement.
Q4: How wire widening helps in optimization ?
Ans: It reduces resistance and improves EM and delay.

13. SUMMARY / REVISION POINTS

✓ Routing optimization improves timing, congestion, power and reliability.
✓ Timing driven routing uses critical path info to reduce delay.
✓ Congestion optimization redistributes routing to remove hotspots.
✓ Common techniques: Rip-up & re-route, layer promotion, widening, buffering, spreading, shielding.
✓ Optimization is iterative and must meet all signoff goals.

14. SIGNOFF CHECKLIST (Post Optimization)

✓ Timing (setup / hold)	✓ Congestion (overflow)
✓ DRC Clean	✓ Antenna Clean
✓ IR Drop	✓ Electromigration
✓ Crosstalk / SI	✓ LVS Clean
✓ Power	✓ Final Extraction & STA

Author : Konda Tharun Vivekananda

POST ROUTE TIMING OPTIMIZATION

1. INTRODUCTION

☑ Post Route Timing Optimization (PRTO) is performed after detailed routing and parasitic extraction. It fixes setup / hold violations and improves timing slack without disturbing design functionality.

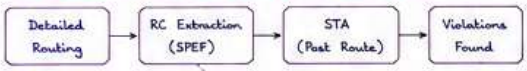
2. DEFINITION

☑ PRTO is the process of modifying routed design by adjusting delays, buffering, sizing or routing in order to meet timing constraints and improve performance.

3. OBJECTIVES

- Remove setup and hold violations.
- Improve timing slack.
- Reduce clock skew and latency variation.
- Meet timing with minimum area / power impact.
- Maintain signal integrity and routability.

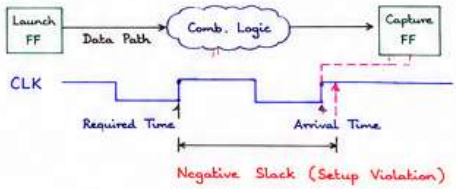
4. POST ROUTE TIMING ANALYSIS CONTEXT



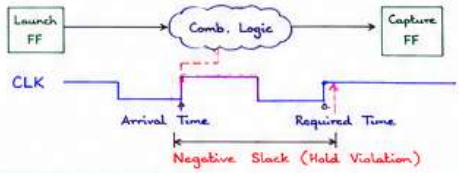
If violations exist → Perform Post Route Timing Optimization

5. TYPES OF TIMING VIOLATIONS (AFTER ROUTE)

a) Setup Violation



b) Hold Violation



KEY POINT

✓ PRTO is performed only after routing and extraction.
✓ Uses real RC parasitics for accurate optimization.
✓ Goal is to achieve positive slack for all paths.

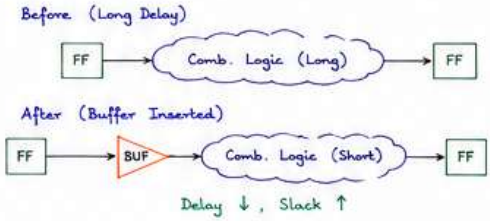
7. TECHNIQUES SUMMARY TABLE

Violation Type	Technique	How it Works	Effect
Setup Violation	Buffer Insertion	Add buffers	Delay ↓, Slack ↑
	Cell Upsizing	Use stronger cells	Delay ↓, Slack ↑
	Logic Optimization	Reduce logic levels	Delay ↓, Slack ↑
	Layer Promotion	Use higher metal	R ↓, Delay ↓
Hold Violation	Buffer Insertion	Add delay	Delay ↑, Slack ↑
	Cell Downsizing	Use smaller cells	Delay ↑, Slack ↑
	Detour Routing	Increase wire length	Delay ↑, Slack ↑

6. FIX METHODS (POST ROUTE TIMING OPTIMIZATION)

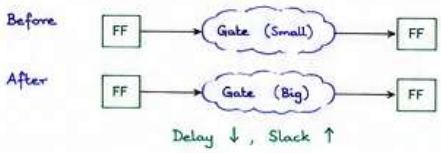
A. SETUP VIOLATION FIXES

① Buffer Insertion
- Insert buffers to reduce data path delay.



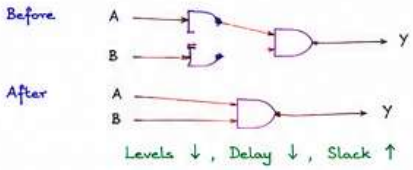
Delay ↓, Slack ↑

② Cell Resizing (Upsizing)
- Upsize gates to reduce cell delay.



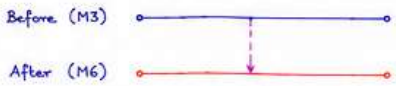
Delay ↓, Slack ↑

③ Logic / Gate Optimization
- Simplify logic to reduce logic levels and delay.



Levels ↓, Delay ↓, Slack ↑

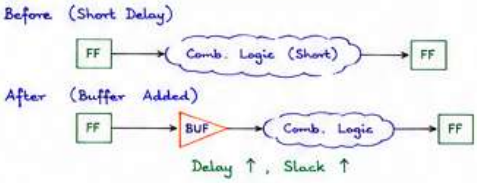
④ Use Higher Metal / Layer Promotion
- Promote critical nets to higher metal layers to reduce R and delay.



R ↓, Delay ↓, Slack ↑

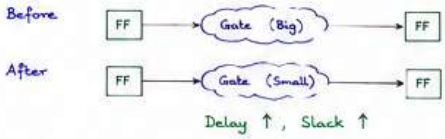
B. HOLD VIOLATION FIXES

① Buffer Insertion in Data Path
- Add small delay using buffer.



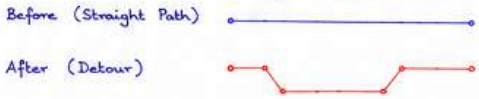
Delay ↑, Slack ↑

② Use Smaller Cells (Downsizing)
- Use smaller drive cells to increase delay.



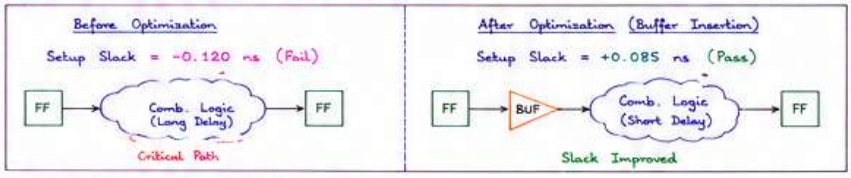
Delay ↑, Slack ↑

③ Detour Routing
- Add detour in routing path to increase delay.

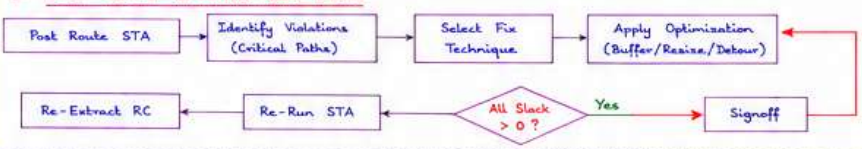


Delay ↑, Slack ↑

8. EXAMPLE – BEFORE & AFTER OPTIMIZATION



9. POST ROUTE OPTIMIZATION FLOW



13. CHECKLIST AFTER PRTO

- Setup fixes reduce delay, Hold fixes increase delay.
- Always check both setup and hold after every fix.
- Avoid excessive buffering → area & power increases.
- Layer promotion improves performance & reliability.
- Iterative optimization is required for timing closure.

13. CHECKLIST AFTER PRTO

✓ All Setup Slack > 0
✓ All Hold Slack > 0
✓ DRC Clean
✓ LVS Clean
✓ Antenna Clean
✓ IR Drop within limit
✓ EM within limit
✓ Final STA Signoff

11. INTERVIEW / VIVA QUESTIONS

Q1. When is Post Route Timing Optimization done ?
A: After routing and RC extraction using SPEF.
Q2. What are the main goals of PRTO ?
A: Remove timing violations and improve slack.
Q3. Name few setup violation fixing techniques.
A: Buffer insertion, cell upsizing, logic optimization, layer promotion.
Q4. Name few hold violation fixing techniques.
A: Buffer insertion, cell downsizing, detour routing.

12. SUMMARY / REVISION POINTS

✓ PRTO improves timing after routing & extraction.
✓ Setup violations: reduce delay (buffers, upsizing, logic opt, layer promotion).
✓ Hold violations: increase delay (buffers, downsizing, detours).
✓ Iterate optimization until all paths have +ve slack.
✓ Final design must meet timing, SI and reliability.

KEY TAKEAWAY

Post Route Timing Optimization ensures that the routed design meets timing requirements reliably with minimum area, power and performance cost.

14. COMMON TOOLS USED

- PrimeTime (STA)
- IC Compiler / Innovus (Opt)
- Tempus, Voltus (Advanced flows)

Author : Konda Tharun Vivekananda

PD – Routing

1. RC EXTRACTION

1.1 DEFINITION

☑ RC Extraction is the process of extracting the parasitic Resistances (R) and Capacitances (C) from the routed layout based on the actual geometry and technology data. The extracted RC values are annotated to the netlist (SPEF format) for accurate timing & SI analysis.

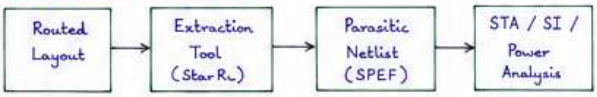
1.2 WHY RC EXTRACTION IS IMPORTANT ?

- Captures the real parasitic effects introduced by routing.
- Accurate timing analysis (delay, slack) using STA.
- Detects SI issues : crosstalk, noise, glitches.
- Essential for sign-off timing, power and reliability.
- Bridges the gap between ideal netlist and physical layout.

1.3 APPLICATIONS

- Sign-off Static Timing Analysis (Post Route STA).
- Timing closure & ECO verification.
- Power analysis (dynamic & leakage).
- Signal Integrity analysis (crosstalk, noise, glitches).
- IR drop & EM analysis require parasitic RC.
- Formal equivalence with extracted netlist.

1.4 EXTRACTION FLOW



Tools : StarRC, Quantus, Raphael, PEX

1.5 ROUTING PARASITICS FUNDAMENTALS

A) RESISTANCE (R)

- Due to the finite resistivity of metal wires.
- $R = \rho \times (L / (W \times T))$

  ρ → resistivity of metal

  L → length, W → width, T → thickness

B) CAPACITANCE (C)

- Capacitance exists between conductors and to substrate.
- Two types :
  1) Ground capacitance (to substrate or well)
  2) Coupling capacitance (between neighboring nets)

Capacitance Type	Symbol	Description
Ground Capacitance	C_g	Capacitance between a net and ground (substrate / well)
Coupling Capacitance	C_c	Capacitance between two neighboring nets on same or adjacent layers
Fringe Capacitance	C_f	Edge & fringe effects due to field spreading

Total Capacitance = $C_g + \Sigma C_c + C_f$

KEY POINT

- ✓ RC extraction makes STA close to silicon behavior.
- ✓ Always use post-route extracted netlist for sign-off.
- ✓ Coupling capacitance can increase delay (crosstalk)
- ✓ Higher metal layers → lower R but coupling still exists.
- ✓ Extraction accuracy depends on layout quality & tech files.

2. COUPLING CAPACITANCE (C_c)

2.1 CONCEPT

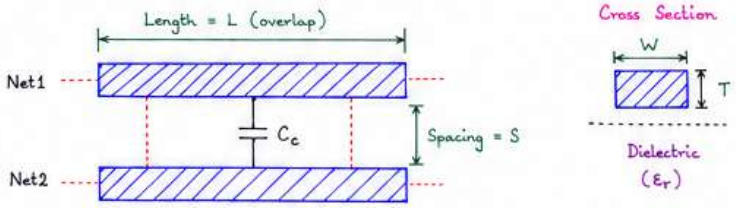
- Occurs between two adjacent nets due to electric field.
- Causes interference, delay variation and glitches.
- Depends on spacing, overlap length and ε_r dielectric.

2.2 TYPES OF COUPLING

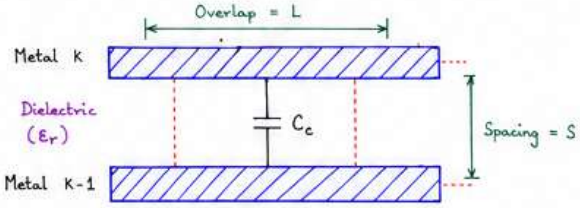
- 1) Intra-layer coupling : between two nets on same layer.
- 2) Inter-layer coupling : between nets on adjacent layers.
- 3) Self-coupling (shielding effect) : when a net is surrounded by shields, C_c reduces.

2.3 COUPLING CAPACITANCE DIAGRAMS

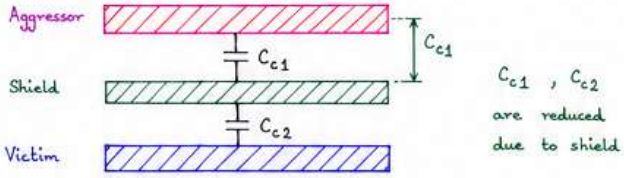
A) Intra-layer Coupling (Same Layer)



B) Inter-layer Coupling (Adjacent Layers)



C) Shielded Net (Reduced Coupling)



3. RC EXTRACTION TECHNIQUES

Technique	Description	Accuracy	Runtime	Usage
Lumped Extraction	Parasitics lumped at pins (R in series, C to ground).	Low	Fast	Early STA, Placement Stage
π (Pi) Model Extraction	Each net represented by π model (R/2 – C – R/2). Better delay accuracy.	Medium	Medium	Mid-stage analysis
Distributed Extraction	Parasitics distributed along the net at many segments.	High	Slow	Sign-off STA, SI, Power
Ceff (Capacitive Effective)	Coupling caps converted to equivalent ground capacitance.	Medium	Fast	Timing estimation with coupling effect

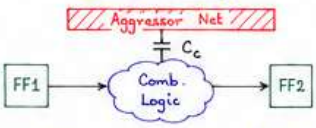
4. EXAMPLE – EFFECT OF COUPLING CAPACITANCE

Without Coupling



Delay = $t_{logic} + t_{net}$ (small)

With Coupling



Delay = $t_{logic} + t_{net}$ (large)

Extra delay due to C_c → Slower transition

5. OUTPUT & SIGN-OFF

- Extraction output is in SPEF format.
- Annotated SPEF + Netlist → used in STA / SI / Power tools.
- Run sign-off checks after extraction.

SPEF Example

```
*D_NET   NET1
*CONN
 *  1    0.000
 *  2    0.000
*RES
   1  2   0.356
*CAP
   1  0   0.0123
   1  2   0.0045
 * END
```

6. PRACTICAL OBSERVATIONS

- Poor routing (narrow, long wires) → high R.
- Closely spaced nets → high C_c → delay ↑.
- Shielding & spacing help reduce C_c.
- Always prefer higher metal layers for critical nets (low R, less C_g).

7. CHECKLIST FOR RC EXTRACTION

- ✓ Use correct tech file & layer map.
- ✓ Ensure all layers are extracted.
- ✓ Check antenna diodes / PG nets.
- ✓ Run consistency & sanity checks.
- ✓ Verify SPEF in STA before sign-off.

8. INTERVIEW / VIVA Qs

Q1. Why RC extraction is required ?

→ To capture real parasitics for accurate timing & SI analysis.

Q2. What is coupling capacitance ?

→ Capacitance between adjacent nets causing crosstalk & delay.

9. SUMMARY POINTS

- RC extraction extracts R & C from layout.
- Important for accurate STA, SI & Power.
- Main parasitics : R, C_g, C_c, C_f.
- Coupling capacitance increases delay and can cause crosstalk & delay.
- Extraction output (SPEF) is used for sign-off verification.

Author : Konda Tharun Vivekananda

PD_Routing

1. RC EXTRACTION OVERVIEW

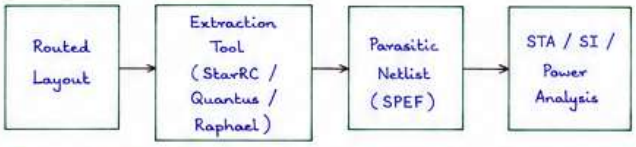
1.1 DEFINITION

RC Extraction is the process of extracting the parasitic Resistances (R) and Capacitances (C) from the routed layout based on actual geometry and technology data. The extracted values are annotated to the netlist (SPEF format) for accurate timing, SI and power analysis.

1.2 WHY RC EXTRACTION IS IMPORTANT ?

- Accurate delay calculation (setup & hold).
- Captures crosstalk, glitch, noise and SI issues.
- Required for sign-off STA, power and reliability.
- Bridges the gap between ideal netlist and real layout.

1.3 EXTRACTION FLOW



Inputs : GDSII / OASIS , Tech File , Rule Deck , Layer Map

Output : SPEF (.spf) annotated to netlist

1.4 FUNDAMENTALS

A) RESISTANCE (R)

- Caused by finite resistivity of metal.
- For a wire segment : $R = \rho \times L / (W \times T)$
  where ρ → resistivity , L → length , W → width , T → thickness
- R is series element along the wire.

B) CAPACITANCE (C)

- Occurs between conductors and between conductor & substrate.
- Two types :
  1) Ground Capacitance (C_g) → to substrate or well.
  2) Coupling Capacitance (C_c) → between neighbouring nets.

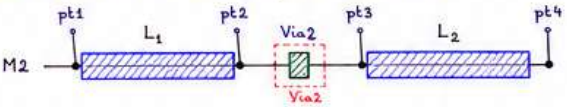
2. RESISTANCE EXTRACTION

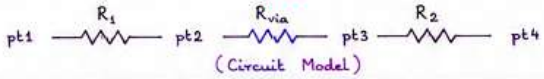
2.1 CONCEPT

- Each metal segment is modeled as a resistor.
- Series resistances are accumulated along the path.
- Contact and via resistances are also included.

2.2 EXAMPLE

Metal path on M2 with 2 segments and a via.





(Circuit Model)

$R_1 = \rho \times L_1 / (W \times T)$, $R_2 = \rho \times L_2 / (W \times T)$, $R_{via} = R_{contact} + R_{via}$

Total R (path) = $R_1 + R_{via} + R_2$

KEY POINT

- ✓ Resistance extraction considers width, thickness, length and sheet resistance of each layer.
- ✓ Higher metal layers → lower R.
- ✓ Vias / contacts add significant resistance.

3. COUPLING EXTRACTION

3.1 COUPLING CAPACITANCE EXTRACTION – DEFINITION

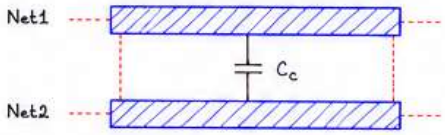
Coupling capacitance is the unwanted capacitance between two adjacent nets due to electric field. It causes crosstalk and affects delay and noise.

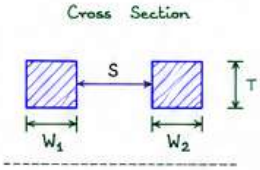
3.2 TYPES OF COUPLING

A) Intra-layer Coupling → between nets on same layer
B) Inter-layer Coupling → between nets on adjacent layers
C) Self Coupling → net and surrounding shields
D) Shielded Coupling → reduced C_c due to shield

3.3 COUPLING CAPACITANCE EXTRACTION – EXAMPLES

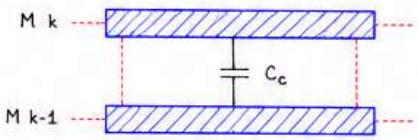
A) INTRA-LAYER COUPLING (Same Layer)

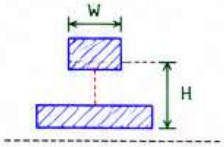




- C_c ∝ Overlap length , Inversely ∝ Spacing (S)

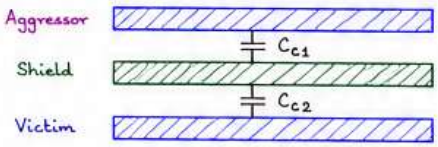
B) INTER-LAYER COUPLING (Adjacent Layers)





- C_c ∝ Overlap area , Inversely ∝ Vertical spacing (H)

C) SHIELDED COUPLING (Reduced Coupling)



Shield reduces effective coupling on victim.

$C_{effective} << C_{without\ shield}$

4. COUPLING CAPACITANCE EXTRACTION DETAILS

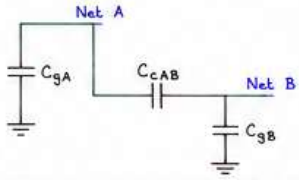
4.1 CAPACITANCE COMPONENTS

C_g = Capacitance to ground (substrate / well)

C_c = Coupling capacitance between two nets

$C_{tot} = C_g + \sum C_c$

Capacitance Network



4.2 COUPLING EFFECTS

- Increases or decreases delay (depends on switching).
- Causes crosstalk noise.
- May change hold and setup slacks.
- Important for SI sign-off.

EXTRACTION TOOLS

- StarRC
- Quantus
- Raphael
- QRC
- Caliber xRC

5. SPEF OUTPUT (Example)

```
*D_NET   N1
*CONN
 * 1      0.000     (Pin N1)
 * 2      0.000     (Pin N2)
*RES
 1  2  2   0.456     (R between 1 and 2)
*CAP
 1      0   0.0123   (Cg at node 1)
 1      2   0.0045   (Cc between 1 and 2)
*END
```

NOTE :
Extracted RC is back annotated to netlist for final timing, SI and power analysis.

6. PRACTICAL OBSERVATIONS

- Accurate extraction depends on layout quality and technology files.
- Wider spacing between nets → lower C_c.
- Shields and higher layers help reduce coupling.
- Always run extraction before sign-off STA / SI.

7. CHECKLIST FOR RC EXTRACTION

- ✓ GDS/OASIS and tech files are correct.
- ✓ Layer map and rule deck are updated.
- ✓ All layers are extracted.
- ✓ Consistency & sanity checks.
- ✓ SPEF is loaded in STA and results verified.

8. INTERVIEW / VIVA Qs

Q1. What is RC extraction ?
→ Extract parasitic R & C from routed layout for analysis.

Q2. What are the types of C ?
→ C_g (to ground) and C_c (coupling between nets).

Q3. How does shield reduce coupling ?
→ It breaks the electric field between aggressor and victim.

9. SUMMARY POINTS

- RC extraction captures real parasitics.
- Resistance depends on L, W, T, ρ.
- Capacitance includes C_g and C_c.
- Coupling affects delay and noise.
- SPEF output is used for sign-off STA, SI and Power.

Author : Konda Tharun Vivekananda

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RC EXTRACTION ENGINES – STAR-RC, QUANTUS, PEGASUS

1. RC EXTRACTION ENGINES – OVERVIEW

RC Extraction engines compute accurate Resistance (R) and Capacitance (C) from the physical layout to create a parasitic model (SPEF) for timing, SI and power analysis.

Commerical tools : StarRC (Synopsys), Quantus (Siemens), Pegasus (Cadence).

KEY POINT

Extraction engines differ in algorithms, accuracy, performance and capacity. Choice depends on design size, accuracy requirement and flow integration.

2. EXTRACTION FLOW (COMMON)

1. INPUT DATA
   GDSII / OASIS, Tech File, .ict/.qrc, Map File
2. LAYOUT READING & PREPARATION
   Layer mapping, hierarchy reading, checks
3. GEOMETRY PROCESSING
   Shape stitching, Boolean, redundancy removal
4. PARASITIC COMPUTATION
   R (Resistive network) + C (Capacitive network)
5. MODEL REDUCTION
   Net reduction, coupling thresholding
6. OUTPUT GENERATION
   SPEF (.spef / .spf), Reports, Logs

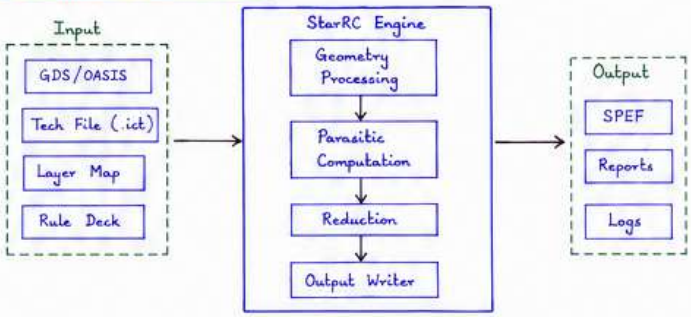
Outputs : SPEF file, RC report, coupling report, summary QoR

Inputs : GDS/OASIS, Tech File (.ict/.qrc), Layer Map, Rules

3. STAR-RC (SYNOPSYS)

- Industry standard extractor from Synopsys.
- Hierarchical, scalable and very fast.
- Uses advanced multi-level solvers and field based techniques.
- Supports block level and full-chip extraction.

3.1 STAR-RC ARCHITECTURE



3.2 STAR-RC FEATURES

- Multi-threading & distributed computing.
- Pattern based & field solver hybrid.
- Accurate coupling, CCS support.
- Excellent capacity for billion transistor designs.

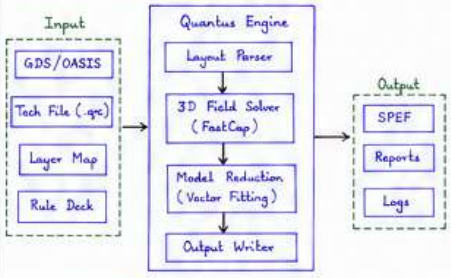
STEPS

1) Read layout & tech files
2) Build geometry database
3) Compute R network
4) Compute C (incl. coupling)
5) Reduce RC network
6) Write SPEF & reports

4. QUANTUS (SIEMENS)

- High accuracy extractor from Siemens EDA.
- Best for sign-off accuracy and advanced nodes.
- Uses field solver (FastCap based) with vector fitting model reduction.
- Supports highly complex 3D structures.

4.1. QUANTUS ARCHITECTURE



4.2 QUANTUS FEATURES

- Very high accuracy (field solver).
- Vector fitting for wideband modeling.
- Excellent coupling accuracy.
- Used for sign-off timing & SI closure.
- More runtime compared to pattern based tools.

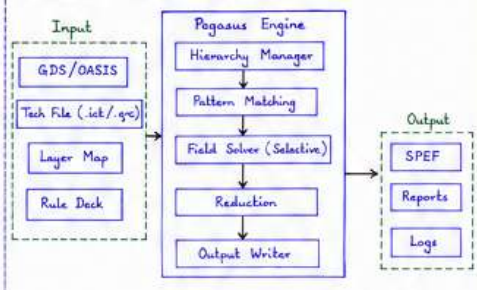
STEPS

1) Read layout & tech
2) Build 3D geometry
3) Field solve for C
4) Compute R network
5) Model reduction
6) Write SPEF

5. PEGASUS (CADENCE)

- Extractor from Cadence.
- Combines speed and accuracy.
- Hierarchical pattern matching with optional field solver refinement.
- Tight integration with Innovus flow.

5.1 PEGASUS ARCHITECTURE



5.2 PEGASUS FEATURES

- Hybrid approach (pattern + field).
- Good balance of accuracy & runtime.
- Selective field solver for critical nets.
- Supports Incremental & ECO extraction.
- Ideal for physical implementation flows.

STEPS

1) Read inputs
2) Pattern match
3) Field solve (if req)
4) Compute R
5) Reduce network
6) Write SPEF

6. ENGINE COMPARISON

Parameter	StarRC	Quantus	Pegasus
Vendor	Synopsys	Siemens	Cadence
Algorithm	Hierarchical Pattern + Field	3D Field Solver (FastCap)	Pattern + Selective Field
Accuracy	High	Very High	High
Runtime	Fast	Slower	Medium
Scalability	Excellent	High	High
Memory Usage	Low – Medium	High	Medium
Best For	Large designs	Sign-off accuracy	Physical Flow Closure
CCS Support	Yes	Yes	Yes
Incremental	Yes	Yes	Yes
Typical Use	Block / Chip Extraction	Sign-off Extraction	Innovus / Tempus Flow

7. EXTRACTION CONSIDERATIONS

- Extraction accuracy depends on tech file, layer map and modeling (pattern vs field).
- Coupling threshold affects runtime and accuracy.
- Use incremental extraction after ECOs.
- Always cross-check reports (max C, max R, coupling %).
- Apply antenna fixes before final extraction.

NOTE

SPEF generated by these engines is used by STA, SI, IR drop and power analysis tools.

8. PRACTICAL OBSERVATIONS

- StarRC → fastest, good balance, industry workhorse.
- Quantus → most accurate, used fook sign-off.
- Pegasus → balanced, seamless with Cadence flow.
- For very large SoC → StarRC preferred.
- For critical timing / SI closure → Quantus.
- For Innovus physical flow → Pegasus.

ENGINE SELECTION GUIDELINE

Large design + speed	→ StarRC
High accuracy sign-off	→ Quantus
Flow integration + balance	→ Pegasus

9. SUMMARY POINTS

- ✓ RC extraction converts layout to accurate R & C model.
- ✓ StarRC, Quantus and Pegasus are leading engines.
- ✓ Algorithms : Pattern based, Field based or Hybrid.
- ✓ Outputs : SPEF, reports, coupling & QoR summary
- ✓ Extraction quality directly impacts timing & SI closure.

VIVA / INTERVIEW Qs

Q1. What is the difference between StarRC and Quantus ?
→ StarRC uses hierarchical pattern + field, faster.
Quantus uses full 3D field solver, more accurate.

Q2. Why is RC extraction important ?
→ It provides accurate parasitics for STA, SI and Power analysis after routing

Author : Konda Tharun Vivekananda

Always run extraction before sign-off STA / SI.

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RC EXTRACTION – SPEF GENERATION , DSPF GENERATION , RSPF GENERATION

1. SPEF GENERATION (Standard Parasitic Exchange Format)

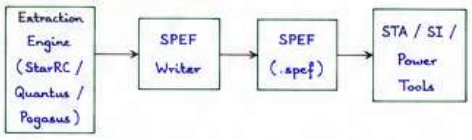
1.1 DEFINITION

SPEF is a parasitic exchange format that stores extracted RC values for timing , SI and power analysis. It contains lumped R and C for each net , pin and coupling information.

1.2 CONTENTS

- Header information
- Library , units and corners
- Capacitors (ground and coupling)
- Resistors (series and contacts)

1.3 SPEF DATA FLOW



1.4 EXAMPLE – SIMPLE SPEF CONTENT

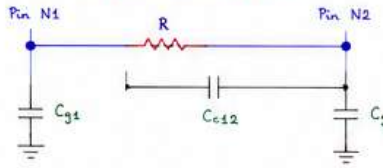
```
*D_NET   net1
*CONN
 P   N1   U1/A
 P   N2   U2/Y
*CAP
 P   N1   2.50e-15
 P   N2   3.10e-15
 C   N2   1.20e-15
*RES
 R   N1   80.5

*END
```

→ Net name
→ Pin connections
→ Ground caps (to gnd)
→ Coupling cap (between pins)
→ Resistance (ohms)

1.5 DIAGRAM – NET PARASITICS (SPEF)

Net with ground and coupling caps



KEY POINTS

- ✓ SPEF is a standard format used across tools
- ✓ Contains lumped R & C values
- ✓ Supports ground and coupling capacitances
- ✓ Used for STA , SI and Power analysis

2. DSPF GENERATION (Detailed Standard Parasitic Format)

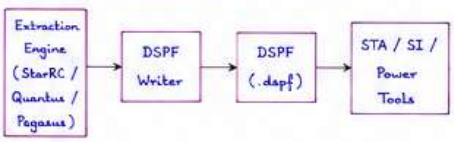
2.1 DEFINITION

DSPF contains detailed node-by-node parasitics with distributed RC network. It preserves more detail compared to SPEF and is used for accurate analysis and debugging.

2.2 CONTENTS

- Detailed node listing
- Segment resistances
- Nodal capacitances
- Coupling capacitances

2.3 DSPF DATA FLOW



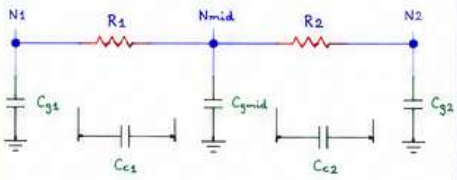
2.4 EXAMPLE – DSPF CONTENT (SNIPPET)

```
*D_NET   net2
*NODE
 P   N1     0.0    0.0
 P   Nmid   10.0   0.0
 P   N2     *
*RES
 R   N1     Nmid    40.2
 R   Nmid   N2      42.8
*CAP
 C   N1     gnd     1.12e-15
 C   Nmid   gnd     2.05e-15
 C   N2     gnd     1.35e-15
 C   N1     Nmid    0.82e-15
 C   Nmid   N2      0.78e-15
*END
```

→ Detailed nodes
→ Segment resistors
→ Ground caps
→ Coupling caps

2.5 DIAGRAM – DISTRIBUTED RC (DSPF)

Distributed RC network



KEY POINTS

- ✓ DSPF keeps node level detail
- ✓ More accurate but larger in size
- ✓ Used for sign-off accuracy and debugging
- ✓ Good for critical nets and SI verification

3. RSPF GENERATION (Reduced Standard Parasitic Format)

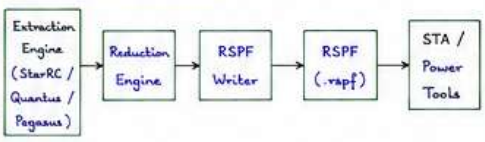
3.1 DEFINITION

RSPF is a reduced parasitic format where the extracted RC network is reduced while preserving delay accuracy. Used to reduce file size and improve runtime in STA.

3.2 CONTENTS

- Reduced nodes
- Equivalent resistors & capacitors
- Preserves delay within specified error

3.3 RSPF DATA FLOW



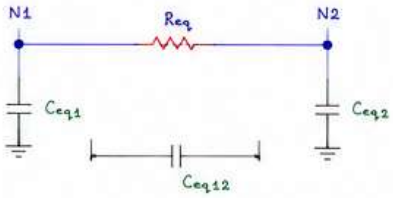
3.4 EXAMPLE – RSPF CONTENT (SNIPPET)

```
*D_NET   net3
*NODE
 P   N1     0.0    0.0
 P   N2     20.0   0.0
*RES
 R   N1     N2     83.6
*CAP
 C   N1     gnd    2.00e-15
 C   N2     gnd    2.12e-15
 C   N1     N2     0.95e-15
*END
```

→ Reduced to few nodes
→ Equivalent R
→ Ground caps
→ Coupling cap

3.5 DIAGRAM – REDUCED RC (RSPF)

Reduced equivalent RC network



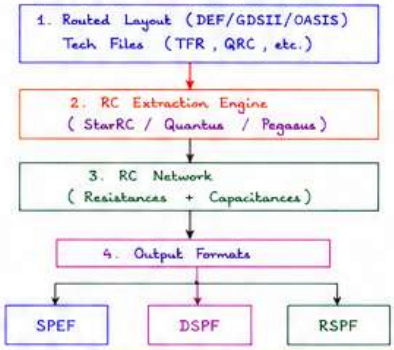
KEY POINTS

- ✓ RSPF reduces network size
- ✓ Maintains delay within user constraint
- ✓ Smaller file → faster STA runtime
- ✓ Used for final sign-off timing closure

4. COMPARISON OF SPEF , DSPF AND RSPF

Parameter	SPEF (Standard)	DSPF (Detailed)	RSPF (Reduced)
Purpose	Standard exchange format	Detailed accuracy & debugging	Reduced for fast runtime
Detail Level	Lumped R & C	Node level detailed RC	Equivalent lumped RC
Accuracy	High	Very High	High (within error limit)
File Size	Medium	Large	Small
Runtime	Medium	Slow	Fast
Use Cases	STA , SI , Power	Sign-off accuracy , critical nets	Final STA , large designs
Generated By	All extraction engines	All extraction engines	Reduction engine after extraction
Example Tools	StarRC , Quantus , Pegasus	StarRC , Quantus , Pegasus	StarRC (RPI) , Quantus , Pegasus

5. EXTRACTION FLOW (WITH FORMATS)



6. PRACTICAL NOTES

- Always extract parasitics after final routing
- Choose DSPF for debugging critical paths
- Use RSPF for sign-off STA to save runtime
- Check extraction summaries and QOR reports
- Ensure tech files and layer maps are correct

7. CHECKLIST

- ✓ Correct Tech Files & Layer Map
- ✓ Routed Layout Clean (DRC Passed)
- ✓ Extraction Run Completed
- ✓ SPEF/DSPF/RSPF Generated
- ✓ Run STA & Verify Results

8. INTERVIEW / VIVA Qs

Q1. What is SPEF ?
→ Standard Parasitic Exchange Format storing lumped R & C.

Q2. Why DSPF is used ?
→ For accurate node level detail and debugging

Q3. Why RSPF is used ?
→ To reduce file size and STA runtime

9. SUMMARY POINTS

- ✓ Extraction engines compute R & C from layout
- ✓ SPEF – standard lumped format
- ✓ DSPF – detailed node level format
- ✓ RSPF – reduced equivalent format
- ✓ All outputs used for STA , SI and power

Author : Konda Tharun Vivekananda

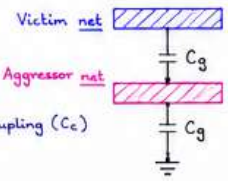
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PARASITICS IMPACT ON TIMING - SETUP & HOLD

1. INTRODUCTION

Parasitics are the unwanted R (resistance) and C (capacitance) introduced during physical implementation. They affect the actual delay of data and clock paths ⟶ impacts setup and hold timing.

Types of parasitics



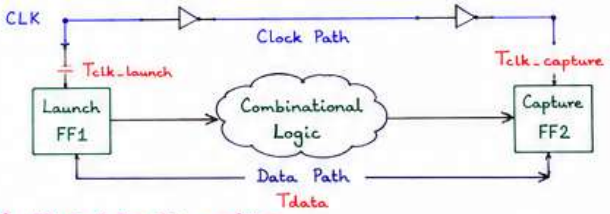
- Gate capacitance (C_{gate})
- Diffusion capacitance (C_{diff})
- Interconnect capacitance (C_{wire}) → ground (C_g) & coupling (C_c)
- Interconnect resistance (R_{wire})
- Via resistance (R_{via}) & Via capacitance (C_{via})

2. WHERE PARASITICS AFFECT TIMING ?

- Data Path : increases propagation delay.
- Clock Path : increases latency and causes skew.
- Coupling : causes crosstalk and extra delay.
- RC increase ⟶ longer delays ⟶ possible setup violation.
  RC increase ⟶ data arrives later at capture flop.

3. PARASITICS IMPACT ON SETUP TIMING

3.1 Setup Timing Model



3.2 Effect of Parasitics on Setup

- R_{wire} ↑ ⟶ resistance ↑ ⟶ delay ↑
- C_{wire-g} ↑ ⟶ load ↑ ⟶ delay ↑
- C_c (coupling) ⟶ extra delay due to crosstalk
  ⟶ Total data arrival time increases.

3.3 Timing Equation

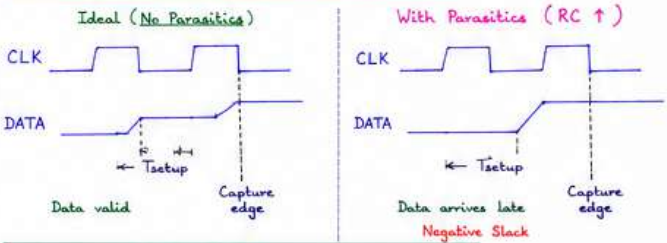
Setup Slack = Required Time - Data Arrival Time

Required Time = (T_{clk_period} - $T_{clk_uncertainty}$) + $T_{clk_capture}$ + T_{setup}

Data Arrival Time = T_{clk_launch} + T_{data}

Slack = (T_{clk_period} - $T_{clk_uncertainty}$) + $T_{clk_capture}$ + T_{setup}
- (T_{clk_launch} + T_{data})

3.4 Setup Timing Diagram - Effect of Parasitics

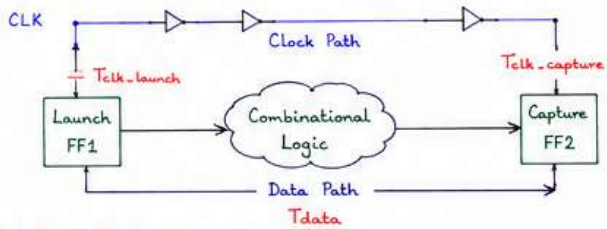


KEY POINTS (SETUP)

- ✓ Parasitics increase data path delay.
- ✓ Coupling capacitance (C_c) adds crosstalk delay.
- ✓ RC increase ⟶ Setup slack reduces.
- ✓ Excess RC ⟶ Setup violation ⟶ Data arrives late.

4. PARASITICS IMPACT ON HOLD TIMING

4.1 Hold Timing Model



4.2 Effect of Parasitics on Hold

- R_{wire} ↑ ⟶ can increase delay (good for hold).
- C_{wire-g} ↑ ⟶ increases delay.
- C_c (coupling) ⟶ may cause data to arrive earlier (if aggressor switch)
  ⟶ reduces delay ⟶ hold violation.

4.3 Timing Equation

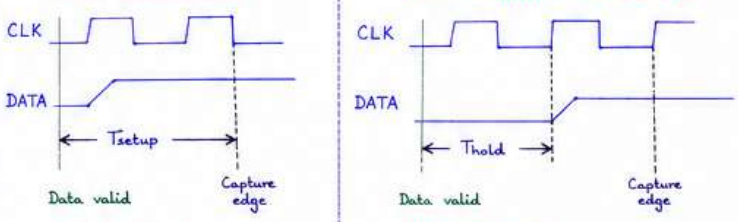
Hold Slack = Data Arrival Time - Required Time

Required Time = $T_{clk_capture}$ + T_{hold} - $T_{clk_uncertainty}$

Data Arrival Time = T_{clk_launch} + T_{data}

Slack = (T_{clk_launch} + T_{data})
- ($T_{clk_capture}$ + T_{hold} - $T_{clk_uncertainty}$)

4.4 Hold Timing Diagram - Effect of Parasitics



Positive Slack

Data arrives early
Negative Slack

5. SUMMARY OF PARASITICS IMPACT

Parameter	Increase In R or C	Effect on Setup	Effect on Hold
Data Path Delay	R ↑, C_g ↑, C_c ↑	Delay ↑ ⟶ Slack ↓	Delay ↑ ⟶ Slack ↑
Coupling Capacitance (C_c)	Aggressor switching	Extra delay ⟶ Slack ↓	Data may arrive early ⟶ Slack ↓
Clock Path Delay	R ↑, C_g ↑	Capture time ↑ ⟶ Slack ↑ (Good)	Required time ↑ ⟶ ⟶ Slack ↓
Skew ($T_{clk_capture}$ - T_{clk_launch})	Due to RC variation	If positive skew ↑ ⟶ Slack ↑	If positive skew ↑ ⟶ Slack ↓

PRACTICAL OBSERVATIONS

- After routing, RC parasitics significantly change delays.
- Setup violations are more common due to increased delays.
- (coupling occur when data path becomes too fast or due to C_c.
- Extraction accuracy directly affects timing accuracy.

6. INTERVIEW / VIVA Qs

Q1. How do parasitics affect setup timing ?
→ They increase data path delay, causing data to arrive late at capture flop, reducing setup slack.

Q2. How do parasitics affect hold timing ?
→ They may reduce delay (especially due to coupling), causing data to arrive early and creating hold violation.

Q3. What is coupling capacitance impact ?
→ It causes crosstalk delay for setup and early arrival for hold depending on aggressor switching.

7. CHECKLIST FOR TIMING WITH PARASITICS

- ✓ Extract RC accurately.
- ✓ Update SPEF / DSPF / RSPF in STA.
- ✓ Check setup & hold slacks.
- ✓ Analyze crosstalk (C_c) impact.
- ✓ Fix violations (buffer, resize, reroute).
- ✓ Re-extract & re-run STA.

8. QUICK REVISION

- Parasitics = R_{wire}, C_g, C_c, R_{via}, C_{via}.
- RC ↑ ⟶ Data delay ↑ ⟶ Setup slack ↓.
- Coupling ⟶ Setup delay ↑, Hold slack ↓.
- Clock path RC ⟶ affects skew ⟶ both timing.
- Always extract & analyze before sign-off.

Author : Konda Tharun Vivekananda

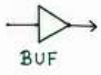
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OPTIMIZATION TECHNIQUES – BUFFER INSERTION & WIRE SIZING

1. BUFFER INSERTION

1.1 DEFINITION

Buffer insertion technique adds buffers along a long or high RC interconnect to break it into smaller segments.
It reduces RC delay, improves slew and helps in setup timing closure.



1.2 WHY BUFFER INSERTION ?

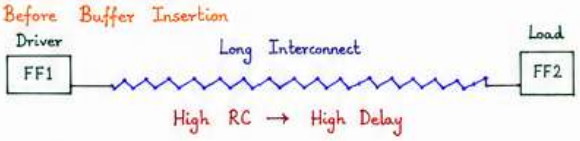
- Reduces Elmore delay of long nets.
- Improves slew rate at the receiver.
- Reduces crosstalk due to lower rise/fall time.
- Improves setup timing slack.

1.3 PROCEDURE

1) Identify long/high RC critical nets from STA report.
2) Determine optimal number and location of buffers.
3) Insert buffers from library.
4) Re-route if required.
5) Re-extract parasitics and re-run STA.

1.4 EXAMPLE DIAGRAM

Before Buffer Insertion



High RC → High Delay

After Buffer Insertion



Long wire → segmented → delay ↓, slew ↑

1.5 ADVANTAGES

- ✓ Reduces delay of long nets.
- ✓ Improves signal integrity.
- ✓ Helps in setup timing closure.
- ✓ Reduces crosstalk.

1.6 DISADVANTAGES

- X Increases area (more cells).
- X Increases dynamic power.
- X More buffers → more leakage.
- X May create congestion.

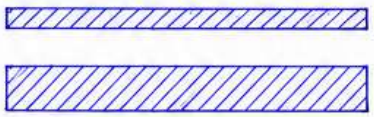
1.7 BUFFER INSERTION GUIDELINES

- Insert buffers on nets with high fanout or long length.
- Keep segment delay roughly equal for balanced tree.
- Avoid inserting buffers on hold critical paths.
- Re-run extraction & STA after any insertion.

2. WIRE SIZING

2.1 DEFINITION

Wire sizing technique increases the width (number of tracks) of metal interconnect to reduce its resistance (R).
It improves delay but increases capacitance (C).



2.2 WHY WIRE SIZING ?

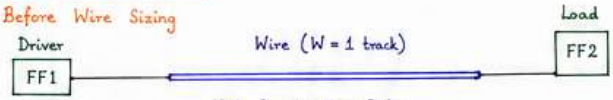
- Reduces interconnect resistance.
- Improves timing for setup critical nets.
- Useful when buffer insertion is not possible.

2.3 PROCEDURE

1) Identify nets with high resistance from STA.
2) Select higher metal layer or increase width.
3) Follow founder Width/spacing rules.
4) Re-route if required.
5) Re-extract parasitics and re-run STA.

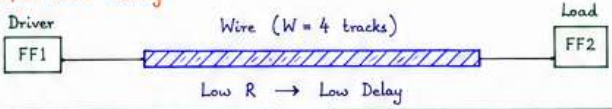
2.4 EXAMPLE DIAGRAM

Before Wire Sizing



High R → High Delay

After Wire Sizing



Low R → Low Delay

2.5 ADVANTAGES

- ✓ Lowers resistance.
- ✓ Reduces delay.
- ✓ Can improve both setup & hold (if timing allows).

2.6 DISADVANTAGES

- X Increases capacitance.
- X More metal resources used.
- X May increase routing congestion.
- X Higher routing capacitance → may hurt hold.

2.7 WIRE SIZING GUIDELINES

- Use higher metal layers for long global nets.
- Increase width gradually, check marginal benefit.
- Avoid over-sizing → unnecesary capacitance.
- Re-check hold slack after sizing.

3. BUFFER INSERTION vs WIRE SIZING

Parameter	Buffer Insertion	Wire Sizing
Main Effect	Reduces RC delay by segmentation	Reduces R by increasing width
Delay Impact	Significant delay reduction	Moderate delay reduction
Capacitance Impact	Slight increase (added buffer Cin)	Increases C → proportional to width)
Area Impact	Area ↑ (more cells)	No cell area increase
Power Impact	Dynamic & leakage ↑	Dynamic power ↑ (due to C)
Best For	Very long nets, high fanout nets	Long nets with high & resistance
Hold Timing	May improve or worsen (depends)	May worsen (↑)
Congestion	Can increase congestion	May increase congestion
SI / Crosstalk	Improves (less slew)	May increase (↑)

4. IMPACT ON SETUP & HOLD

Setup Path



More RC (R,C) → Data Arrival Time ↑ → Setup slack ↓

Hold Path



More RC can delay data → Data arrives later
Hold slack ↑

5. PRACTICAL OBSERVATIONS

- Use both techniques together for best closure.
- Insert buffers first, then optimize wire sizing.
- Always re-run extraction & STA after changes.
- Consider power, area and congestion while optimizing.

6. CHECKLIST

- ✓ Check timing report (setup & hold violations).
- ✓ Identify long / high RC nets.
- ✓ Try buffer insertion → re-run STA.
- ✓ Try wire sizing → re-run STA.
- ✓ Re-extract parasitics.
- ✓ Final sign-off timing closure.

7. INTERVIEW / VIVA Qs

Q1. What is buffer insertion ?
→ Adding buffers on long nets to reduce RC delay and improve timing.

Q2. What is wire sizing ?
→ Increasing metal width to reduce resistance and improve timing.

Q3. Which is better: buffer or sizing ?
→ Depends on trade-off between delay, power, area and hold timing.

8. SUMMARY POINTS

- ✓ Buffer insertion segments long nets.
- ✓ Wire sizing reduces resistance.
- ✓ Both techniques impact setup & hold.
- ✓ Always verify with extraction & STA.
- ✓ Balance timing, power, area & congestion.
- ✓ Final goal → timing closure.

Author : Konda Tharun Vivekananda

PD _ Routing

EXTRACT REPORT ANALYSIS – STARRC.LOG, STARRC_SUMMARY.RPT, STARRC_UNMATCH.RPT, STARRC_OPEN.RPT

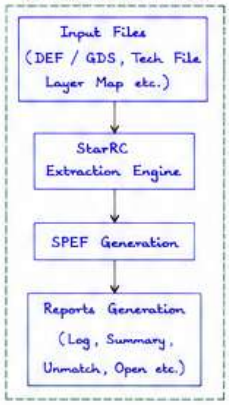
1. OVERVIEW

After RC extraction using StarRC, several report files are generated. Analysis of these reports is important to ensure extraction accuracy and to identify issues like unmatched pins, open nets, etc.

Main reports :

- Starrc.log
- Starrc_summary.rpt
- Starrc_unmatch.rpt
- Starrc_open.rpt

RC EXTRACTION FLOW



2. STARRC.LOG (EXTRACTION LOG FILE)

2.1 Purpose

It is the main log file which contains complete extraction process information, messages, warnings and errors.

2.2 Key Information

- Run start & end time
- Input files read
- Technology & layer map details
- Extraction settings
- Number of nets, pins, instances
- Memory usage and CPU time
- Warnings / Errors (DRC, open, unmatch)
- SPEF writing status

2.3 Sample Snippet (Illustration)

```
StarRC v2023.06                                   ┐ Run information
Started : Mon May 20 10:15:23 2024                ┘
Reading design.def ..... Done                     ┐
Reading tech.tch ..... Done                       ┘
Total Instances :      1,25,640                   ┐
Total Nets :           3,42,118                   │ Design statistics
Total Pins :           6,84,236                   │
Extraction Mode :      Coupled                    │
Multithreading :       8 threads                  ┘
...
SPEF file written :    design.spef                ┐
Warnings : 12                                     │ Warnings / Errors
Errors : 0                                        ┘
CPU time : 00:12:34   Memory : 6.2 GB             ┐ Completion status
Finished : Mon May 20 10:27:57 2024               ┘
```

2.4 How to Analyze

- Check errors = 0 (must be zero).
- Check warnings – investigate if critical.
- Verify SPEF file size is reasonable.
- Check memory & CPU time for performance.

KEY POINT

- ✓ Log file is first file to check after extraction.
- ✓ Any error must be fixed before using SPEF for STA.
- ✓ Warnings may indicate potential accuracy issues

3. STARRC_SUMMARY.RPT (SUMMARY REPORT)

3.1 Purpose

Provides a summary of extracted parasitics and design coverage.
Very useful to check extraction completeness and data quality.

3.2 Key Information

- Counts of nets, pins, instances
- Parasitic elements (R, C, Coupling C)
- RC totals and averages
- Coupling statistics
- Extraction coverage
- Top N largest nets / Couplings

3.3 Sample Table (Illustration)

Parameter	Value	
Total Instances	1,25,640	Design counts
Total Nets	3,42,118	
Total Pins	6,84,236	
Total Resistances	18,45,67,892	Parasitic totals
Total Capacitances (C to G)	32,14,66,233	
Total Coupling Caps (C_c)	12,78,34,567	
Average Net Resistance	53.92 ohms	Averages & Extremes
Average Net Capacitance	94.13 fF	
Max Net Resistance	8.46 Kohms	
Max Net Capacitance	12.81 pF	
Extraction Coverage (Pins)	99.87 %	Coverages & Quality
Unmatched Pins	116	
Open Nets	42	

3.4 How to Analyze

- High coverage (> 99.5%) is good.
- Large number of unmatched/open → check other reports.
- Abnormally high C or R → verify layer mapping / rules.
- Compare totals with previous runs for consistency.

4. STARRC_UNMATCH.RPT (UNMATCHED PINS REPORT)

4.1 Purpose

Lists pins that could not be matched to technology (lib) or have no physical shape mapping.
These pins are ignored in extraction → leads to inaccuracy.

4.2 Contents

- Pin name and instance
- Reason for unmatch
- Suggested action

4.3 Sample Snippet (Illustration)

Instance	Pin	Net	Reason	Action
U12345	A	net_3456	No layer mapping	Fix tech file
U12789	Y	net_6789	Pin not in lib	Check lib
U13001	VDD	net_vdd	Unconnected pin	Tie or ignore
U13222	ZN	net_1122	Wrong cell mapping	Check LEF/Lib

KEY POINTS

- ✓ Unmatched pins reduce extraction accuracy.
- ✓ Fix tech file / layer map / library.
- ✓ Re-run extraction after fixing.

5. STARRC_OPEN.RPT (OPEN NETS REPORT)

5.1 Purpose

- Reports nets that are electrically → unmatch what make digital open (no valid connection.
- Helps once using net agressor setup.

5.2 Contents

3.3 Sample Snippet (Illustration)

Net Name	Connected Pins	Reason	Suggested Action
net_open_1	U123/A	Only one pin connected	Check routing / ECO
net_open_2	—	No pins connected	Remove or tie
clk_test	U456/CLK	Port not connected	Connect or ignore

KEY POINTS

- ✓ Open nets are ignored in SPEF.
- ✓ May cause functional or timing issues
- ✓ Fix connectivity and re-run extraction.

6. SUMMARY – ALL REPORTS

Report File	Purpose	Key Checks	Impact if Ignored
Starrc.log	Run information	Errors = 0, Warnings	May use wrong SPEF
Starrc_summary.rpt	Extraction summary	Coverage %, Totals, Averages	Inaccurate timing
Starrc_unmatch.rpt	Unmatched pins	Unmatched count = 0 (ideal)	Pins ignored → inaccuracy
Starrc_open.rpt	Open nets	Open net count = 0	Nets missing → wrong RC

7. PRACTICAL OBSERVATIONS

- Always check log first for errors.
- Fix unmatch and open issues iteratively.
- Compare summary with previous runs to detect abnormal changes.
- Ensure high extraction coverage before using SPEF for STA.

8. CHECKLIST

- ✓ Check Starrc.log → no errors
- ✓ Review Starrc_summary.rpt
- ✓ Unmatched pins = 0
- ✓ Open nets = 0
- ✓ SPEF size reasonable
- ✓ Re-run STA and verify timing

9. INTERVIEW / VIVA Qs

Q1. Why is extract report analysis important ?
→ To ensure accurate parasitics and reliable timing analysis.

Q2. What happens if unmatched pins are high ?
→ These pins are ignored → R/C missing → timing inaccuracy.

Q3. What is open net ?
→ Net with one or no pin connected.

10. SUMMARY POINTS

- ✓ Starrc.log gives run status & errors.
- ✓ Summary report gives overall stats.
- ✓ Unmatch report → fix library / tech issues.
- ✓ Open report → fix connectivity.
- ✓ Clean reports → accurate extract → reliable STA.

Author : Konda Tharun Vivekananda

Always check reports before STA !

PD_Routing

SIGN OFF EXTRACTION – INTRODUCTION, REQUIREMENTS & CHECKLIST

1. INTRODUCTION

Sign off Extraction is the final and most accurate RC extraction performed after detailed routing. It captures all parasitics including coupling, crosstalk and accurate resistances which are used for sign-off STA, SI, Power and reliability analysis.

Goal : Generate sign-off quality parasitic data for tape-out that meets timing, power & SI closure.

2. REQUIREMENTS FOR SIGN OFF EXTRACTION

- 100% routed design (DRC clean).
- Correct technology files (QRC / TLU+).
- Up-to-date process corners (RC corners).
- Accurate layer map & via definitions.
- Power / ground nets properly connected.
- Antenna, fill and shielding in final routed database.
- Sufficient memory, CPU & disk space.
- Extraction setup (include/exclude rules, netlist, etc.).

3. SIGN OFF EXTRACTION FLOW



Reports generated : Starrc.log , Starrc_summary.rpt , Starrc_unmatch.rpt , Starrc_open.rpt

4. STARRC.LOG (EXTRACTION LOG FILE)

4.1 Purpose

Contains complete information about extraction run including settings, run time, memory usage, warnings and errors.

4.2 Key Contents

- Run start & end time
- Input files & technology
- Layer map & units
- Extraction modes & options
- Net / pin / instance counts
- Memory & CPU usage
- Warnings, Errors and their summary
- SPEF file write status

Sample Snippet

```
StarRC v2023.06
Run started : Mon May 20 10:15:23 2024
Technology  : tsmcN7_QRC
Mode        : Coupled
Multithreading : 8 threads

Design Summary
Instances   : 1,25,640
Nets        : 3,42,118
Pins        : 6,84,236

Extraction completed
SPEF file written : design.spef
Warnings : 12      Errors : 0
CPU time : 00:12:34    Memory : 6.2 GB
Run finished : Mon May 20 10:27:57 2024
```

4.3 How to Analyse

- ✓ Ensure Errors = 0
- ✓ Check for important warnings
- ✓ Verify instance/net/pin count is reasonable
- ✓ Confirm SPEF written successfully
- ✓ Check memory & runtime for future tuning

KEY POINTS
- ✓ Log file is the first check for extraction health.
- ✓ Warnings may indicate potential accuracy issues.
- ✓ Errors must be fixed before sign-off.

5. STARRC_SUMMARY.RPT (SUMMARY REPORT)

5.1 Purpose

Gives a summarized view of extracted parasitics and design coverage. Very useful for quick quality check.

5.2 Key Information

- Total instances, nets, pins
- Total R, C, coupling C
- Average / max / min values
- Extraction coverage (pins, nets)
- Top N largest nets / couplings

5.3 Sample Table (Illustration)

Parameter	Value	
Total Instances	1,25,640	Design Counts
Total Nets	3,42,118	
Total Pins	6,84,236	
Total Resistance (R)	18,45,67,892	Parasitic Totals
Total Capacitance (C to G)	32,14,66,233	
Total Coupling Cap (Cc)	12,78,34,567	
Average Net Resistance	53.92 ohm	Averages & Extremes
Average Net Capacitance	94.13 fF	
Max Net Resistance	8.46 Kohm	
Max Net Capacitance	12.81 pF	
Pin Extraction Coverage	99.87 %	Coverage
Net Extraction Coverage	99.91 %	

5.4 How to Analyse

- ✓ Coverage should be ~100%
- ✓ Check max R, max C for outliers
- ✓ Compare with previous runs
- ✓ Verify top large nets for optimization.

6. STARRC_UNMATCH.RPT (UNMATCHED PINS / INSTANCES)

6.1 Purpose

Lists pins or instances that could not be matched to the technology library or have incorrect mapping.

6.2 Contents

- Instance / pin name
- Library / cell reference
- Reason for mismatch
- Suggested action

6.3 Sample Table (Illustration)

Instance	Pin	Library/Cell	Reason	Action
U12345	A	NAND2_X1	Pin not in lib	Check Lib
U12789	Y	NOR2_X1	No physical shape	Map / Create
U13001	VDD	VDD	Unconnected pin	Tie or ignore
U13222	ZN	INV_X1	Wrong cell mapping	Fix LEF/Lib

6.4 How to Analyse

- ✓ Unmatched pins reduce accuracy and may cause wrong RC.
- ✓ Fix library or layer map and re-run extraction.
- ✓ Ensure all power pins mapped correctly.

KEY POINTS
- ✓ Zero unmatched pins is goal.
- ✓ Improves RC accuracy and STA reliability

7. STARRC_OPEN.RPT_FOOMS

- Net name, Capt ve. legal.
- Pins not connected.
- Number of opens with previous extraction run.
- Fix unmatch/open before sign-off.
- Re-run STA after fixes.

Net Name	Open Pins	Reason
– pat	U1234/A , U5678/Y	2 Disconnected
– test	U456/CLK	1 Port not ignore
rop_bates	U789/D , U987/Q	2 Missing via Fix

8. SIGN OFF EXTRACTION CHECKLIST

S.No.	Check Item	Description	Status (✓ / X)
1	DRC Clean	Routed design has 0 DRC violations	
2	100% Routing	No unrouted / partially routed nets	
3	Library & Tech File	Correct QRC/tech file, layer map	
4	RC Corners	All required RC corners setup	
5	Power / Ground	All PG nets connected properly	
6	Fill / Shielding	Final fill and shielding done	
7	Antenna	Antenna checks passed	
8	Run Log	Starrc.log has Errors = 0	
9	Coverage	Pin & Net coverage ~100%	
10	Unmatched	Starrc_unmatch.rpt has 0 entries	
11	Open Nets	Starrc_open.rpt has 0 entries	
12	Parasitic Quality	Check max R, max C for outliers	
13	SPEF/DSPF/RSPF	Files generated successfully	
14	Sign-off Use	Ready for STA / SI / Power sign-off	

7. PRACTICAL OBSERVATIONS

- Always read log file first.
- Don't ignore warnings.
- Compare summary with previous extraction run.
- Fix unmatch/open before sign-off.

10. INTERVIEW / VIVA Qs

Q1. Why is sign off extraction important ?
→ It provides accurate parasitics for final STA, SI and Power closure.

Q2. What is unmatched pin ?
→ Pin not mapped to any layer/cell in tech library.

Q3. What is open net ?
→ Net with no valid electrical connection.

11. SUMMARY POINTS

- ✓ Sign off extraction ensures accurate RC.
- ✓ Starrc.log shows run status & errors.
- ✓ Summary report gives totals, averages and coverage.
- ✓ Unmatch report must be zero.
- ✓ Open nets must be fixed.
- ✓ Only then extraction is ready for sign-off.

12. QUICK TIPS

- Run extraction on clean, final routed design.
- Use correct RC corners for sign-off.
- Fix DRC, antenna, unmatch, open issues early.
- Validate all reports before using in STA.
- Keep good log for documentation.

Author : Konda Tharun Vivekananda

PD _ Routing

POST ROUTE ANALYSIS - INTRODUCTION, WHY IT IS CRITICAL?, APPLICATIONS, POST ROUTE STA

1. INTRODUCTION

Post Route Analysis is the complete verification performed after detailed routing using extracted parasitics (SPEF/DSPF/RSPF). It ensures that the final routed design meets timing, power and physical correctness requirements.

2. WHY IT IS CRITICAL ?

- Routing introduces real interconnect R & C → affects delay.
- Unseen violations may appear after extraction.
- Ensures timing closure before tape-out.
- Detects DRC/LVS/antenna/EM/IR violations.
- Improves silicon reliability and yield.

3. APPLICATIONS

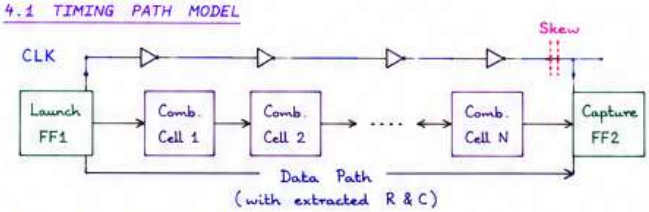
- Final STA sign-off for setup & hold.
- Power integrity analysis (IR drop, EM).
- Signal integrity (crosstalk, noise).
- Physical verification (DRC, LVS, Antenna).
- Generate accurate reports for tape-out.

4. POST ROUTE STA - OVERVIEW



Extracted parasitics + RC corners → Accurate delay calculation

4.1 TIMING PATH MODEL



4.2 SETUP TIMING EQUATION

$$\text{Setup Slack} = \text{Required Time} - \text{Data Arrival Time}$$

$$\text{Required Time} = T_{clk} - T_{uncertainty} - T_{clk\text{-}to\text{-}q(launch)} - T_{setup(capture)}$$

$$\text{Data Arrival Time} = T_{clk\text{-}to\text{-}q(launch)} + T_{comb(RC)} + T_{net(RC)}$$

$$\text{Slack} = T_{clk} + T_{uncertainty} - (T_{clk\text{-}to\text{-}q} + T_{comb} + T_{net})$$
$$= T_{clk} - (T_{uncertainty} + T_{setup} + T_{comb} + T_{clk\text{-}to\text{-}q})$$

Positive Slack (> 0) → Pass Negative Slack (< 0) → Violation

4.3 HOLD TIMING EQUATION

$$\text{Hold Slack} = \text{Data Arrival Time} - \text{Required Time}$$

$$\text{Required Time} = T_{clk\text{-}to\text{-}q(launch)} + T_{hold(capture)} - T_{clk\text{-}skew}$$

$$\text{Data Arrival Time} = T_{clk\text{-}to\text{-}q(launch)} + T_{comb(min)} + T_{min(net)}$$

$$\text{Slack} = (T_{clk\text{-}to\text{-}q} + T_{comb(min)} + T_{min(net)} - (T_{clk\text{-}to\text{-}q} + T_{hold})$$
$$= T_{comb(min)} + T_{min(net)} + T_{clk\text{-}skew} - T_{hold}$$

Positive Slack (> 0) → Pass Negative Slack (< 0) → Violation

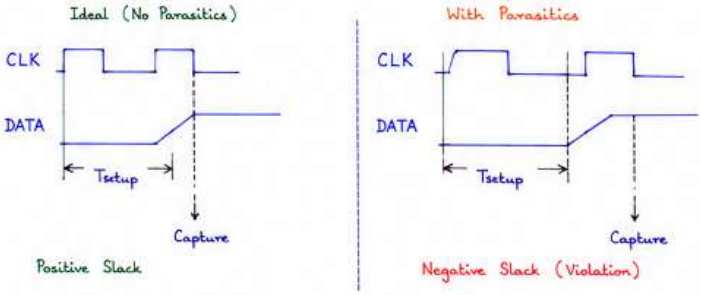
KEY POINTS

- ✓ Post route STA uses extracted parasitics for accurate timing.
- ✓ Extracted R & C can increase or limit density.
- ✓ Both setup and hold must have positive slack.
- ✓ Reports from STA are used for sign-off decision.

5. POST ROUTE STA - EFFECT OF PARASITICS

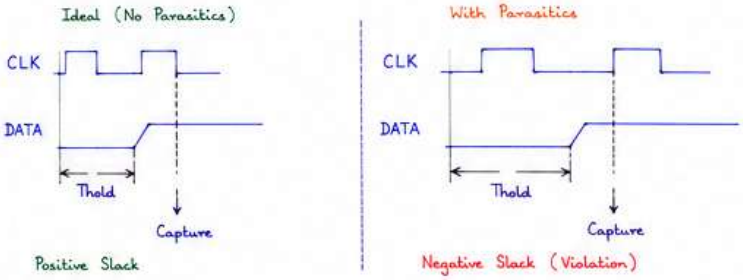
5.1 Effect on Setup Timing

- Interconnect R & C increase → data path delay increases.
- Coupling C may also increase delay.
- More delay → data arrives late → setup slack ↓



5.2 Effect on Hold Timing

- Interconnect R & C increase → delay increases.
- Coupling C may reduce delay (early arrival).
- Less delay → data arrives early → hold slack ↓



6. COMMON POST ROUTE STA REPORTS

Report	Purpose	Key Information
Timing Summary	Overall timing status	WNS, TNS, # violations, path groups
Setup Report	Setup paths	Worst setup paths & slack
Hold Report	Hold paths	Worst hold paths & slack
Path Report	Detailed path	Cell delays, net delays, slack
Skew Report	Clock analysis	Skew, latency, uncertainty
Critical Path Report	Longest path	Max delay path information

7. POST ROUTE ANALYSIS CHECKLIST

✓	Extraction completed successfully	(SPEF/DSPF/RSPF ready)
✓	Extracted RC files read in STA tool	(Check log for errors)
✓	All libraries & PVT corners included	(min, max, typical)
✓	Setup timing analyzed	(WNS ≥ 0, no violations)
✓	Hold timing analyzed	(WNS ≥ 0, no violations)
✓	Worst paths reviewed	(Setup & Hold)
✓	Clock tree & skew analyzed	(Skew within budget)
✓	DRC clean	(0 violations)
✓	Antenna check	(Within limit)
✓	IR Drop / EM check	(Within limit)
✓	LVS clean	(0 mismatches)
✓	Final sign-off criteria met	(Ready for tape-out)

8. PRACTICAL OBSERVATIONS

- Some nets meet timing pre-route but fail post-route due to RC increase.
- Coupling can cause early or late arrival depending on aggressor switching.
- Always run both setup & hold with all corners.
- Path based reporting helps in optimization.

9. INTERVIEW / VIVA Qs

Q1. Why post route STA is important ?
→ It uses actual extracted parasitics to verify final timing before tape-out.

Q2. What can cause timing to degrade after routing ?
→ Increased interconnect R & C and crosstalk effects.

Q3. What is WNS ?
→ Worst Negative Slack in timing paths.

10. SUMMARY POINTS

- ✓ Post route analysis verifies final routed design accuracy.
- ✓ Extracted parasitics affect delay & skew.
- ✓ Both setup & hold must be closed.
- ✓ Multiple checks ensure silicon reliability.
- ✓ Final sign-off only after all checks pass.

11. QUICK TIPS

- Check extraction log for errors first.
- Verify SPEF area & instance count.
- Use correct RC corners in STA.
- Review worst paths and optimize.
- Always maintain documentation.

Author : Konda Tharun Vivekananda

PD _ Routing

POST ROUTE ANALYSIS IN PRIME TIME - STEP BY STEP ANALYSIS

1. INTRODUCTION

Post Route Analysis (PRA) in PrimeTime is the final STA performed after detailed routing and parasitic extraction. It uses SPEF (or DSPF/RSPF) with accurate RC to verify that the design meets all timing requirements.

Output :

- Slack (setup/hold)
- Worst paths
- Timing reports
- Sign-off decision

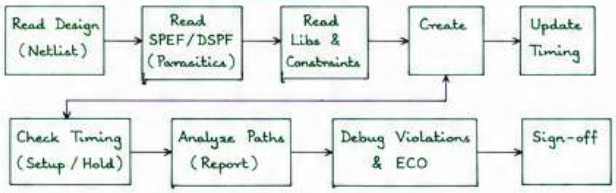
2. WHY IT IS CRITICAL ?

- Uses real R & C from routed design.
- Detects real violations due to routing parasitics.
- Validates timing closure before tape-out.
- Ensures silicon reliability and performance.
- Necessary for SI, EM/IR and power sign-off.

3. APPLICATIONS

- Final STA sign-off for setup & hold.
- Timing ECO validation.
- Slack improvement tracking.
- What-if analysis (voltage, derates, corners).
- Gate level path debugging.

4. POST ROUTE STA FLOW IN PRIMETIME



5. STEP BY STEP ANALYSIS IN PRIMETIME

5.1 Step 1 : Read Design (Netlist)

```
read_verilog design.v        or    read_db design.db
```

→ Loads gate-level netlist.

5.2 Step 2 : Read Parasitics (SPEF/DSPF/RSPF)

```
read_spef design.spef        -scale 1.0
```

→ Annotates extracted R & C to the design.

5.3 Step 3 : Read Libraries

```
read_liberty -lib slow.lib
read_liberty -lib fast.lib
```

→ Loads timing libraries for required corners.

5.4 Step 4 : Read Constraints

```
read_sdc constraints.sdc
```

→ Loads clocks, I/O delays, & various exceptions, loads, etc.

5.5 Step 5 : Create Clocks

```
create_clock -name CLK -period 1.0 [get_ports clk]
```

→ Defines clock source and period.

5.6 Step 6 : Update Timing

```
update_timing
```

→ Propagates delays with RC parasitics.

5.7 Step 7 : Check Timing

```
report_timing -delay max -max_paths 10        (Setup)
report_timing -delay min -max_paths 10        (Hold)
```

→ Generates worst setup & hold paths.

5.8 Step 8 : Analyze Reports

- Identify negative slack paths.
- Check path components : logic / net / clock.
- Note violations, locations, and reasons.

5.9 Step 9 : Debug Violations & ECO

- Use report_* to understand root cause.
- Perform ECO : buffer insertion, resizing, reroute.
- Re-extract parasitics if ECO changes routing.

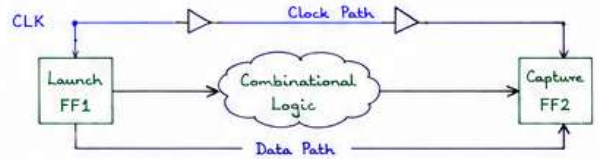
5.10 Step 10 : Sign-off Timing

- All required checks clean.
- Slack ≥ 0 for all paths.
- Save final reports for tape-out.

KEY POINT

PRA is the final and most accurate timing verification step before tape-out. Only when PRA is clean → sign-off.

6. TIMING PATH MODEL (AFTER ROUTING)



7. EFFECT OF PARASITICS ON DELAYS

- Interconnect R ↑ → delay ↑.
- Coupling C (Cc) ↑ → delay ↑.
- Wire C (Cw) ↑ → delay ↑.

⇒ Total data path delay increases.

8. SETUP & HOLD TIMING EQUATIONS (POST ROUTE)

8.1 Setup Slack

Setup Slack = Required Time − Data Arrival Time

Required Time = Tclk − Tuncertainty − Tclk-to-q(launch) − Tsetup (capture)

Data Arrival Time = Tclk-to-q(launch) + Tcomb (with RC) + Trc (net)

Slack = Tclk − Tuncertainty − Tsetup − (Tcomb + Trc)

(Positive → Pass, Negative → Violation)

8.2 Hold Slack

Hold Slack = Data Arrival Time − Required Time

Required Time = Tclk-to-q(launch) + Thold (capture)

Data Arrival Time = Tclk-to-q(launch) + Tcomb (with RC) + Trc (net)

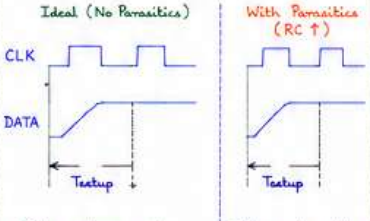
Slack = (Tcomb + Trc) − Thold

(Positive → Pass, Negative → Violation)

9. WAVEFORM VIEW (POST ROUTE)

9.1 Setup Timing

(Ideal Vs With Parasitics)

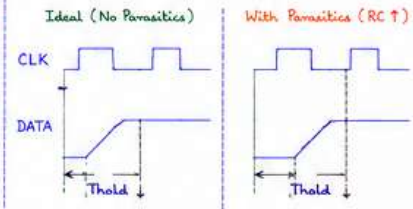


Data arrives early — Positive Slack

Data arrives late — Negative Slack

9.2 Hold Timing

(Ideal Vs With Parasitics)



Data arrives late enough — Positive Slack

Data arrives too early — Negative Slack

10. MOST USED PRIME TIME REPORTS (POST ROUTE)

Report Command	Purpose	Important Information
report_timing -max	Worst setup paths	Slack, path delay, logic/net/cap
report_timing -min	Worst hold paths	Slack, path delay, early arrival
report_timing_summary	Overall timing summary	WNS, TNS, # violations
report_clock -attributes	Clock information	Period, latency, uncertainty, skew
report_area	Area summary	Cell area, total area
report_qor	QoR summary	Timing, power, area snapshot
report_constraint	Check constraints	Unconstrained paths, warnings
report_annotated_delay	Check annotated delays	RC annotation status
check_timing -verbose	Quick check	Warnings, errors, cleanup
report_power	Power analysis	Total, leakage, dynamic

11. PRA CHECKLIST SUMMARY

- ✓ Netlist loaded correctly
- ✓ SPEF / DSPF / RSPF annotated
- ✓ Libraries (PVT & corners) loaded
- ✓ SDC constraints loaded
- ✓ Clocks created and updated
- ✓ No unconstrained / unconnected paths
- ✓ Setup & hold reports checked
- ✓ WNS ≥ 0, TNS ≥ 0
- ✓ No critical violations
- ✓ DRC clean routed design
- ✓ LVS, IR drop, EM, Antenna clean
- ✓ Final sign-off criteria met

12. PRA PRACTICAL OBSERVATIONS

- Routing parasitics dominate delays.
- Cross-talk (Cc) can significantly increase delay.
- Poor DRC clean routing → more delay & violation.
- Even small RC change can flip hold slack.
- Timing closure is iterative : Route → Extract → STA → Fix.

13. INTERVIEW / VIVA Qs

Q1. What is Post Route Analysis ?
→ Final STA after routing with extracted RC.

Q2. Why PRA is critical ?
→ Uses real RC, detects real violations, ensures silicon reliability.

Q3. What are main reports used in PRA ?
→ report_timing, report_timing_summary, report_clock, report_qor, etc.

14. SUMMARY POINTS

- ✓ PRA validates routed design timing.
- ✓ Extracted RC makes analysis accurate.
- ✓ Setup & hold both must be checked.
- ✓ Violations are fixed through ECO / reroute.
- ✓ Clean PRA is mandatory for sign-off.

15. QUICK TIPS

- Always use correct SPEF scale.
- Check unconstrained paths first.
- Examine path components deeply.
- Validate after every ECO.
- Keep final reports for documentation.

Author : Konda Tharun Vivekananda

PD _ Routing

POST ROUTE ANALYSIS - SETUP TIMING CLOSURE & HOLD TIMING CLOSURE

1. INTRODUCTION

After routing and RC extraction, Post Route Analysis is performed in PrimeTime using SPEF/DSPF/RSPF.
It verifies that final design meets timing for both setup and hold across all corners and scenarios.

2. SETUP TIMING CLOSURE

2.1 Definition

Setup timing closure means achieving non-negative setup slack for all timing paths.

Setup Slack = Required Time - Data Arrival Time

Slack ≥ 0 → Pass , Slack < 0 → Violation

2.2 Issues Faced (Setup)

- Long data path delay
- High routing parasitics (R & C)
- High fanout nets and loading
- High clock latency on capture path
- High clock uncertainty
- Poor placement (long interconnect)

2.3 Setup Analysis in PrimeTime (Steps)

(1)	Read Design & Constraints →	read_verilog / read_db read_sdc
(2)	Read Parasitics →	read_spef / read_dspf / read_rspf
(3)	Link Design →	link_design top
(4)	Set Analysis Mode →	set_timing_derate , set_operating_conditions
(5)	Update Timing →	update_timing
(6)	Check Timing →	report_timing -delay max -max_paths 10
(7)	Analyze Violations →	report_timing_summary , report_qor
(8)	Fix Violations (ECO) →	Buffer insertion, resizing, placement opt

2.4 Techniques to Resolve Setup Violations

- Buffer Insertion on long data path / high fanout nets
- Upsizing / Cell Resizing
- Local / Global Placement Optimization
- Reduce unnecessary logic / logic restructuring
- Reduce clock latency on capture side (CTS optimization)
- Useful skew - advance clock at capture or delay at launch
- Increase Vdd or use faster cells (if power budget allows)

3. HOLD TIMING CLOSURE

3.1 Definition

Hold timing closure means achieving non-negative hold slack.

Hold Slack = Required Time - Required Time

Slack ≥ 0 → Pass , Slack < 0 → Violation

3.2 Issues Faced (Hold)

- Short data path delay
- Low routing parasitics
- High clock skew (capture clock early)
- Aggressive buffer insertion on data path
- Improper cell resizing

3.3 Hold Analysis in PrimeTime (Steps)

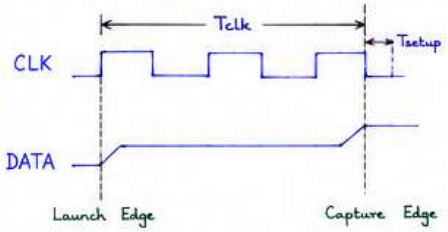
(1)	Read Design & Constraints →	read_verilog / read_db read_sdc
(2)	Read Parasitics →	read_spef / read_dspf / read_rspf
(3)	Link Design →	link_design top
(4)	Set Analysis Mode →	set_timing_derate , set_operating_conditions
(5)	Update Timing →	update_timing
(6)	Check Timing →	report_timing -delay min -max_paths 10
(7)	Analyze Violations →	report_timing_summary , report_qor
(8)	Fix Violations (ECO) →	Add delay, buffer resize, CTS tuning

3.4 Techniques to Resolve Hold Violations

- Buffer insertion in clock path (delay capture clock)
- Add delay in data path (buffer / inverter / wire detour)
- Use smaller / slower cells
- Increase routing resistance
- Useful skew - advance launch clock or delay capture clock
- Avoid over-aggressive cell downsizing

4. TIMING PATH MODELS

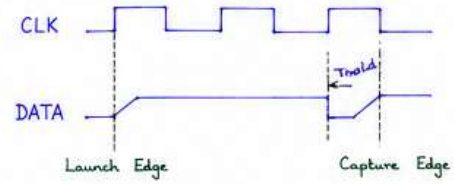
4.1 Setup Path (Launch FF → Capture FF)



Tclk = Clock period
Tcq = Clk to Q delay
Tcomb = Logic delay
Troute = Routing delay
Tsetup = Setup time
Tskew = Skew (Cap - Launch)
Tunc = Uncertainty

Required Time = Tclk + Tskew - Tsetup - Tunc
Data Arrival Time = Tcq + Tcomb + Troute
Setup Slack = Required Time - Data Arrival Time

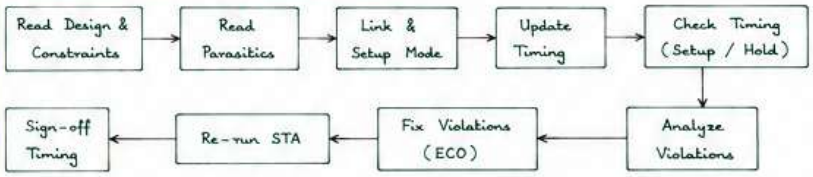
4.2 Hold Path (Launch FF → Capture FF)



Thold = Hold time
Tccq = Clk to Q (min)
Tcomb(min) = Min logic delay
Troute(min) = Min routing delay
Tskew = Skew (Cap - Launch)
Tunc = Uncertainty

Required Time = Tskew + Thold + Tunc
Data Arrival Time = Tccq + Tcomb(min) + Troute(min)
Hold Slack = Data Arrival Time - Required Time

5. PRA ANALYSIS FLOW IN PRIMETIME



6. COMPARISON - SETUP vs HOLD

Parameter	Setup Timing	Hold Timing
Requirement	Data must arrive before capture edge	Data must be stable after capture edge
Slack Formula	Req Time - Arrival Time	Arrival Time - Req Time
Critical Paths	Long paths	Short paths
Major Issues	High delay, high RC, high skew	Low delay, low RC, high skew
Common Fixes	Buffer, upsizing, placement opt, reduce cap, useful skew	Add delay, buffer in clock path, downsizing, useful skew
Impact	Affects maximum frequency	Affects minimum delay
Reports	report_timing -delay max	report_timing -delay min

7. PRA CHECKLIST

- ✓ Design & SDC loaded correctly
- ✓ Parasitics (SPEF/DSPF/RSPF) annotated
- ✓ All libraries & PVT corners loaded
- ✓ Setup timing analyzed (WNS ≥ 0)
- ✓ Hold timing analyzed (WNS ≥ 0)
- ✓ Timing reports reviewed
- ✓ DRC, LVS, Antenna, IR Drop clean
- ✓ Clock tree & skew within target
- ✓ ECO changes re-verified
- ✓ Final sign-off criteria met

8. KEY POINTS

- PRA ensures final routed design meets setup and hold requirements
- Parasitics and skew have major impact on timing
- Setup closure improves maximum frequency
- Hold closure ensures functional correctness
- Iterative analysis and ECO are required for timing closure

9. PRA PRACTICAL OBSERVATIONS

- Long wires after routing increase delay and reduce setup slack
- Buffer insertion improves setup but may hurt hold
- Hold violations usually appear after setup fixes
- Useful skew is powerful in both setup and hold closure

10. INTERVIEW / VIVA Qs

Q1. What is Post Route Analysis ?
→ STA after routing with extracted RC

Q2. Why is PRA important ?
→ To ensure accurate timing before tape-out considering real parasitics

Q3. How do you fix setup violation ?
→ Insert buffer, upsize, place opt, reduce clock latency, useful skew

Q4. How do you fix hold violation ?
→ Add delay, buffer in clock path, use smaller cells, useful skew

11. SUMMARY POINTS

- ✓ PRA is final STA with extracted parasitics
- ✓ Two targets - Setup closure & Hold closure
- ✓ Setup slack = Req - Arrival (≥ 0)
- ✓ Hold slack = Arrival - Req (≥ 0)
- ✓ Parasitics, skew, uncertainty impact timing
- ✓ Buffer insertion, sizing, placement and useful skew help in closure
- ✓ All sign-off checks must pass before tape-out

12. QUICK TIPS

- Always run both max and min analysis
- Check worst paths, not just WNS
- Verify across all PVT corners
- Keep reports organized
- Document ECO and re-run STA

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MCMM (MULTICORNER MULTIMODE ANALYSIS)

1. DEFINITION

MCMM is a static timing analysis methodology where the circuit is analyzed across multiple Process, Voltage, Temperature (PVT) corners and multiple Operating Modes to ensure timing closure under all possible real-world conditions.

2. WHY MCMM IS REQUIRED ?

- Circuit behavior varies with P, V, T and operating mode.
- A design closing in one condition may fail in another.
- Guarantees silicon reliability & performance.
- Required for sign-off STA, low power & DVFS designs.

3. CORNERS TO CHECK (PVT CORNERS)

Corner Type	Process (P)	Voltage (V)	Temperature (T)	Speed
SS	Slow	Low	High	Slowest
SF	Slow	Nominal	High	Slow
TT	Typical	Nominal	Nominal	Typical
FF	Fast	Nominal	Low	Fast
FS	Fast	High	Low	Fastest
FS_HV	Fast	High	High	Leakage check
SS_LV	Slow	Low	Low	Hold critical

Common practice → SS, SF, TT, FF, FS.

4. MODES TO CHECK (OPERATING MODES)

- Functional modes (different feature sets, mux selections)
- Logic poving summary, & calatme)
- Scan mode (scan chains enis.
- Test / BIST modes
- DVFS modes (different frequency bins)

5. PARAMETERS TO FOCUS IN MCMM

1. Setup Slack (max path delay)
2. Hold Slack (min path delay)
3. Negative Slack count & WNS / TNS
4. Critical path groups & endpoints
5. Clock skew, clock latency, uncertainty
6. Transition & Capacitance on critical nets
7. Power (IR drop, EM) across corners
8. Area impact for fixes

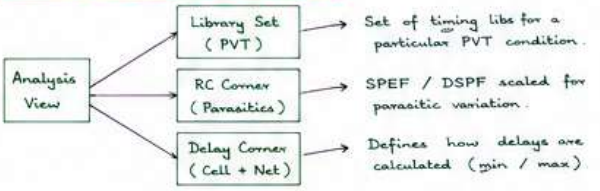
KEY POINT

- ✓ Worst corner may change for setup & hold.
- ✓ Never sign-off with single PVT corner.
- ✓ Always run MCMM at post route.

6. MCMM ANALYSIS FLOW

Create Libraries (PVT) → Define RC Corners → Create Analysis Views → Define Modes → Set Constraints per view → Run MCMM STA → Analyze Results & Fix

7. VIEW HIERARCHY IN PRIME TIME



8. MCMM EXAMPLE

Mode	Views (Corners)	Purpose
FUNC	SS, TT, FF	Functional timing sign-off
SCAN	SS, FF	Scan shift timing
LOW_POWER	SS_LV, TT	Sleep / retention timing
DVFS_LOW	SS, TT	Low frequency bin
DVFS_HIGH	FF, FS	High frequency bin

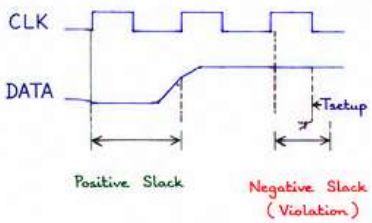
KEY POINTS

- ✓ More modes → better coverage
- ✓ More corners → higher confidence
- ✓ Balance runtime & accuracy.

9. SETUP TIMING - MCMM ANALYSIS

9.1 Definition

Setup timing checks that data arrives at capture FF before the required time.



9.2 Issues Faced

- Long data path delay.
- High routing R & C.
- High fanout nets.
- High logic depth.
- High clock latency / skew.
- Low drive strength cells.
- Poor placement.

9.3 Analysis in PrimeTime

- report_timing -max → Worst setup paths.
- report_timing_summary → WNS, TNS, # violations.
- report_clock_timing → Clock latency & skew.
- report_constraint → Constraint violations.
- report_design -histogram → Path delay distribution.

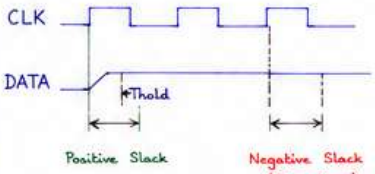
9.4 Techniques to Resolve (Setup)

1. Buffer insertion
2. Upsize / swap to faster cells
3. Logic restructuring / reduce logic
4. Pipeline stages (add registers)
5. Reduce fanout / add buffers
6. Optimize placement & routing
7. Gate sizing / reduce load

10. HOLD TIMING - MCMM ANALYSIS

10.1 Definition

Hold timing ensures that data remains stable at capture FF for minimum hold time.



10.2 Issues Faced

- Short data path delay.
- Low routing R & C.
- Fast (high drive) cells.
- Less logic depth.
- Low clock skew (capture early).
- Improper buffer insertion.

10.3 Analysis in PrimeTime

- report_timing -min → Worst hold paths.
- report_timing_summary → WNS (min), # violations.
- report_clock_timing → Clock skew (capture early).
- report_hold → Hold slack details.
- report_transition → Fast transitions.

10.4 Techniques to Resolve (Hold)

1. Buffer insertion on data path
2. Use smaller size cells
3. Add delay cells / inverters
4. Increase routing (detours)
5. Adjust clock skew (CTS tuning)
6. Use hold fixing buffers (ECO)
7. Increase wire resistance

11. COMPARISON - SETUP vs HOLD

Parameter	Setup Timing	Hold Timing
Check	Data arrives before required time	Data must remain after required time
Path Type	Max delay path	Min delay path
Critical Corner	Slow corner (SS, Low V, High T)	Fast corner (FF, High V, Low T)
Slack Equation	Slack = Req Time - Arr Time	Slack = Arr Time - Req Time
Violation	Slack < 0 → Late	Slack < 0 → Early
Fix Focus	Reduce delay	Increase delay

12. ADVANTAGES OF MCMM

- ✓ Accurate timing sign-off across all conditions.
- ✓ Prevents silicon failures in worst case.
- ✓ Captures real-world variation.
- ✓ Improves yield and reliability.
- ✓ Supports low power & DVFS designs.

13. PRACTICAL OBSERVATIONS

- Worst setup corner ≠ worst hold corner.
- MCMM runtime increases with modes/views.
- Proper constraint per mode is critical.
- Always review worst paths from all critical views.
- Sign-off only after all views pass.

14. MCMM PRE CHECKLIST

- ✓ All PVT corners defined
- ✓ All operating modes defined
- ✓ Constraints applied per mode
- ✓ Views created correctly
- ✓ MCMM STA run completed
- ✓ Worst paths analyzed
- ✓ ECO fixes validated
- ✓ Final sign-off achieved

15. INTERVIEW / VIVA Qs

Q1. Why MCMM is important in sign-off ?
→ To ensure design works in all real world PVT and modes.

Q2. Which corners are critical for setup and hold ?
→ Setup → slow corners.
→ Hold → fast corners.

Q3. What reports are used in MCMM?
→ report_timing -max / -min, report_timing_summary etc.

16. SUMMARY POINTS

- ✓ MCMM analyzes design across PVT corners and modes.
- ✓ Setup closure removes late paths.
- ✓ Hold closure removes early paths.
- ✓ Different corners are critical for setup and hold.
- ✓ Accurate MCMM sign-off ensures silicon reliability.

17. QUICK TIPS

- Always verify worst paths for all views.
- Don't ignore hold in fast corners.
- Use ECO sparingly and wisely.
- Maintain mode based constraints.
- Keep documentation updated.

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PD _ Routing

POST ROUTE SIGNAL INTEGRITY ANALYSIS - INTRODUCTION, WHY IT IS CRITICAL?, FIXES AND ADVANTAGES

1. INTRODUCTION

Post Route Signal Integrity (PSI) analysis verifies that all signals maintain correct voltage levels and timing integrity after routing considering parasitics and crosstalk.
It ensures reliable operation of the final silicon.

2. WHY POST ROUTE SI IS CRITICAL ?

- Aggressive scaling → smaller noise margins.
- High interconnect density → more coupling.
- Fast switching → SSN and IR noise.
- Protects against functional failures in silicon.
- Mandatory for high performance and low power designs.

3. MAJOR SI CONCERNS

SI Concern	Description	Impact
Crosstalk Noise	Coupling between adjacent aggressor and victim nets	Delay, glitches, functional errors
Simultaneous Switching Noise (SSN)	Ground / Power bounce due to simultaneous switching	Timing shift, false triggers
IR Drop	Voltage drop on power distribution network	Slow devices, timing violations
Ground Bounce	Noise on ground due to return current	Increased delay, hold failures
Electromigration (EM)	High current density in interconnect	Reliability failure, open circuits
Antenna Effect	Charge build up during plasma etch	Oxide damage, leakage

4. SI ANALYSIS FLOW (POST ROUTE

Extract RC (SPEF) → Build SI Model → Define Scenarios → Run SI Analysis → Analyse Results → Apply Fixes → Re-run & Sign-off

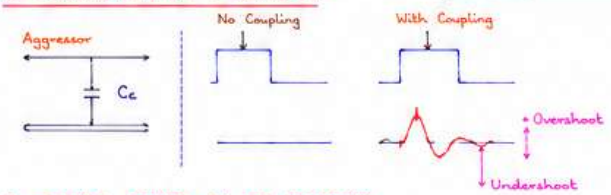
5. TYPES OF SI ANALYSIS

- Crosstalk (Next, Fext, Pext)
- SSN / IR Drop analysis
- Ground Bounce analysis
- Signal Integrity checks (overshoot / undershoot)
- EM and Antenna checks

KEY POINT

SI analysis must be performed on the final routed design with extracted parasitics to ensure real-world accuracy.

6. CROSSTALK NOISE - CONCEPTS



7. ISSUES CAUSED BY SI PROBLEMS

- Setup / Hold timing violations.
- Functional failures (glitches, wrong logic value).
- Increased leakage and power.
- Reliability degradation (EM, antenna).
- Yield loss and silicon re-spins.

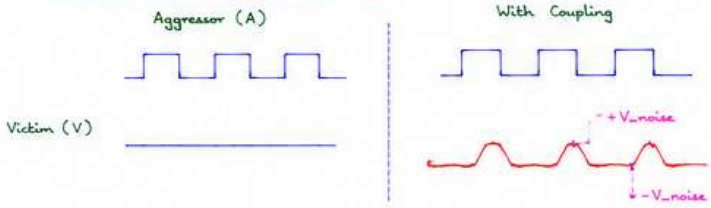
8. SI FIXES AND TECHNIQUES

Fix / Technique	SI Problem Addressed	Description
Insertion	Crosstalk, Delay	Strong driver reduces net sensitivity.
Shielding	Crosstalk	Add shield net tied to GND/VDD.
Wire Spacing	Crosstalk	Increase spacing between aggressor and victim.
Wire Sizing	Delay, Noise	Increase width to reduce R.
Decoupling Caps	SSN, IR Drop	Place decap near high switching blocks.
Power Network Opt.	IR Drop, SSN	Add stripes, rings, vias.
Cell Legalisation	Hold Violations	Shorten data path.
Use Low Swing Buffers	EM, Crosstalk	Reduce transition and current.

9. ADVANTAGES OF POST ROUTE SI ANALYSIS

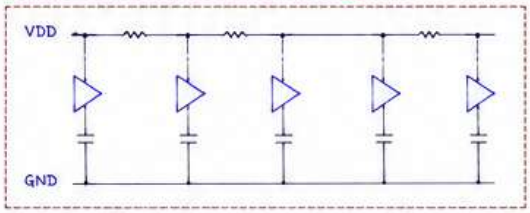
- ✓ Ensures functional correctness of design.
- ✓ Improves timing closure confidence.
- ✓ Reduces risk of silicon failures.
- ✓ Improves reliability and product quality.
- ✓ Avoids costly re-spins and delays.

10. CROSSTALK ANALYSIS - EXAMPLE



Noise amplitude ∝ C_c, Slew of aggressor
∝ $\frac{1}{\text{spacing}}$, victim impedance

11. SSN AND IR DROP - CONCEPT



Simultaneous switching causes current surge → voltage drop on power rails.
(SSN / IR DROP)

12. PARAMETERS TO FOCUS IN SI ANALYSIS

- Peak Crosstalk Noise (Next, Fext, Pext).
- Noise Margin (NMH, NML).
- Overshoot / Undershoot levels.
- SSN (ΔV on VDD / GND).
- IR Drop (ΔV on VDD).
- Delay impact due to noise.
- EM Current Density.
- Antenna Ratio.

SIGN-OFF CRITERIA

- ✓ All noise < Noise margin (WNS ≥ 0).
- ✓ SSN within limit.
- ✓ IR Drop within limit.
- ✓ EM within limit.
- ✓ Antenna ratio within limit.
- ✓ No functional violations.

13. SI REPORTS (POST ROUTE)

Report	Purpose	Key Information
Crosstalk Report	Noise between nets	Victim net, aggressor, noise type, amplitude.
SSN / IR Drop Report	Power integrity	ΔV on VDD / GND, location, severity.
SI Summary	Overall status	Worst noise, WNS, failing paths.
Detailed Waveform	Noise investigation	Voltage waveforms at critical nodes.

14. PRACTICAL OBSERVATIONS

- Aggressive clock nets are highly sensitive to crosstalk.
- Narrow spacing on long parallel nets causes large noise.
- Decaps must be placed close to high switching blocks.
- Shield nets must be tied properly to avoid floating.
- Always re-run SI after each fix.

15. INTERVIEW / VIVA Qs

Q1. Why is post route SI analysis important ?
→ To ensure that routed design works correctly in real silicon under noise and variation.

Q2. What are main SI issues ?
→ Crosstalk, SSN, IR Drop, Ground Bounce, EM, Antenna.

Q3. How to reduce crosstalk ?
→ Increase spacing, add shielding, buffer insertion, reduce slew, increase wire width.

Q4. What is SSN ?
→ Voltage noise on power / ground due to simultaneous switching of many cells.

16. SUMMARY POINTS

- ✓ Post route SI analysis ensures noise free and reliable design.
- ✓ Crosstalk, SSN, IR Drop and other effects impact timing & function.
- ✓ Multiple fixes are available to improve SI.
- ✓ SI analysis must be an iterative process with re-run and
- ✓ Final sign-off only after all SI criteria are met.

17. QUICK TIPS

- Always check worst case scenarios.
- Maintain good spacing and shielding.
- Use enough decaps for SSN control.
- Keep power network robust.
- Document all SI fixes and results.

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POST ROUTE POWER ANALYSIS - INTRODUCTION, WHY IT IS CRITICAL?, FIXES AND ADVANTAGES

1. INTRODUCTION

Post Route Power Analysis (PRPA) is performed after detailed routing and RC extraction to estimate accurate power (Dynamic, Leakage and Total) and check power distribution integrity such as IR Drop and EM.

2. WHY POST ROUTE POWER ANALYSIS IS CRITICAL ?

- Accurate power estimation only with real parasitics.
- Detects power hot spots and over-current regions.
- Ensures IR Drop and EM within safe limits.
- Helps in power budget closure and battery life.
- Ensures functional reliability and timing stability.

3. TYPES OF POWER

- Dynamic Power → Switching due to signal activity.
- Leakage Power → Due to transistor leakage.
- Total Power → Dynamic + Leakage.

$$P_{total} = P_{dynamic} + P_{leakage}$$

4. POWER COMPONENTS

Component	Description
Internal Power	Power inside standard cells and macros.
Switching Power	Charging / discharging of load capacitances.
Clock Power	Clock tree buffers and interconnect switching.
Short Circuit Power	During input transitions in gates.
Leakage Power	Subthreshold, gate-Oxide and junction leakage.

5. POST ROUTE POWER ANALYSIS FLOW



6. INPUTS FOR PRPA

- Post Route Netlist (DEF annotated).
- Extracted Parasitics (SPEF / DSPF / RSPF).
- Switching Activity (SAIF / VCD / FSDB).
- Liberty files (with power tables).
- IR Drop [illegible].
- Power intent (UPF / CPF).

7. CORNERS TO CHECK (PVT)

Corner Type	Process (P)	Voltage (V)	Temperature (T)
SS_LV_125C	Slow	Low	125°C
SS_LV_-40C	Slow	Low	-40°C
TT_NV_25C	Typical	Nominal	25°C
FF_HV_125C	Fast	High	125°C
FF_NV_-40C	Fast	High	-40°C

Analyze all setup and hold corners for power and IR / EM.

8. PARAMETERS TO FOCUS

1. Total Power (mW)
2. Dynamic Power (mW)
3. Leakage Power (mW)
4. Power Density (mW / mm²)
5. IR Drop (mV)
6. EM Current Density (A / mm²)
7. Peak Switching Activity
8. Hot Spot Cells / Nets

KEY POINT

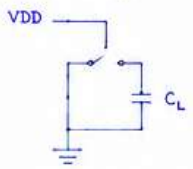
- ✓ Post route numbers are final and sign-off numbers.
- ✓ Power varies with activity and PVT conditions.
- ✓ IR Drop and EM must be checked in all modes.

9. REPORTS GENERATED

Report	Contents
Power Summary	Total, Dynamic, Leakage, Internal, Switching power.
Power by Hierarchy	Block wise power breakdown.
Power by Type	Clock, Internal, Switching, Leakage.
Power by Corner	Power values for each PVT corner.
IR Drop Report	Voltage drop, worst point, % drop.
EM Report	Current density, violations, unsafe nets.

10. POWER ESTIMATION CONCEPT

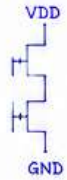
Dynamic Power (Switching)



$$P_{dyn} = \alpha \cdot C_L \cdot V^2 \cdot f$$

- α → activity factor
- C_L → Load Capacitance
- V → Supply Voltage
- f → Frequency

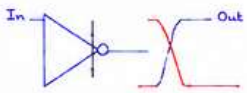
Leakage Power (Static)



$$P_{leak} = I_{leak} \cdot VDD$$

- I_{leak} → total leakage current
- VDD → Supply Voltage

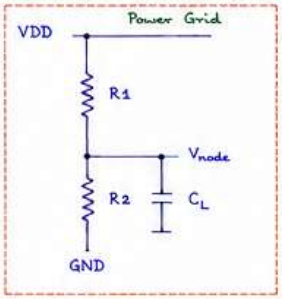
Short Circuit Power (During Transition)



$$P_{sc} = I_{sc(avg)} \cdot VDD$$

- $I_{sc(avg)}$ → avg short circuit current

11. IR DROP ANALYSIS - CONCEPT



- Current through R causes voltage drop.
- $V_{node} = VDD - I \cdot (R1 + R2)$
- Excessive IR Drop → timing failures, noise, functional errors.

IR DROP LIMITS (Typical)

- ≤ 5% of VDD for core
- ≤ 10% for non-critical
- Follow tech / foundry limits.

12. EM (ELECTROMIGRATION) - CONCEPT



High Current → Metal stress → Void formation → Open

EM LIMIT

- $J < J_{safe}$ from foundry
- Check for all power and ground nets.

13. FIXES / TECHNIQUES TO REDUCE POWER

1. Reduce Switching Activity → Clock gating, operand isolation, FSM encoding.
2. Reduce Capacitance → Buffer sizing, wire sizing, remove unnecessary loads.
3. Multi-VDD / Power Gating → Shut down idle blocks.
4. Use Low Power Cells → High-Vt cells for non-critical paths.
5. Optimize Clock Tree → Minimize clock power, useful skew.
6. Logic Optimization → Remove redundant logic, constant propagation.

14. FIXES FOR IR DROP AND EM

1. Wider Power / Ground Wires → Reduce resistance.
2. Add More Power Stripes / Rings → Improve current distribution.
3. Add Decap Cells → Reduce transient voltage drop.
4. Reduce Switching Power → Follow power reduction fixes.
5. Reduce Current Density → Increase width / parallel paths.

15. ADVANTAGES OF POST ROUTE POWER ANALYSIS

- ✓ Provides accurate and sign-off power numbers.
- ✓ Helps in power budget closure and meeting targets.
- ✓ Ensures IR Drop and EM safety → reliable design.
- ✓ Improves performance, battery life and yield.
- ✓ Identifies hot spots and helps in optimization early.
- ✓ Supports low power and high performance SoC / ASIC designs.

16. PRACTICAL OBSERVATIONS

- Power numbers increase after routing due to real parasitics.
- IR Drop is higher in long power delivery paths.
- Clock power is major component in many designs.
- Decap and power grid planning is key.

17. INTERVIEW / VIVA Qs

Q1 Why PRPA is important ?
→ To get accurate power after routing with real parasitics and activity.

Q2 What are power types ?
→ Dynamic, Leakage and Total.

Q3 What causes IR Drop ?
→ Voltage drop due to current through resistance.

Q4 How to reduce power ?
→ Reduce activity, capacitance, use low power techniques.

18. QUICK TIPS

- Always use real SPEF and SAIF for PRPA.
- Check all PVT corners.
- Verify IR Drop and EM.
- Focus on hot spots first.
- Sign-off only after all limits are met.

Author : Konda Tharun Vivekananda

PD _ Routing

ECO FUNDAMENTALS – INTRODUCTION, REASONS, PRINCIPLES, TYPES OF ECO, POST ECO VALIDATION

1. INTRODUCTION

ECO (Engineering Change Order) is the process of making late design changes on the implemented physical design while preserving timing, power, area and physical sign-off.

2. REASONS FOR ECO

- Fix timing violations (setup / hold / slack).
- Fix functional issues / logic bugs.
- Change connectivity / metal fixes.
- Add / remove features.
- Improve power / IR drop / EM.
- Address DRC, antenna, LVS or ERC issues.
- Customer / spec changes.

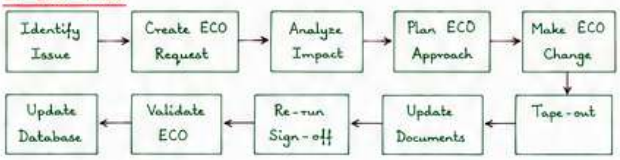
3. ECO PRINCIPLES

- ✓ Make minimum changes to the design.
- ✓ Preserve timing, power, and area as much as possible.
- ✓ Avoid large disturbance to routing / placement.
- ✓ Maintain physical sign-off (DRC, LVS, SI, IR, EM).
- ✓ ECO should be localized and traceable.

4. TYPES OF ECO

ECO Type	Description	Typical Use	Scope	Impact
Timing ECO	Fix timing violations	Setup / Hold improvement	Local / Global	Medium
Logic ECO	Fix functional errors	Bug fix / wrong logic	Local	High
Metal ECO	Fix routing / connectivity	Opens, shorts, wrong nets	Local	Low
DRC ECO	Fix DRC violations	Spacing, width, via, etc.	Local	Low
Antenna ECO	Fix antenna violations	High antenna nets	Local	Low
IR / EM ECO	Fix power issues	IR drop / EM violations	Local / Global	Medium
Feature ECO	Add / remove features	Spec modifications	Global	High

5. ECO FLOW



6. ECO APPROACHES

- Local ECO : Small localize change (buffer, inverter, re-route).
- Global ECO : Multiple changes across design.
- Incremental ECO : Minimal re-run of tools.
- Full ECO : Re-run major steps (CTS, Route etc.) if impact high.

7. ECO TEMPLATE ECO OPERATIONS

Operation	Used To
Buffer Insertion	Reduce delay, fix setup.
Buffer Removal	Reduce unnecessary delay / power.
Inverter Insertion	Fix hold, polarity / logical fix.
Gate Sizing	Improve timing / drive strength.
Net Re-route	Fix opens, shorts, DRC, antenna.
Via Addition / Move	Fix DRC / connectivity.
Cell Movement	Fix placement / congestion.
Tie Cell Addition	Fix floating nets, connect tie-high / low.
Power Cell Addition	Fix IR drop / EM issues.

8. ECO IMPACT CONSIDERATIONS

- Timing impact (setup, hold, slack).
- Power impact (dynamic, leakage, IR, EM).
- Area impact.
- Congestion impact.
- Physical verification impact (DRC, LVS, Antenna, SI).
- Tool runtime and schedule impact.

KEY POINT

A good ECO fixes immediate issue without creating new violations or degrading overall design quality.

9. EXAMPLE – TIMING ECO (SETUP)



Slack < 0 (Setup VIOLATION)

Fix Options :
1. Insert Buffer
2. Upsize Cells
3. Reduce Logic
4. Rebalance Path
5. Optimize CTS

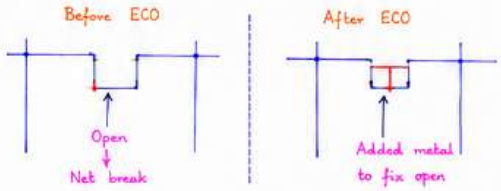
10. EXAMPLE – HOLD ECO



Slack < 0 (Hold VIOLATION)

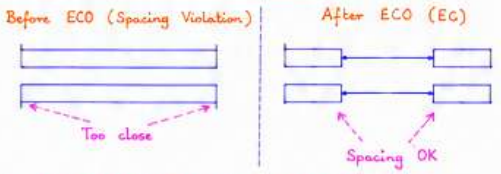
Fix Options :
1. Insert Delay
2. Use Smaller Cells
3. Add Inverter
4. Re-route (longer)
5. Optimize CTS

11. EXAMPLE – METAL ECO (OPEN NET)



Fix Options :
1. Add metal wire
2. Move via
3. Re-route
4. Add Jumper

12. EXAMPLE – DRC ECO (SPACING)



Fix Options :
1. Move wire
2. Widen spacing
4. Add spacing

13. POST ECO VALIDATION – CHECKLIST

Check	Description
✓ Timing	Re-run STA (Setup / Hold, WNS, TNS).
✓ DRC	Re-run DRC to ensure no new violations.
✓ LVS	Re-run LVS to ensure connectivity correctness.
✓ IR Drop	Check IR drop within limits.
✓ EM	Check EM on critical nets.
✓ Antenna	Check antenna violations.
✓ Power	Verify power numbers after ECO.
✓ SI	Check noise, crosstalk, SSN if required).
✓ Physical	Check congestion, routing utilization.
✓ ECO Report	Document ECO changes and results.

ECO VALIDATION FLOW

Apply ECO Change → Update DB / DEF → Incremental STA → Run Sign-off Checks → ECO Sign-off

14. ADVANTAGES OF ECO

- Fixes issues late in the flow bout re-starting full design.
- Reduces respin risk and saves cost.
- Preserves most of the design effort already done.
- Improves quality and reliability.
- Provides flexibility to handle late changes.

15. CHALLENGES OF ECO

- ECO may create new violations.
- ECO impact prediction is complex.
- Too many ECOs increase risk.
- Tool runtime can increase.
- Balancing timing, power and DRC.

16. ECO GOOD PRACTICES

- ✓ Analyze impact before applying ECO.
- ✓ Use minimum change to fix the issue.
- ✓ Keep ECO localized and traceable.
- ✓ Re-run only necessary steps (incremental).
- ✓ Always re-verify all sign-off criteria.
- ✓ Maintain ECO log / version control.

ECO METRICS TO TRACK

- No. of ECOs
- ECO turnaround time
- Timing improvement (WNS/TNS)
- Power impact (ΔP)
- No. of new violations
- ECO closure status

17. PRACTICAL OBSERVATIONS

- Most ECOs are timing and metal related.
- Buffer insertion is most common ECO fix.
- Local fix with incremental STA is preferred.
- CTS changes can fix many hold / setup issues.

18. INTERVIEW / VIVA Qs

Q1. What is ECO in Physical Design ?
→ To te changes to fix issues without restarting full flow.

Q2. What are types of ECO ?
→ Timing, Logic, Metal, DRC, Antenna, IR/EM, Feature ECO.

Q3. Why is post ECO validation important ?
→ To ensure ECO fixed the issue without causing new violations.

19. SUMMARY POINTS

- ✓ ECO is done to fix late issues in physical design.
- ✓ Types : Timing, Logic, Metal, DRC, Antenna, IR/EM, Feature.
- ✓ Principles : Minimum change, preserve quality, traceability.
- ✓ Always perform post ECO validation for all sign-off checks.
- ✓ Proper ECO flow / reduces risk and saves cost.

20. QUICK TIPS

- Backup design before ECO.
- Use ECO commands smartly.
- Run incremental STA.
- Check all sign-off after ECO.
- Document every ECO.
- Don't ignore power / IR impact.

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PD _ Routing

ROUTING SIGN OFF - INTRODUCTION, WHY IT IS CRITICAL?, COMMON VALIDATION AND FIXES

1. INTRODUCTION

Routing sign off ensures that the final routed design meets all physical, electrical and manufacturing requirements before tape-out. It checks the routed layout for rule compliance, correctness, reliability and performance.

2. WHY ROUTING SIGN OFF IS CRITICAL ?

- Ensures manufacturability of the final routed chip.
- Avoids costly silicon re-spins and metal fixes.
- Prevents unexpected failures (shorts, opens, reliability).
- Guarantees timing, power and SI requirements.
- Meets foundry and packaging requirements.

3. ROUTING SIGN OFF COVERAGE

- ✓ Design Rule Compliance
- ✓ Electrical Rule Compliance
- ✓ Layout vs Schematic Consistency
- ✓ Reliability & Integrity Checks
- ✓ Timing / Power / SI Checks
- ✓ Physical Correctness & Connectivity

4. COMMON VALIDATIONS (CHECKS)

Check	Purpose	Tool / Method	Sign-off Criteria
DRC	Check design rules	Calibre/ICV	0 errors
LVS	Layout vs schematic matched	Calibre/ICV	0 mismatches
ERC	Check electrical rule violations	Calibre	0 ERC errors
Antenna	Prevent gate oxide damage ?	Calibre	0 antenna violations
Metal Density	Meet density rules (CMP)	Calibre	Within min/max
IR Drop	Check voltage drop on power grid	Voltus/RedHawk	Within limit
EM (ElectroMig.)	Check current density in metal	Voltus/RedHawk	Within limit
Timing (Post Route)	Setup / Hold closure	PrimeTime	WNS / TNS ≥ 0
SI (Signal Int.)	Check noise, crosstalk, SSN, reflections	PrimeSI	Within limit
Connectivity	All nets connected correctly	Calibre/ICV	0 opens / shorts
Via Coverage	Check via rules & stacking	Calibre/ICV	No via errors

5. DRC CHECKS - KEY AREAS

- Minimum width, spacing, enclosure, extension.
- Layer spacing and via rules.
- NDR (Non-Default Routing) rule compliance.
- Pin access, off-grid, overlap.
- Cut spacing and via array rules.

6. ELECTRICAL RULE CHECKS (ERC)

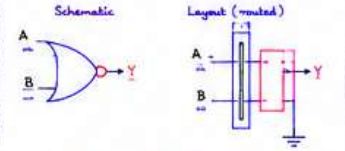
- Check shorts between nets.
- Unconnected pins / floating nets.
- Power / ground short issues.
- Multi-source nets.
- Pin direction conflicts.

KEY POINT

- ✓ Sign-off means ZERO violations.
- ✓ All reports must be clean.
- ✓ Always sign-off with real parasitics.

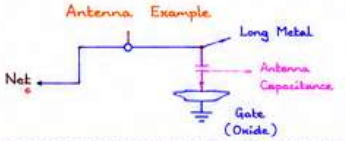
7. LVS CHECKS

- Compares routed layout with schematic netlist.
- Reports missing, extra or mismatched devices.
- Checks connectivity of all nets.
- Must be 100% clean.



8. RELIABILITY CHECKS

- Antenna Rule Check
- Electromigration (EM)
- Via Reliability
- Metal Density rules
- Stress / CMP friendly layout



9. TIMING / POWER / SI CHECKS

(a) Timing Check

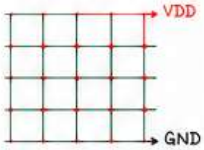
Post route STA



- Check Setup & Hold.
- WNS / TNS ≥ 0.
- Check all critical paths.

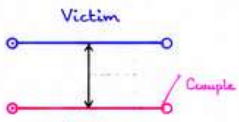
(b) Power Check

Power Grid



- IR Drop within limit.
- EM within limit.
- No power shorts.

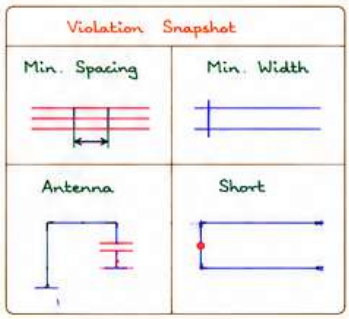
(c) SI Check



- Crosstalk noise.
- SSN / ground bounce.
- Reflections / ringing.

10. COMMON VIOLATIONS FOUND

- Minimum spacing / width violations.
- Metal density over / under.
- Antenna ratio exceeded.
- Via rule violations.
- Off-grid / off-cut issues.
- Pin access violations.
- IR Drop / EM violations.
- Timing violations.
- LVS mismatches / opens / shorts.



11. FIXES / CORRECTIVE ACTIONS

Violation Type	Fix / Technique
DRC Violations	Adjust width / spacing, use jogs, layer change, follow design rules.
LVS Mismatch	Fix missing / extra devices, reconnect nets.
Antenna Violation	Add antenna diodes / gate diodes, insert buffer / break long metal.
IR Drop	Add power straps / rings, widen power lines, add vias, improve power grid.
EM Violation	Increase width / parallel metal, add vias, improve current distribution.
Timing Violation	Buffer insertion, upsizing, gate resizing, re-route critical nets.
Crosstalk / SI	Increase spacing, shield nets, layer change, re-route aggressor / victim nets.
Pin Access / Off-grid	Use vias, jogs, re-align to track/grid.
Metal Density	Add fillers or reduce metal density.

12. ROUTING SIGN OFF FLOW



12. ADVANTAGES OF ROUTING SIGN OFF

- ✓ Ensures 100% manufacturable routed layout.
- ✓ Avoids silicon re-spins and metal fixes.
- ✓ Ensures timing, power, reliability and SI closure.
- ✓ Improves yield and product quality.
- ✓ Meets foundry, package and customer requirements.

SIGN-OFF MEANS

- ✓ 0 DRC / ERC / LVS errors
- ✓ Timing Clean (WNS ≥ 0)
- ✓ Power / IR / EM Clean
- ✓ SI within limits
- ✓ All reports archived

14. POST SIGN OFF REPORTS (KEY)

- DRC Summary Report
- LVS Clean Report
- ERC Summary
- Antenna Summary
- IR Drop Report
- EM Report
- Timing (STA) Report
- SI Report
- Metal Density Report
- Power Summary Report

15. PRACTICAL OBSERVATIONS

- Always fix DRC first, then LVS / ERC.
- Do not ignore antenna & density even if timing is clean.
- Use real parasitics for final sign-off.
- Verify critical nets manually (clocks, resets, buses).
- Keep sign-off environment consistent till tape-out.

15. INTERVIEW / VIVA Qs

Q1. What is routing sign off ?

→ Final verification of routed layout for all physical & electrical checks before tape-out

Q2. What are key sign-off checks ?

→ DRC, LVS, ERC, Antenna, IR, EM, Timing, SI, Connectivity

17. QUICK TIPS

- Fix root cause, not just symptoms.
- Use proper layer planning.
- Follow grid and design rules.
- Re-run all checks after every fix.
- Sign-off only after clean reports.

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PD _ Routing

a) DRC SIGN OFF – INTRODUCTION, WHY IT IS CRITICAL?, COMMON VALIDATION AND FIXES.

b) LVS SIGN OFF – INTRODUCTION, WHY IT IS CRITICAL?, COMMON VALIDATION AND FIXES.

a) DRC SIGN OFF

1. INTRODUCTION

Design Rule Check (DRC) sign off verifies that the final routed layout obeys all foundry design rules before tape-out. It ensures that every geometric and spacing rule is satisfied for a manufacturable chip.

2. WHY DRC SIGN OFF IS CRITICAL ?

- Prevents fabrication defects and yield loss.
- Ensures manufacturability of the silicon.
- Avoids metal shorts, opens and spacing violations.
- Meets foundry qualification requirements.
- Required for tape-out and reliable product.

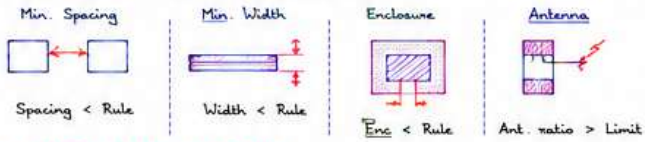
3. COMMON DRC VALIDATIONS (CHECKS)

Check Type	Purpose	Typical Violations Checked
Spacing Rules	Ensure min. distance between objects	M1-M1, M1-Via, Via-Via, M1-Poly, Metal-Diffusion
Width Rules	Ensure min. line width	Metal width, Via size, Poly width
Enclosure Rules	Ensure proper overlap	Via enclosure by metal, Gate enclosure by diffusion
Extension Rules	Ensure min. extension beyond contact	Metal extension over via, Poly extension over active
Layer Rules	Ensure valid layer usage	Correct layer mapping, No illegal layers
Density Rules	Ensure density constraints	Metal density, CMP density
Antenna Rules	Prevent antenna charging	Antenna ratio violations
Edge / Cut Rules	Ensure edges are valid	Min. distance to die edge, Min. notch/slot size

4. DRC CHECK FLOW



5. COMMON DRC VIOLATIONS – EXAMPLES



6. FIXES FOR DRC VIOLATIONS

- Increase spacing / width.
- Add jogs / detours to increase spacing.
- Resize wires and vias.
- Add dummy fill / shielding for density.
- Add antenna diodes or insert buffers.
- Move objects away from die edge.
- Re-route / Layer change to meet rules.

KEY POINT
- ✓ DRC clean layout = primary sign-off
- ✓ Zero violations in all rule decks are mandatory.

7. DRC SIGN OFF METRICS

Metric	Description
Total Violations	Total number of DRC violations.
By Rule Count	Violations grouped by rule type.
By Layer Count	Violations grouped by layer.
Waived Violations	Rules waived with justification.
Sign-off Status	DRC Clean (0 violations / No waivers).

8. ADVANTAGES OF DRC SIGN OFF

- ✓ Ensures manufacturable and defect free layout.
- ✓ Improves yield and reduces re-spins.
- ✓ Satisfies foundry and customer requirements.
- ✓ Reduces cost and tape-out risk.
- ✓ Provides confidence in final physical implementation.

b) LVS SIGN OFF

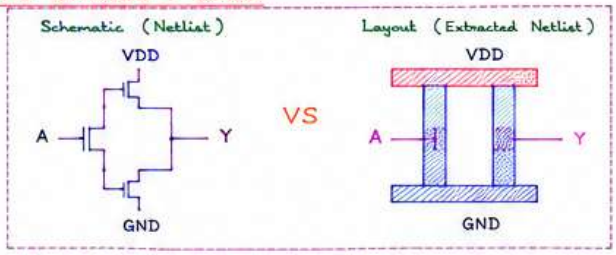
1. INTRODUCTION

Layout Versus Schematic (LVS) sign off verifies that the final layout matches the original schematic at the transistor level. It ensures correct connectivity and no missing or extra devices.

2. WHY LVS SIGN OFF IS CRITICAL ?

- Ensures functional correctness of the silicon.
- Detects missing / extra devices and wrong connections.
- Prevents functional failures and product recalls.
- Required for tape-out and product reliability.
- Confirms that layout implements the intended design.

3. LVS COMPARISON – CONCEPT

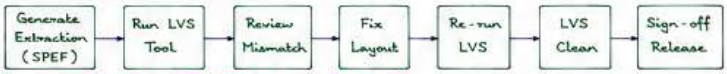


LVS tool compares both netlists and reports mismatches.

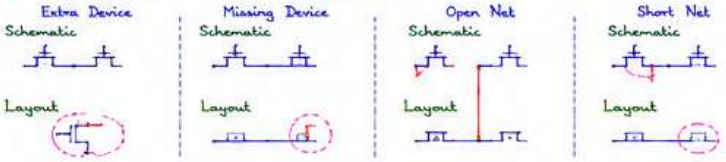
4. COMMON LVS VALIDATIONS (CHECKS)

Check Type	Purpose	Typical Errors Found
Device Check	Verify all devices are present	Missing device, extra device, wrong device size
Connectivity Check	Verify connectivity of all nets	Open nets, short nets, wrong connections
Pin Check	Verify I/O pins	Missing pins, swapped pins, wrong direction
Net Names Check	Verify net naming	Wrong net names, unconnected nets
Model / Cell Check	Verify correct device models	Wrong model, wrong cell mapping.
Power / Ground Check	Verify VDD / GND connectivity	Floating power/ground, shorts

5. LVS SIGN OFF FLOW



6. COMMON LVS MISMATCHES – EXAMPLES



7. FIXES FOR LVS MISMATCHES

- → Add missing devices.
- → Remove extra devices.
- → Correct wrong connections.
- → Fix opens / shorts in nets.
- → Correct pin connections and directions.
- → Update device sizes / models.

KEY POINT
- ✓ LVS clean ensures functional equivalence between layout and schematic.
- ✓ Zero mismatches are mandatory.

8. ADVANTAGES OF LVS SIGN OFF

- ✓ Ensures functional correctness of final silicon.
- ✓ Prevents costly silicon bugs and re-spins.
- ✓ Guarantees reliability and customer satisfaction.
- ✓ Meets tape-out and foundry requirements.
- ✓ Builds confidence in final product.

9. DRC vs LVS COMPARISON

Parameter	DRC	LVS
Full Form	Design Rule Check	Layout Versus Schematic
Purpose	Check geometry rules	Check electrical equivalence
Checks	Spacing, Width, Enclosure, Density, Antenna, etc.	Devices, Connectivity, Pins, Models, Power/Ground, etc.
Input	Layout	Layout + Schematic (Netlist)
Output	Rule Violations	Netlist Mismatches
Sign-off Criteria	0 Violations	0 Mismatches
Tool Examples	Calibre DRC, Pegasus, Assura, PVS	Calibre LVS, Pegasus LVS, Assura, LVS

10. PRACTICAL OBSERVATIONS

- Always achieve DRC clean before LVS.
- Fixing opens/shorts early reduces LVS efforts.
- Use consistent pin and net naming.
- Antenna and density should be addressed before sign-off.
- Final reports must be archived for audit.

11. INTERVIEW / VIVA Qs

Q1. Why is DRC sign off important ?
→ To ensure manufacturable layout meeting all design rules.

Q2. What are common DRC checks ?
→ Spacing, Width, Enclosure, Density, Antenna, etc.

Q3. Why is LVS sign off important ?
→ To ensure layout matches schematic functionally.

Q4. What are common LVS errors ?
→ Missing/extra devices, opens, shorts, wrong connections.

12. QUICK TIPS

- Run DRC frequently during routing.
- Keep DRC violations at zero.
- Run LVS after major layout changes.
- Always fix opens and shorts first.
- Verify power/ground connectivity.
- Maintain clean reports and waivers (if any) with justification.

Author : Konda Tharun Vivekananda

50. TIMING & MCMM SIGN OFF

a) TIMING SIGN OFF

1) INTRODUCTION

- Timing Sign Off is the final timing verification to ensure the design meets all setup and hold requirements.
- Performed on the final routed design with extracted parasitics (SPEF).
- Uses Static Timing Analysis (STA) to check all timing paths across the design.

2) WHY IT IS CRITICAL?

- Ensures chip will function correctly at the target frequency.
- Prevents functional failures due to timing violations.
- Guarantees design robustness across PVT variations.
- Mandatory for tape-out and silicon success.

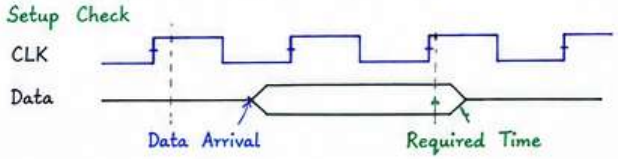
3) COMMON VALIDATIONS

Check Item	Description
Setup Timing	All setup paths must have positive slack.
Hold Timing	All hold paths must have positive slack.
WNS / TNS	Check Worst Negative Slack and Total Negative Slack.
Clock Domain Crossing (CDC)	Verify synchronization and proper constraints.
False Paths	Ensure false paths are correctly defined.
Multicycle Paths	Validate multicycle path constraints.
I/O Timing	Check input/output delays and timing.
Clock Uncertainty	Verify clock uncertainty settings.
Derates	Apply OCV/AOCV/POCV derates.
Parasitic Check	Ensure SPEF is properly back-annotated.
Exception Check	Verify all timing exceptions.

4) COMMON FIXES

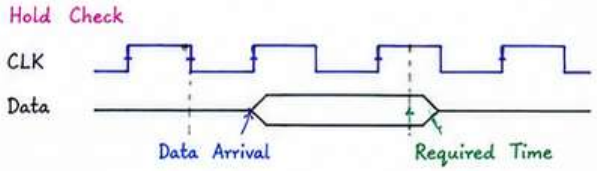
- Setup Violation → Buffering, upsizing, restructuring, logic optimization.
- Hold Violation → Buffer insertion, resizing, routing detour.
- High Delay Path → Reduce logic depth, optimize placement, buffering.
- High Fanout → Add buffer tree.
- Clock Skew → CTS optimization, balance clock tree.
- Congested Area → Re-route, spread cells, improve placement.
- Poor Constraints → Fix false path / multicycle / I/O constraints.
- OCV / Derate Issues → Adjust derate settings, analyze critical paths.

5) TIMING DIAGRAM (SETUP & HOLD)



Setup Slack = Required Time - Arrival Time

Slack > 0 → PASS Slack < 0 → FAIL



Hold Slack = Arrival Time - Required Time

Slack > 0 → PASS Slack < 0 → FAIL

KEY POINT

★ Timing Sign Off ensures all paths meet setup and hold requirements with extracted parasitics under correct constraints before tape-out.

b) MCMM SIGN OFF

1) INTRODUCTION

- MCMM (Multi-Corner Multi-Mode) Sign Off verifies timing across multiple PVT corners, modes and constraints.
- Ensures design robustness under all operating conditions.
- Captures worst-case scenarios for both setup and hold.

2) WHY IT IS CRITICAL?

- Guarantees performance across chip operating range.
- Prevents silicon failures due to PVT/mode variations.
- Required for advanced process nodes with high variations.
- Ensures reliable timing sign off under all real-world conditions.

3) MCMM SCENARIOS

PVT Corners	Modes	Constraints
• SS (Slow-Slow)	• Functional Mode	• Setup
• FF (Fast-Fast)	• Scan Mode	• Hold
• SF (Slow-Fast)	• Low Power Mode	• I/O
• FS (Fast-Slow)	• Test Mode	• Multi-cycle
• TT (Typical-Typical)	• At-Speed / Shift Mode	• False Path

MCMM Analysis = PVT Corners × Modes × Constraints

All combinations must be analyzed and signed off.

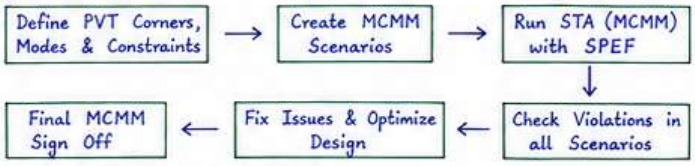
4) COMMON VALIDATIONS

Check Item	Description
All Corners	Verify setup and hold across all PVT corners.
All Modes	Check all functional, scan, low power, test modes.
All Constraints	Validate constraints for every scenario.
WNS / TNS	Ensure positive slack in all scenarios.
OCV / AOCV / POCV	Apply derates for each scenario.
CDC in MCMM	Verify CDC paths across all modes and corners.
Mode Transitions	Check proper constraints during mode changes.
Power States	For low power, verify retention and isolation.
I/O in MCMM	Validate I/O timing for all scenarios.

5) COMMON FIXES

- Corner Violations → Upsize cells, buffering, optimize logic.
- Mode Specific Issues → Add constraints, fix mode logic.
- Hold Failures (Fast) → Add delay, buffers, route detour.
- Setup Failures (Slow) → Reduce logic depth, upsizing.
- OCV Issues → Optimize critical paths, reduce variation.
- Constraint Issues → Correct constraints, false path, multicycle path.

6) MCMM FLOW



7) PRACTICAL OBSERVATIONS

- MCMM Sign off catches issues hidden in single corner analysis.
- Worst case corner for setup is usually SS, for hold is FF.
- Always keep constraints clean and consistent across modes.
- POCV provides more accurate analysis for advanced nodes.

8) VIVA QUESTIONS

Q1. Why is Timing Sign Off important?

Ans: It ensures the design meets all setup and hold requirements before tape-out to guarantee silicon success.

Q2. Why is MCMM Sign Off necessary?

Ans: It verifies timing across multiple PVT corners, modes and constraints to ensure robustness in all conditions.

9) QUICK REVISION POINTS

- Timing Sign Off uses extracted parasitics and final constraints.
- It validates setup, hold, paths, exceptions and derates.
- MCMM Sign Off covers all PVT corners, modes and constraints.
- Ensures design robustness and prevents silicon failures.
- Final sign off must show positive slack in all scenarios.

PD _ Routing

a) Routing Sign Off

1. INTRODUCTION

- Routing sign off verifies that the final routed design meets all physical, electrical and design rule requirements.
- It ensures corting connectivity, required RC limits, no DRC violations and good wiat integrity.

2. WHY IT IS CRITICAL ?

- Avoids routing failures and shorts / opens.
- Ensures timing, power and signal integrity targets are met.
- Prevents metal density, EM and antenna issues.
- Required for tape-out and manufacturing readiness.

3. COMMON VALIDATION AND FIXES

Validation	Purpose	Common Issues	Fixes
DRC Check	Check routing rule compliance	Spacing, width, via violations	Change track, add spacing, resize wire
Connectivity Check	Ensure all nets are connected	Open / short nets	Re-route or add vias
Parasitic Check	Verify RC values (SPEF)	High R/c, coupling	Use better layers, reduce length
Timing Check	Verify setup / hold constraints	High R/cc, violation	Fast beipt / buffer, upsize
Congestion Check	Check routing density	High congestion	High dayers, re-route
SI Check	Validate signal criiating	EN1 crosstalk	Shielding, spacing, layer change

4. DOVANTAGES

- Improves reliability and performance.
- Reduces risk of functional failures.
- Ensures manufacturability and sign-off readiness.
- Avoidd costly ECOs after tape-out.

b) Electromigration Sign Off

1. INTRODUCTION

- Electromigration (EM) sigi off ensures metal lines can carry the required current without damage over time.
- It checks current censity, temperature and lifetime reliability of interconnects.

2. WHY IT IS CRITICAL ?

- Prevents metal voids and open circuits.
- Avoids early failures and reduced lifetime.
- Ensures long term reliability of the chip.
- Essential for high current and long life designs.

3. COMMON VALIDATION AND FIXES

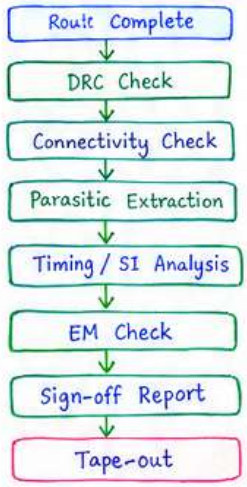
Validation	Purpose	Common Issues	Fixes
Current Density Check	Ensure current < allowed limit	High current density	Increase width, add vias, use lower layers
EM Analysis (Static/Dynamic)	Check for EM stress hotspots	Hotspots at corners, via, narrow lines	Re-route, add vias, redike current
Temperature Check	Verify temperature limits	High temp regions	Improve cooling, optimize routing
Sign-off / Report	Final EM report and lifetime	Failing segments	Fix and re-analyse until clean

4. ADVANTAGES

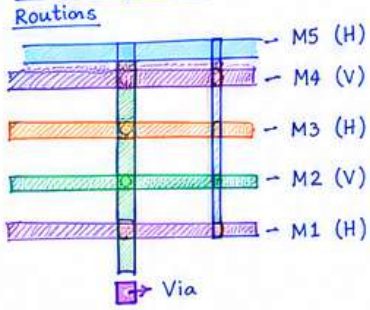
- Improves reliability and lifetime.
- Prevents metal migration and void formation.
- Reduces risk of open/short circuits.
- Ensures long term Performance.

DIAGRAMS

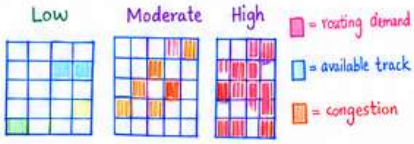
1. Routing Sigmt Of Flow



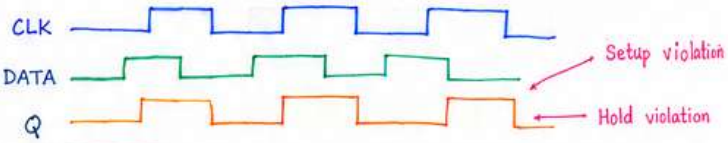
2. Metal Layers & Routions



3. Congestion Example



4. Timing Impact



KEY POINT

Routing sign off ensures correct connectivity, meets DRC, timing, SI and EM limits for a reliable and manufacturable design.

COMPARISON TABLE

Parameter	DRC	Fixs
Purpose	Check design rules	Check Schetibc is. layout
Input	GDS-DEF, tech file	GDS, netlist, LEF/DEF
Output	DRC report	LVS report
Main concern	Spacing, width, via	Connectivity, device match
Typical issue	Rule violations	Open/short, mismatch
Sign-off relevance	Mandatory	Mandatory

6. NTELEWIK / VIVA QS

Q1. Why is routing sign of important ?

→ Ensures connectivity, meets DRC, timing, SI and EM requirements for a reliable design.

Q2. How does EM affect routing ?

→ Causes metal voids and opens, reducing lifetime and reliability.

PRACTICAL OBSERVATION

- Routing must satisfy connectivity as well as DRC constraints.
- Higher layers improve congestion and SI.
- EM and SI checks are critical for high current nets.
- Sign-off avoids costlf ECOs later.

SUMMARY / REVISION POINTS

1. Why is is wor of ipdiuncts & s D. CDC tinoat as comeclald bocatus, fier Sayrfcy an tking iaurad REPts. Chor-tine callating; coalitiics of Emgréat bid tpck concéd:and Mofcan, reatlus. Ragerlóra, wuliaes oetrents.
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5. Routixes lange comguestiins and podlacer and iimling.

5. SUMMARY / REVISION POINTS

1. Routing offB verifis connectivity, tmating, SI and EM.
2. DRC prevents rule violations like spacing and width issues.
3. EVS ensures cavout matches the netlist.
4. Euices validats current density and lifetime.
5. Propcrg tang assignment and via usage reduce congestion.
6. Sign-off ensures a manufacturable and reliable design.

Author : Konda Tharun Vivekanada

PD _ Routing

MANUFACTURING CHECKS – INTRODUCTION, WHY IT IS CRITICAL?, COMMON VALIDATION AND FIXES.

1. INTRODUCTION

Manufacturing checks are final sign-off verifications that ensure the routed layout can be fabricated in the target foundry without any process or reliability violations and can be manufactured with high yield.

2. WHY IT IS CRITICAL ?

- Ensures manufacturability and high yield.
- Prevents costly respins and tape-out delays.
- Satisfies foundry design rules and reliability requirements.
- Reduces risk of functional failures in silicon.
- Ensures compliance with foundry/technology sign-off criteria.

3. MANUFACTURING CHECK FLOW

Final Routed Layout → Run Checks → Analyze Violations → Fix Issues → Re-run Checks → Sign-off Release

4. MANUFACTURING CHECKS OVERVIEW

Check	Purpose	Typical Tool	Output
DRC	Checks design rule violations	Calibre DRC, Pegasus, Assura, PVS	DRC Report
LVS	Compares layout vs schematic	Calibre LVS, Pegasus LVS, Assura LVS	LVS Report
ERC	Checks electrical rule violations	Calibre xRC, Assura ERC	ERC Report
Antenna	Checks antenna ratio violations	Calibre PERC, Pegasus	Antenna Report
DFM	Ensures manufacturability	Calibre DFM, Yield Enhancer	DFM Report
Density	Checks metal density rules	Calibre Density, StarRC	Density Report
CMP	Checks CMP density requirement	Calibre CMP, PERC	CMP Report
Reliability	EM, IR, Stress, TDDB etc.	Calibre xACT, RedHawk	Reliability Report

5. DRC (DESIGN RULE CHECK)

- Verifies that the layout adheres to all the geometric rules defined in the technology LEF/DRC deck.
- Checks spacing, width, enclosure, overlap, extension, notch, min area, layer-to-layer rules etc.

COMMON DRC VIOLATIONS

Violation Type	Description	Fix / Technique
Min. Width	Wire/space width < min. allowed	Increase width
Min. Spacing	Spacing between shapes < rule	Increase spacing
Enclosure	Via/Contact not enclosed properly	Increase enclosure
Min. Area	Area of shape < min. area	Add area (fill or merge)
Extension	Layer not extended enough	Increase extension
Notch / Slot	Notch length < min. rule	Remove/notch using fill
Layer Jump	Direct connection between non-adjacent layers	Add intermediate layer or via

6. LVS (LAYOUT VS SCHEMATIC)

- LVS compares the extracted circuit from layout with the given schematic/netlist.
- Ensures connectivity and device correctness.
- Any mismatch leads to functional failure.

COMMON LVS ERRORS

Error Type	Description	Fix / Technique
Missing Device	Device in schematic not found in layout	Add missing device
Extra Device	Extra device present in layout	Remove extra device
Open Net	Net open in layout	Fix connection
Short Net	Two nets shorted together	Remove short
Pin Mismatch	Pin names / directions mismatch	Rename / correct pins
Device Mismatch	Device type/size mismatch	Correct device

7. ERC (ELECTRICAL RULE CHECK)

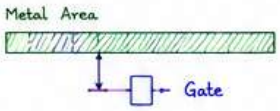
- Checks electrical connectivity rules not covered in LVS.
- Detects floating nets, multiple drivers, pin type conflicts.
- Ensures safe electrical connections.

COMMON ERC VIOLATIONS

Violation	Description	Fix
Floating Net	Net not connected to any driver	Connect driver
Multiple Drivers	More than one driver on a net	Remove extra driver
Pin Type Conflict	Input pin shorted to output pin	Fix connection
Power/GND Short	Power shorted with GND	Remove short
Unconnected Pin	Some pins not connected	Connect or tie-off

8. ANTENNA CHECK

- Checks antenna ratio (metal area to gate area) violations.
- High antenna ratio can damage gate oxide during fabrication.
- Fix: Add antenna diode or increase gate area.



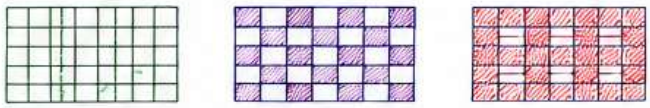
$$\text{Antenna Ratio} = \frac{\text{Metal Area (connected to gate)}}{\text{Gate Area}}$$

9. DFM (DESIGN FOR MANUFACTURABILITY)

- Ensures the layout can be manufactured with high yield.
- Checks complex patterns, forbidden structures, notch, slot,
- min metal density, color conflicts etc.
- Fix: Modify offending patterns, add fill, follow foundry DFM rules.

10. METAL DENSITY CHECK

- Checks that metal density in any window is within the specified range to maintain CMP performance.
- Low density → dishing, High density → erosion.
- Fix: Add metal fill or remove excess metal.



11. RELIABILITY CHECKS (EM, IR, STRESS, TDDB)

Check	Purpose	Fix / Technique
EM (Electromigration)	Ensures current density is within limit	Increase width, add parallel routes, reduce current
IR Drop	Checks voltage drop on power grid	Add straps, increase width, add vias, fix power network
Thermal	Checks temperature hotspots	Improve cooling, increase metal width, add vias
Stress / Mechanical	Checks stress due to pattern density	Add fill, balance density
TDDB / NBTI	Checks device lifetime reliability	Use larger devices, lower stress

12. COMMON FIX STRATEGIES

- Increase width / spacing / enclosure to meet rules.
- Add vias, change layer, or add intermediate layers.
- Add metal fill to fix density and DFM issues.
- Insert antenna diodes or increase gate area.
- Optimize power grid to fix IR drop and EM.
- Modify layout pattern to avoid forbidden structures.
- Iterate until all checks are clean.

KEY POINT

- ✓ Manufacturing checks ensure that the design is manufacturable, reliable and meets all foundry sign-off requirements.
- ✓ Clean reports lead to high yield and successful tape-out.

13. MANUFACTURING SIGN-OFF CHECKLIST

- ✓ DRC Clean
- ✓ LVS Clean
- ✓ ERC Clean
- ✓ Antenna Clean
- ✓ Density in Range
- ✓ CMP Density Clean
- ✓ DFM Clean
- ✓ EM / IR / Thermal Clean
- ✓ Stress Clean
- ✓ Reliability Clean
- ✓ No Waiver Pending
- ✓ Reports Reviewed
- ✓ Sign-off Released

14. INTERVIEW / VIVA Qs

Q1. Why are manufacturing checks important ?
→ They ensure the routed layout can be manufactured with high yield without functional failures.

Q2. What is the purpose of DRC ?
→ To verify that the layout follows all geometric design rules of the technology.

Q3. What causes antenna violation ?
→ Large metal area connected to gate without proper diode or sufficient gate area.

15. PRACTICAL OBSERVATIONS

- → Fixing DRC and density early saves much rework later.
- → Metal fill improves yield and reliability.
- → Always run all checks before final sign-off.
- → Clean sign-off reports are mandatory for tape-out.

Author : Konda Tharun Vivekananda

PD _ Routing

ROUTING CLOSURE CHECKLIST

Physical verification checklist , Timimg signoff checklist ,
Power and Reliability checklist , Functional correctness checklist .

1. PHYSICAL VERIFICATION CHECKLIST

Ensures the routed layout obeys all foundry rules , is clean and free from physical violations .

Check Item	What to Verify	Tool / Method	Sign-off Criteria
☐ DRC	All design rule violations	Calibre DRC, Pegasus DRC	Zero DRC violations
☐ LVS	Layout vs schematic equivalence	Calibre LVS, Pegasus LVS	LVS Matched (0 mismatches)
☐ ERC	Electrical rule violations	Calibre xRC, Assura ERC	Zero ERC violations
☐ Antenna	Antenna ratio violations	Calibre PERC, Pegasus	Antenna OK
☐ DFM	Manufacturability checks	Calibre DFM, Yield Enhancer	DFM clean
☐ Metal Density	Metal density in windows	Calibre Density, StarRC	Within min-max range
☐ CMP	CMP density requirement	Calibre CMP, PERC	Within spec
☐ Litho / Pattern Checks	Forbidden patterns, min feature , notch	Calibre xACT, LithoCheck	All patterns clean
☐ Connectivity	Open / short / bridges	Calibre Connect, Pegasus	100% nets connected
☐ Shield / Spacing	Shield rules and spacing checks	Calibre SI, Assura	No violations
☐ ESD / Latch-up	ESD structures and latch-up rules	Calibre PERC, xRC	ESD & latch-up clean

2. TIMING SIGNOFF CHECKLIST

Ensures all timing constraints are met across PVT and all configured conditions .

Check Item	What to Verify	Tool / Method	Sign-off Criteria
☐ Setup Timing	No negative setup slack	PrimeTime / Tempus	$WNS \geq 0$, $TNS = 0$
☐ Hold Timing	No negative hold slack	PrimeTime / Tempus	$WNS_{hold} \geq 0$, $TNS_{hold} = 0$
☐ Clock Skew	Skew within allowed limit	PrimeTime / Tempus	Skew ≤ target limit
☐ Clock Latency	Clock latency within range	PrimeTime / Tempus	Within design budget
☐ Max Delay	Max delay paths within budget	PrimeTime / Tempus	$WNS_{max} \geq 0$
☐ Min Delay	Min delay paths within budget	PrimeTime / Tempus	Hold $WNS \geq 0$
☐ False Path	All false paths properly defined	PrimeTime / Tempus	No unintended paths
☐ Multicycle Path	MCP constraints correct	PrimeTime / Tempus	Paths honored correctly
☐ I/O Timing	Input/output delays met	PrimeTime / Tempus	$WNS_{IO} \geq 0$
☐ OCV / AOCV	Variation derates applied	PrimeTime / Tempus	Derates enabled
☐ SDF Back Annotation	SDF annotated timing clean	PrimeTime / Tempus	No new violations

KEY POINT
100% timing closure across all corners , modes , derates and constraints is mandatory before tape-out .

3. POWER AND RELIABILITY CHECKLIST

Ensures the design is power efficient and reliable under all operating conditions .

Check Item	What to Verify	Tool / Method	Sign-off Criteria
☐ IR Drop	Voltage drop within limits	Voltus / RedHawk	IR Drop ≤ spec (e.g., 5%)
☐ EM (Electromigration)	Current density within limit	Voltus EM / RedHawk EM	No EM violations
☐ Metal Usage	Metal width and rules	DRC / DM	Within metal rules
☐ Power Consumption	Total power within budget	PrimePower / PowerArtist	Within power target
☐ Dynamic Power	Switching power within budget	PrimePower	Within dynamic budget
☐ Leakage Power	Leakage within limit	PrimePower	Within leakage budget
☐ Hot Spot	Local temperature within limit	Voltus / Thermal	Temp ≤ spec
☐ Antenna Reliability	Antenna ratio within limit	Calibre PERC	Antenna OK
☐ Signal Integrity	SI margin within limit	PrimeTime SI / SI tools	SI slack ≥ 0
☐ Aging / NBTI / PBTI	Aging impact within limit	PrimeTime PX	Aging slack ≥ 0

KEY POINT
Power , IR , EM and thermal checks ensure chip reliability , long life and safe operation in silicon .

4. FUNCTIONAL CORRECTNESS CHECKLIST

Ensures the routed design matches intent and functionality is not affected .

Check Item	What to Verify	Tool / Method	Sign-off Criteria
☐ Connectivity	All nets connected as intended	Netlist Compare / Calibre Connect	100% nets connected
☐ Equivalence Check	Routed netlist vs reference	Formality / Conformal	Equivalence matched
☐ Logic ECO Review	All ECOs correctly implemented	FEC / Manual Review	ECOs verified clean
☐ Scan Chain	Scan connectivity intact	DFT Tools / TetraMAX	Scan chains clean
☐ Clock Tree	No unintended clock gating	CTS Report / PrimeTime	Clock function intact
☐ Tie Cells	Tie high / low cells correct	DRC / LVS / Netlist Check	No floating nets
☐ Power Intent	UPF / CPF rules followed	UPF Check / Conformal	Power intent clean
☐ IO Cells	IO connections and constraints	IO Report / PrimeTime	All IO timing clean

KEY POINT
Functional correctness gives confidence that silicon behaves exactly as intended .

5. OVERALL ROUTING CLOSURE SUMMARY

- ✓ All physical verification checks are clean .
- ✓ All timing checks are clean across all corners and modes .
- ✓ Power , IR drop , EM and thermal are within limits .
- ✓ Signal integrity and antenna are clean .
- ✓ No DRC , LVS , ERC , DFM or reliability violations .
- ✓ Functional equivalence and connectivity are 100% clean .
- ✓ Reports reviewed and sign-off obtained .
- ✓ Design is ready for tape-out .

6. INTERVIEW / VIVA Qs

Q1. Why is routing closure important ?
→ It ensures the routed design meets all physical , timing , power , reliability and functional requirements for silicon success .

Q2. What happens if IR drop is not within limit ?
→ It can cause circuit malfunction , timing failures and even permanent damage .

Q3. What is the purpose of equivalence check ?
→ To ensure the routed netlist is functionally identical to the reference design .

7. QUICK TIPS

- Run all checks in all corners and modes .
- → Always fix root cause , not just symptoms .
- → Use automation for repeatable sign-off .
- Cross verify reports before final sign-off .
- Keep ECOs minimal and well verified .
- Clean sign-off reports are mandatory .

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PD _ Routing

FINAL ROUTING VALIDATION FLOW

- DEFINITION , STEPS FOLLOW IN FLOW , COMMON ISSUES AND FIXES

1. DEFINITION

Final Routing Validation ensures that the fully routed design meets all physical , electrical , timing , reliability and manufacturability requirements before tape-out . It is the last checkpoint to guarantee a clean , sign-off quality layout .

2. STEPS FOLLOW IN VALIDATION FLOW

1. Final Routed Design (DEF / GDS)
2. Design Rule Check (DRC)
3. Layout Vs Schematic Check (LVS)
4. Electrical Rule Check (ERC)
5. Antenna Check
6. Parasitic Extraction (RC Extraction)
7. Timing Signoff (STA / MCMM)
8. Power Analysis (IR Drop / EM / Power)
9. Reliability Checks (EM / Hotspot / NBTI / PBTI)
10. Signal Integrity Sign-off
11. DFM / Manufacturability Checks
12. Functional Equivalence Check
13. Sign-off Reports & Review

→ Tape-out (GDSII / OASIS Release)

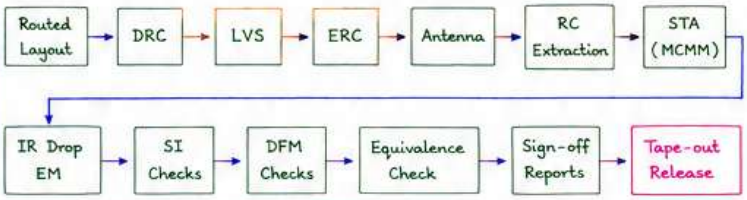
3. DETAILS OF EACH STEP

1. Final Routed Design : Latest DEF/GDS with all ECOs.
2. DRC : Checks layout against foundry design rules.
3. LVS : Compares layout vs schematic for correctness.
4. ERC : Verifies electrical connectivity & rules.
5. Antenna Check : Ensures antenna ratio within limit.
6. RC Extraction : Extract R,C parasitics from routed layout.
7. Timing Signoff : STA across all modes , corners , scenarios.
8. Power Analysis : IR drop , EM current , power numbers.
9. Reliability Checks : EM , Hotspot , NBTI , PBTI , Aging.
10. SI Sign-off : Signal integrity checks for crosstalk , noise.
11. DFM Checks : Metal density , CMP , litho , min feature etc.
12. Functional Equivalence : Ensures logic behavior unchanged.
13. Sign-off Review : All reports reviewed and approved.

4. COMMON ISSUES AND FIXES

Stage	Common Issues	Fixes / Techniques
DRC	Spacing , width , enclosure , notch , min-area violations	Auto DRC fix , resize , add fill , adjust spacing
LVS	Missing / extra device , short / open nets	Check connectivity , fix device , re-route
ERC	Floating nets , multiple drivers , pin conflicts	Fix floating , add tie cell , resolve driver conflicts
Antenna	High antenna ratio	Add antenna diode , increase gate area
Timing	Setup / Hold violations , high negative slack	Buffer insertion , resize , re-route , opt design
IR Drop	Voltage drop beyond limit	Add power strap , widen rails , add vias
EM	High current density	Enlarge metal width , add vias , layer change
SI	Crosstalk , noise , glitch , ringing	Increase spacing , shield , re-route , slow down
DFM	Metal density , CMP dishing , min feature	Add fill , tweak density , follow DFM rules
Functionality	Logic mismatch after ECO	Run equivalence check , fix netlist / layout

5. EXAMPLE : FINAL VALIDATION FLOW DIAGRAM



KEY POINT

- ✓ Final routing validation guarantees that the design is correct , reliable , manufacturable and ready for tape-out .
- ✓ No critical violations should be left .
- ✓ All sign-off reports must be clean and approved .

6. BENEFITS

- Ensures first pass silicon success .
- Avoids costly respins and schedule delays .
- Improves reliability and product life .
- Ensures compliance with foundry sign-off criteria .
- Builds confidence in final tape-out .

7. PRACTICAL OBSERVATION

- Always run all checks in final stage .
- Fix root cause , not just symptoms .
- Review worst mode / corner results .
- Cross verify reports before sign-off .
- Clean sign-off is mandatory for tape-out .

8. INTERVIEW / VIVA Qs

Q1. Why is final routing validation important ?
→ It ensures the routed design meets all physical , electrical , timing , reliability and manufacturability requirements before tape-out .

Q2. What happens if DRC violations exist in final layout ?
→ Fabrication may fail , leading to yield loss and costly respins .

Q3. What is equivalence check in final validation ?
→ It verifies that the routed netlist matches the reference netlist to ensure functional correctness .

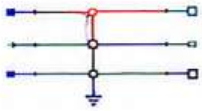
SUMMARY / REVISION POINTS

1. Final validation is the last and most important sign-off step .
2. It includes DRC , LVS , ERC , Antenna , STA , IR , EM , SI , DFM and Eq check .
3. All violations must be fixed before report sign-off .
4. Clean sign-off reports ensure successful tape-out .
5. Multiple corners , modes and scenarios must be verified .

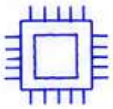
9. SIGN-OFF CHECKLIST (QUICK VIEW)

Physical Verification	Timing Signoff	Power & Reliability	Signal Integrity	Functional Correctness	Sign-off
✓ DRC Clean	✓ Setup Clean (All modes)	✓ IR Drop within limit	✓ Crosstalk Clean	✓ Netlist Equivalence OK	✓ All Reports Clean
✓ LVS Clean	✓ Hold Clean (All modes)	✓ EM within limit	✓ Noise Within Limit	✓ ECOs Implemented	✓ Waivers Closed
✓ ERC Clean	✓ No Critical Paths	✓ Hotspot within limit	✓ Glitch / Ringing OK	✓ Scan Chain OK	✓ Peer Review Done
✓ Antenna OK	✓ WNS ≥ 0	✓ Power within budget	✓ Shielding OK	✓ Clock Tree OK	✓ Sign-off Approval
✓ DFM Clean	✓ TNS ≤ 0	✓ Aging (NBTI/PBTI) OK	✓ Signal Quality OK	✓ No Floating Nets	✓ Tape-out Ready

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PD _ Routing



ADVANCED ROUTING AND INDUSTRY PRACTICES

1. INTRODUCTION

- Advanced routing techniques and industry best practices are used to achieve high performance, low power, reliable and manufacturable routed designs.
- It addresses congestion, RC control, IR drop, SI, EM and ensures sign-off quality for tape-out.

2. ADVANCED ROUTING TECHNIQUES

① Layer Assignment Strategy
- Critical nets → top thick metal layers (low R).
- Non-critical nets → lower layers.
- Balance wirelength, resistance and congestion.

② Track and Grid Optimization
- Adjust routing tracks per layer based on demand.
- Use non-default grids to reduce congestion.
- Preferred direction routing for efficient usage.

③ Multi-Pattern / Multi-Track Routing
- Use multi-patterning (color aware) to avoid litho conflicts.
- Multi-track for wide buses to reduce coupling and RC.

④ Shielding and Spacing Optimization
- Shield aggressor nets with GND / PWR wires.
- Maintain min spacing to reduce crosstalk.
- Use double / triple shielding for sensitive nets.

⑤ Layer Promotion & Detouring
- Promote congested nets to higher layer.
- Detour wires intelligently using router cost functions.

⑥ RC Aware and SI Aware Routing
- Consider extracted RC during detailed routing.
- Control crosstalk, slew, delay and reflection.
- Use SI aware cost metrics in router.

3. INDUSTRY BEST PRACTICES

→ Establish clear routing guidelines and layer usage plan.
→ Early congestion prediction and mitigation.
→ Incremental ECO friendly routing flow.
→ Timing, SI, IR drop feedback during routing iterations.
→ Re-use routing from similar blocks / IPs.
→ Automate verification and sign-off flows.
→ Maintain design for manufacturability (DFM) compliance.

KEY POINT
Advanced routing ensures optimal wirelength, minimal RC, better signal integrity, reduced power and high manufacturability with correct sign-off closure.

4. ROUTING CONSIDERATIONS

Consideration	Why Important
Congestion	High congestion causes detours and delay.
RC Delay	Affects performance and timing closure.
Crosstalk (SI)	May cause noise, glitches and setup/hold fails.
IR Drop	Voltage drop affects speed and reliability.
Electromigration	High current density leads to failure.
Antenna Effect	Causes oxide breakdown during fabrication.
DFM / Lithography	Ensure patterns are manufacturable.
ECO Friendliness	Allow easy fixes in later stages.

5. ROUTING FLOW WITH ADVANCED PRACTICES



* Feedback loops between routing and optimization.
* Use RC & SI aware routing engines.
* Iterate until all sign-off criteria are met.

6. COMPARISON : CONVENTIONAL vs ADVANCED ROUTING

Parameter	Conventional Routing	Advanced Routing
Layer Usage	Mostly lowest layers	Optimized layer assignment
RC Consideration	Less RC aware	RC aware routing
SI Handling	Basic spacing rules	SI aware, shielding, crosstalk control
Congestion	More congestion, detours	Congestion driven optimization
Optimization	Minimal post route opt.	Aggressive post route opt.
Lithography	Basic DRC clean	Multi-patterning, DFM clean
ECO Support	Difficult to modify	ECO friendly routing flow

7. COMMON CHALLENGES AND SOLUTIONS

→ High Congestion : Layer promotion, width adjustment, detour.
→ Crosstalk Issues : Shielding, spacing, layer separation.
→ RC Delay High : Use higher metals, buffer insertion.
→ IR Drop : Wider power straps, add vias, upsize rails.
→ EM Violations : Increase width, add vias, reduce current.
→ Antenna Violations : Add diodes, increase gate area.

8. INDUSTRY TOOLS (EXAMPLES)

- Global & Detailed Routing : Innovus, ICC2, Fusion, Aprisa
- SI Analysis : PrimeRoute SI, RedHawk SI
- IR Drop / EM : Voltus, RedHawk, Helix
- Parasitic Extraction : StarRC, Calibre xRC, Quantus
- Sign-off Verification : Calibre, Pegasus, Questa DRC

Synopsys
Cadence
Siemens
Ansys

9. PRACTICAL OBSERVATIONS

- Start with good floorplan & placement to avoid routing congestion.
→ Keep critical nets short and on top metal layers.
→ Always run verification after every major iteration.
→ Early detection of issues saves time in final sign-off.

10. INTERVIEW / VIVA Qs

Q1. What is advanced routing ?
→ Techniques used to optimize wirelength, RC, SI, power and reliability.

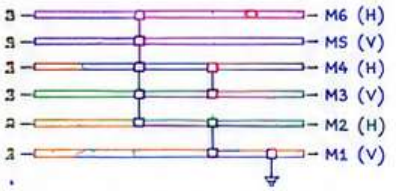
Q2. Why use shielding in routing ?
→ To reduce crosstalk and improve signal integrity.

Q3. Why is layer assignment important ?
→ To balance congestion, RC and performance.

11. SUMMARY / REVISION POINTS

1. Advanced routing improves performance, power, SI and reliability.
2. Techniques include layer assignment, shielding, multi-patterning, RC/SI aware routing.
3. Industry practices focus on congestion, DFM, ECO friendliness and early feedback.
4. Verification and sign-off at every stage ensures tape-out success.
5. Correct routing decisions reduce costly re-spins and improve product quality.

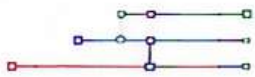
Metal Layer Stack (Example)

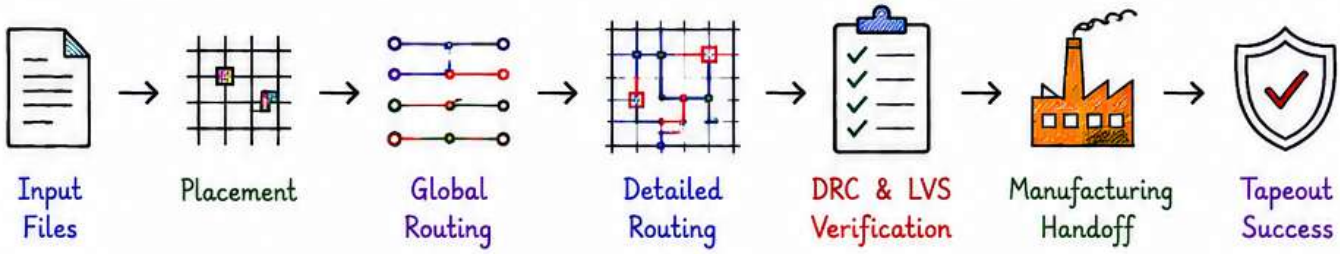


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PHYSICAL DESIGN ROUTING

THANK YOU!

Thank you for your time, interest and support in exploring Routing in Physical Design.

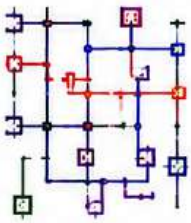
Hope these notes help you understand the concepts clearly and work confidently with Routing and GDS II files.

KEY TAKEAWAYS

- ☆ Routing connects all placed cells using metal layers to form a complete design.
- ☆ Global routing finds the path and guides detailed routing.
- ☆ Accurate routing ensures DRC clean, LVS clean and high reliability.
- ☆ Good routing = Good silicon.

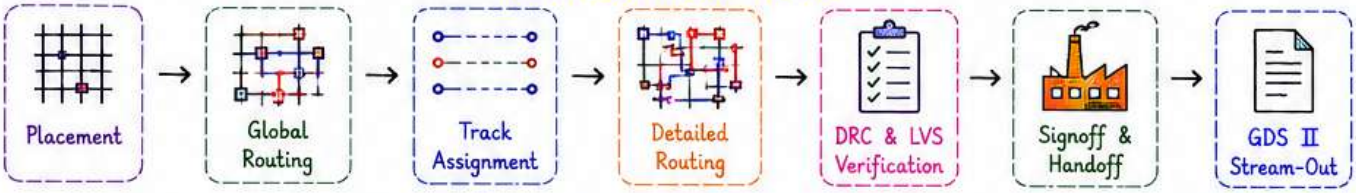
WHAT GOOD ROUTING DELIVERS

- ✓ Clean DRC and LVS
- ✓ Correct connectivity
- ✓ Optimized wirelength & area
- ✓ Better performance & reliability
- ✓ Robust chip & smooth tapeout
- ✓ Happy Foundry ☺



> "A clean, verified layout is not the end, It is the beginning of a successful silicon."

ROUTING FLOW RECAP



Keep learning, keep building, keep innovating!
Wishing you all the very best in your journey of Physical Design!

CLK

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VDD
GND